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Abstract

Using hundreds of significant anomalies as testing portfolios, this paper compares the performance of major empirical asset pricing models. The q -factor model and a closely related five-factor model are the two best performing models among a long array of models. The q -factor model outperforms the five-factor model in factor spanning tests and in explaining momentum and profitability anomalies, but the five-factor model has an edge in explaining value-versus-growth anomalies. Investment and profitability, not liquidity, are the key driving forces in the broad cross section of expected stock returns.

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1 Introduction

This paper compares the performance of a large array of empirical asset pricing models in explaining hundreds of significant anomalies in the broad cross section. We consider the classic models, including the Capital Asset Pricing Model (CAPM), the Fama-French (1993) three-factor model, the Carhart (1997) four-factor model, and the Pastor-Stambaugh (2003) model that adds their liquidity factor to the three-factor model. In addition, we consider newly proposed factor models including the Hou-Xue-Zhang (2015) q -factor model and the Fama-French (2015) five-factor model.

In factor spanning tests, from January 1967 to December 2014, the q -factor alphas of the Fama-French (2015) RMW (robustness-minus-weak profitability) and CMA (conservative-minus-aggressive investment) factors are 0.04% and 0.01% per month ($t = 0.42$ and 0.32). However, the five-factor alphas of the investment and ROE factors in the q -factor model are 0.12% and 0.45% ($t = 3.35$ and 5.6), respectively. As such, RMW and CMA seem to be noisy versions of the q -factors. The q -factor model also explains the Carhart (1997) momentum factor, UMD. The average return of UMD is 0.67% ($t = 3.66$), but its q -factor alpha is only 0.11% ($t = 0.43$). In contrast, the five-factor model cannot explain UMD, with an alpha of 0.69% ($t = 3.11$).

The q -factor model and the five-factor model seem to be the best performing models in explaining anomalies. Across the 161 significant anomalies with New York Stock Exchange (NYSE) breakpoints and value-weighted returns, the average magnitude of the high-minus-low alphas is 0.26% per month in the q -factor model and 0.37% in the five-factor model. The number of significant high-minus-low alphas is 46 in the q -factor model and 84 in the five-factor model. The number of rejections by the Gibbons, Ross, and Shanken (1989, GRS) test is 107 in the q -factor model and 108 in the five-factor model. Across the 216 significant anomalies with all-but-micro breakpoints and equal-weighted returns, the average magnitude of the high-minus-low alphas is 0.26% in the q -factor model and 0.38% in the five-factor model. The number of significant high-minus-low alphas is 66 in the q -factor model and 128 in the five-factor model. However, the number of rejections by

the GRS test is lower in the five-factor model, 151, than 172 in the q -factor model.

The q -factor model outperforms the five-factor model as well as the Carhart (1997) model that contains a momentum factor in explaining momentum. Across the 37 significant momentum anomalies with NYSE breakpoints and value-weighted returns, the average winner-minus-loser alpha is 0.26% per month in the q -factor model, 0.3% in the Carhart model, and 0.65% in the five-factor model. The number of significant winner-minus-loser alphas is nine in the q -factor model, which is lower than 18 in the Carhart model and 35 in the five-factor model. The q -factor model also outperforms the five-factor model in the profitability category. The two models are largely comparable in the investment, intangibles, and trading frictions categories, but the five-factor model has an edge in the value-versus-growth category, benefited from having the value factor, HML.

Adding the Pastor-Stambaugh (2003) liquidity factor to the three-factor model adds little explanatory power. Across the 161 significant anomalies with NYSE breakpoints and value-weighted returns, the three-factor and Pastor-Stambaugh models have the identical average magnitude of the high-minus-low alphas, 0.49% per month, and the same mean absolute alpha across all the deciles, 0.144%. Across the 216 significant anomalies with all-but-micro breakpoints and equal-weighted returns, the two models have virtually identical average magnitudes of the high-minus-low alphas, 0.55–0.56%, and the same mean absolute alpha, 0.142%. In all, economic fundamentals including investment and profitability, not liquidity, dominate the broad cross section of expected returns.

Workhorse factor models for estimating expected stock returns are of immense importance, both in academic research and investment management practice (Ang 2014). Our key insight is that the q -factor model and the closely related five-factor model are the two best performing models among a long array of models and across a vast universe of testing assets. In addition, the q -factor model outperforms the five-factor model in factor spanning tests and in explaining momentum and profitability anomalies, but the five-factor model has an edge in explaining value-versus-growth anomalies. Our work has important implications for stock valuation, capital budgeting, mutual

fund performance evaluation, and investment management, among many other applications.

The rest of the paper is organized as follows. Section 2 reports factor spanning tests. Section 3 compares factor models in explaining hundreds of anomalies. Finally, Section 4 concludes.

2 Factor Spanning Tests

Monthly returns are from the Center for Research in Security Prices (CRSP) and accounting information from the Compustat Annual and Quarterly Fundamental Files. Financial firms and firms with negative book equity are excluded. We follow Hou, Xue, and Zhang (2015, 2017) in constructing the q -factors (see Appendix A for details). The sample is from January 1967 to December 2014. We construct the market factor, MKT, as value-weighted market returns minus one-month Treasury bill rates from CRSP. The data of SMB and HML in the three-factor model, SMB, HML, RMW, and CMA in the five-factor model, as well as UMD are from Kenneth French's Web site. The data of the Pastor-Stambaugh liquidity factor, LIQ, are from Robert Stambaugh's Web site.

Table 1 reports factor spanning tests in the sample from January 1967 to December 2014 (the sample of LIQ starts in January 1968). Panel A shows that the size, investment, and ROE factors in the q -factor model earn on average 0.32%, 0.43%, and 0.56% per month ($t = 2.42, 5.08$, and 5.24), respectively. The investment and ROE factor premiums cannot be explained by the Carhart model, with alphas of 0.29% ($t = 4.57$) and 0.51% ($t = 5.58$), or the Pastor-Stambaugh model, with alphas of 0.35% ($t = 5.73$) and 0.75% ($t = 7.61$), respectively. The loadings of q -factor returns on LIQ are close to zero. Finally, the five-factor model cannot explain the q -factor premiums either, with alphas of 0.12% ($t = 3.35$) and 0.45% ($t = 5.6$), respectively.

Panel B shows that SMB, HML, RMW, and CMA earn on average 0.26%, 0.36%, 0.27%, and 0.34% per month ($t = 1.92, 2.57, 2.58$, and 3.63), respectively. The Carhart alphas of RMW and CMA are 0.33% ($t = 3.31$) and 0.19% ($t = 2.83$), and their Pastor-Stambaugh alphas are 0.34% ($t = 3.19$) and 0.24% ($t = 3.71$), respectively. Most important, the q -factor model explains the

average RMW and CMA returns, leaving tiny alphas of 0.04% ($t = 0.42$) and 0.01% ($t = 0.32$), respectively. As such, RMW and CMA are likely noisy versions of the q -factors. The q -factor model also explains the HML return, with an alpha of 0.03% ($t = 0.28$).

Panel C shows that UMD is on average 0.67% per month ($t = 3.66$). The q -factor model has a small alpha of 0.11% ($t = 0.43$). The ROE-factor loading is 0.91 ($t = 5.59$). The five-factor model cannot capture UMD, with an alpha of 0.69% ($t = 3.11$). The RMW loading is only 0.25 ($t = 1.23$). The Pastor-Stambaugh model cannot explain UMD either, with a large alpha of 0.89% ($t = 5.25$). Panel D shows that the LIQ premium is 0.42% ($t = 2.81$). None of the other factor models can explain LIQ, and all leave significantly positive alphas for LIQ.

Panel E reports pairwise correlations for the factors. The investment factor has a high correlation of 0.69 with HML, and the ROE factor has a high correlation of 0.49 with UMD. Both are highly significant. The investment factor has an almost perfect correlation of 0.91 with CMA, but the ROE factor has a lower correlation of 0.68 with RMW. LIQ is largely orthogonal to all the other factors. Its correlations with the other factors are all economically small and statistically insignificant.

3 Explaining Anomalies in the Broad Cross Section

We turn our attention to explain anomalies in the broad cross section. We describe the testing portfolios in Section 3.1, discuss the overall performance of factor models in Section 3.2, and report factor regressions, including alphas in Section 3.3 as well as factor loadings in Section 3.4.

3.1 Testing Portfolios

We use the large data library consisting of 437 anomalies from Hou, Xue, and Zhang (2017). Table A1 provides the list of anomaly variables (Hou et al. detail variable definition and portfolio construction). There are 57, 68, 38, 78, 100, and 96 variables across the momentum, value-versus-growth, investment, profitability, intangibles, and trading frictions categories, respectively. With NYSE breakpoints and value-weighted returns (NYSE-VW), Hou et al. document 161 significant anoma-

lies at the 5% level, including 37, 31, 27, 33, 26, and 7 variables across the six categories, respectively.

Fama and French (2015) argue that value-weighted portfolio returns can be dominated by a few big stocks, but the most serious challenges for asset pricing models are in small stocks, which value-weighted portfolios tend to underweight. To address this concern, we also form testing deciles with all-but-micro breakpoints and equal-weighted returns (ABM-EW). We exclude microcaps from the NYSE-Amex-NASDAQ universe, use the remaining stocks to calculate breakpoints, and then equal-weight all the stocks within a given decile to give small stocks sufficient weights in the portfolio. By construction, microcaps are excluded. In this set of deciles, we find 216 anomaly variables with significant high-minus-low decile returns, including 49, 38, 36, 47, 29, and 16 across the momentum, value-versus-growth, investment, profitability, intangibles, and trading frictions categories, respectively. Table A2 in Appendix B details the remaining insignificant anomalies.

3.2 Overall Performance

We examine six return factor models, including the CAPM, the Fama-French (1993) three-factor model, the Carhart (1997) model, the Pastor-Stambaugh (2003) model, the Fama-French (2015) five-factor model, and the q -factor model. We use four measures of overall performance, including the average magnitude of the high-minus-low alphas, the number of significant high-minus-low alphas at the 5% level, the mean absolute alpha across all the anomaly deciles, and the number of the sets of anomaly deciles across which a factor model is rejected by the GRS test.

From Panel A of Table 2, across the 161 significant anomalies with NYSE-VW, the average magnitude of the high-minus-low alphas is 0.26% per month in the q -factor model, in contrast to 0.36% in the Carhart model and 0.37% in the five-factor model. The number of significant high-minus-low alphas is 46 in the q -factor model, which is lower than 84 in the five-factor model and 94 in the Carhart model. The mean absolute alpha across all the deciles is 0.122% in the q -factor model, in contrast to 0.126% in the Carhart model and 0.13% in the five-factor model. Finally, the number of rejections by the GRS test is 107 in the q -factor model, 108 in the five-factor model, and 119 in the Carhart model.

From Panel B, across the 216 significant anomalies with ABM-EW, the average magnitude of the high-minus-low alphas is 0.26% per month in the q -factor model, which is lower than 0.38% in the five-factor model and 0.42% in the Carhart model. The number of significant high-minus-low alphas is 66 in the q -factor model, which is lower than 128 in the five-factor model and 154 in the Carhart model. However, the five-factor model has the lowest mean absolute alpha across all the testing deciles, 0.115%, in contrast to 0.145% in the q -factor model and 0.171% in the Carhart model. The number of rejections by the GRS test is also lowest in the five-factor model, 151, in contrast to 172 in the q -factor model and 183 in the Carhart model.

In the descending ranking of overall performance, the next models are the three-factor model and the Pastor-Stambaugh model. Surprisingly, the Pastor-Stambaugh liquidity factor adds little explanatory power in the broad cross section. Across the 161 significant anomalies with NYSE-VW, the three-factor and Pastor-Stambaugh models have the identical average magnitude of the high-minus-low alphas, 0.49% per month, and the same mean absolute alpha, 0.144%. Adding the liquidity factor reduces the number of significant high-minus-low alphas slightly from 116 to 113, and the number of rejections by the GRS test from 128 to 126. Across the 216 significant anomalies with ABM-EW, the two models have similar magnitudes of the high-minus-low alphas, 0.55–0.56%, and the same mean absolute alpha, 0.142%. Adding the liquidity factor reduces the number of significant high-minus-low alphas from 185 to 184, and the number of rejections by the GRS test from 173 to 172. In all, the evidence is consistent with Table A2, which shows that a vast majority of trading frictions variables are insignificant in the broad cross section.

Not surprisingly, the CAPM is ranked at the bottom. The average magnitude of the high-minus-low alphas is 0.56% per month with NYSE-VW, and 0.67% with ABM-EW. Across the two sets of testing deciles, almost all the anomalies with significant high-minus-low average returns also have significant CAPM alphas, 152 out of 161 (or 94%) and 212 out of 216 (or 98%), respectively. As such, using the significance of the CAPM alphas for the high-minus-low deciles to select significant anomalies would yield largely similar results as using the significant of average returns as the yardstick.

3.2.1 Performance by Category

Across different categories, the q -factor model outperforms the five-factor model in explaining momentum and profitability anomalies. The two models are largely comparable in the investment, intangibles, and trading frictions categories, but the five-factor model has an edge in the value-versus-growth category. In the momentum category, with NYSE-VW, the average magnitude of the winner-minus-loser alphas across the 37 significant anomalies is 0.26% per month in the q -factor model, which is lower than 0.3% in the Carhart model and 0.65% in the five-factor model. The number of significant alphas is nine in the q -factor model, which is even lower than 18 in the Carhart model that includes UMD as an explanatory factor. Almost all the alphas, 35 out of 37, in the five-factor model are significant. The mean absolute alpha across all the momentum deciles is 0.11% in the q -factor model, which is close to 0.109% in the Carhart model, but lower than 0.16% in the five-factor model. The number of rejections by the GRS test is 25 in the q -factor model, which is close to 27 in the Carhart model, but lower than 35 in the five-factor model.

With ABM-EW, the average magnitude of the winner-minus-loser alphas across the 50 significant anomalies is 0.28% per month in the q -factor model, in contrast to 0.31% in the Carhart model and 0.61% in the five-factor model. The number of significant alphas is 14 in the q -factor model, which is lower than 27 in the Carhart model and 44 in the five-factor model. The mean absolute alpha across all the deciles is 0.133% in the q -factor model, which is lower than 0.14% in the Carhart model and 0.155% in the five-factor model. The number of rejections by the GRS test is 37 in the q -factor model, in contrast to 34 in the Carhart model and 43 in the five-factor model.

In the value-versus-growth category, the average magnitude of the high-minus-low alphas across the 31 significant anomalies with NYSE-VW is 0.23% per month in the q -factor model and 0.25% in the Carhart model, but only 0.13% in the five-factor model. The number of significant high-minus-low alphas is six in the q -factor model and ten in the Carhart model, but only two in the five-factor model. The five-factor model also has the lowest mean absolute alpha, 0.093%, in contrast to

0.121% in the q -factor model and 0.118% in the Carhart model, as well as the lowest number of rejections by the GRS test, ten, in contrast to 18 in the q -factor model and 15 in the Carhart model. The relative performance of the q -factor model improves with ABM-EW. Across the 38 significant anomalies, the average magnitude of the high-minus-low alphas is 0.19%, which is close to 0.18% in the five-factor model, and lower than 0.36% in the Carhart model. The number of significant alphas is only two in the q -factor model, but seven in the five-factor model and 22 in the Carhart model.

In the investment category, the average magnitude of the high-minus-low alphas across the 27 significant anomalies with NYSE-VW is 0.19% per month in the q -factor model, which is lower than 0.22% in the five-factor model and 0.28% in the Carhart model. Seven high-minus-low alphas are significant in the q -factor model, in contrast to 11 five-factor alphas, and 17 Carhart alphas. The mean absolute alphas across the deciles are largely comparable: 0.099% in the q -factor model, 0.09% in the five-factor model, and 0.115% in the Carhart model. The number of rejections by the GRS test is 17 in the q -factor model, which is close to 16 in the five-factor model, but lower than 24 in the Carhart model. The evidence with ABM-EW is largely similar. The average magnitude of the high-minus-low alphas is lower in the q -factor model than the five-factor model, 0.28% versus 0.35%, but the mean absolute alpha is higher, 0.136% versus 0.094%.

In the profitability category, the average magnitude of the high-minus-low alphas across the 33 significant anomalies with NYSE-VW is 0.23% per month in the q -factor model, which is lower than 0.39% in the five-factor model and 0.52% in the Carhart model. The number of significant high-minus-low alphas is nine in the q -factor model, in contrast to 23 in the five-factor model and 29 in the Carhart model. The mean absolute alpha across the deciles is also the lowest in the q -factor model, 0.121%, in contrast to 0.139% in the Carhart model and 0.161% in the five-factor model. The number of rejections by the GRS test is 20 in the q -factor model, which is lower than 26 in the five-factor model and 30 in the Carhart model. With ABM-EW, the average magnitude of the high-minus-low alphas across 47 significant anomalies is 0.22%, in contrast to 0.37% in the five-factor model and 0.55% in the Carhart model. The number of significant high-minus-low alphas is 11 in the

q -factor model, in contrast to 27 in the five-factor model and 39 in the Carhart model. However, the mean absolute alpha is 0.141% in the q -factor model, which is higher than 0.116% in the five-factor model, but lower than 0.18% in the Carhart model. The number of rejections by the GRS test is 38 in the five-factor model, which is lower than 42 in the q -factor model and 45 in the Carhart model.

The q -factor and five-factor models are largely comparable in the remaining categories. The average magnitude of the high-minus-low alphas across the 26 significant intangibles anomalies with NYSE-VW is 0.41% per month, which is close to 0.39% in the five-factor model, but lower than 0.49% in the Carhart model. This average magnitude is higher in the q -factor model than the five-factor model across the seven trading frictions anomalies with NYSE-VW, 0.24% versus 0.2%, but lower across the 16 significant anomalies with ABM-EW, 0.12% versus 0.18%. However, the five-factor model has a mean absolute alpha of only 0.08% across the 16 deciles, in contrast to 0.152% in the q -factor model, although the difference is smaller, 0.081% versus 0.102%, with NYSE-VW.

3.3 Alphas

We detail factor regressions, with this subsection on alphas, and the next subsection on betas. Table 3 shows, for the 161 significant anomalies with NYSE-VW, the high-minus-low alphas and their t -statistics, as well as mean absolute alphas across a given set of deciles and the corresponding GRS p -values. Table 4 reports the results for the 216 significant anomalies with ABM-EW. To save space, we restrict the scope of our discussion to the two best performing models, which are the q -factor and five-factor models. For momentum, we also discuss the Carhart model.

3.3.1 Momentum

Columns 1–37 in Table 3 present the results for the 37 significant momentum anomalies with NYSE-VW, and columns 1–50 in Table 4 for the 50 significant variables with ABM-EW. The q -factor model outperforms the Carhart model, which in turn outperforms the five-factor model in this category.

For standardized unexpected earnings (Sue) with NYSE-VW, the average return of the high-

minus-low decile is significant only at the 1-month horizon (Sue1), 0.47% per month ($t = 3.42$). The q -factor model captures this average return, with a tiny alpha of 0.05% ($t = 0.4$). In contrast, the Carhart alpha is 0.43% ($t = 3.61$), and the five-factor alpha is 0.51% ($t = 3.69$). With ABM-EW, both Sue1 and Sue6 are significant, with average returns of 0.84% ($t = 6.31$) and 0.4% ($t = 3.59$), respectively. The q -model alphas are 0.37% ($t = 3.5$) and 0.00% ($t = 0.03$), which are smaller than the Carhart alphas of 0.74% ($t = 6.21$) and 0.36% ($t = 3.6$), and the five-factor alphas of 0.84% ($t = 6.73$) and 0.43% ($t = 4.06$), respectively. However, all the models are still rejected by the GRS test.

For prior 6-month returns at the 1-, 6-, and 12-month horizons (R^6_1 , R^6_6 , and R^6_{12}), the winner-minus-loser average returns with NYSE-VW are significant, 0.6%, 0.82%, and 0.55% per month ($t = 2.04, 3.49$, and 2.9), respectively. The Carhart alphas are -0.26% , 0.08% , and 0.09% ($t = -1.31, 0.79$, and 0.9), and the q -factor alphas are -0.04% , 0.24% , and 0.16% ($t = -0.1, 0.78$, and 0.75), in contrast to the large and significant five-factor alphas, 0.73% , 0.97% , and 0.77% ($t = 2.11, 3.5$, and 3.93), respectively. With ABM-EW, the average returns are 1.06% , 0.91% , and 0.56% , all of which are significant. The Carhart alphas are 0.18% , 0.04% , and 0.01% , and the q -factor alphas are 0.38% , 0.08% , and 0.01% , all of which are within one standard error from zero. In contrast, the five-factor alphas are 1.13% , 0.91% , and 0.68% ($t = 3.26, 2.8$, and 2.91), respectively.

Several alternative measures of earnings momentum deliver stronger results than the more popular Sue, including cumulative abnormal returns around earnings announcement (Abr), revisions in analysts' earnings forecasts (Re), and change in analysts' forecasts (dEf).¹ At the 1-month horizon, the winner-minus-loser average returns from sorting on Abr, Re, and dEf with NYSE-VW are 0.74% , 0.81% , and 1.03% per month ($t = 5.85, 3.28$, and 4.65), respectively. the Carhart alphas are 0.63% , 0.52% , 0.76% ($t = 4.62, 2.61$, and 3.85), the q -factor alphas 0.66% , 0.11% , and 0.64% ($t = 4.49, 0.45$, and 2.81), and the five-factor alphas 0.85% , 0.88% , and 1.22% ($t = 6.12, 3.46$, and 5.23), respectively. With ABM-EW, the winner-minus-loser average returns are 0.95% , 0.76% , and

¹dEf is the month-to-month change in the consensus mean forecast of earnings per share, whereas Re is the 6-month moving average of prior changes in analysts' earnings forecasts scaled by share price.

1.2% ($t = 8.67, 4.01, \text{ and } 6.23$), the Carhart alphas 0.87%, 0.45%, and 0.98% ($t = 8.6, 2.66, \text{ and } 5.81$), the q -factor alphas 0.85%, 0.24%, and 0.95% ($t = 5.66, 1.43, \text{ and } 4.54$), and the five-factor alphas 1.01%, 0.82%, and 1.37% ($t = 8.13, 4.53, \text{ and } 6.57$), respectively.

We also examine several new momentum variables, including customer momentum (Cm) (Cohen and Frazzini 2008), as well as supplier industries momentum (Sim) and customer industries momentum (Cim) (Menzly and Ozbas 2010). At the 1-month horizon, the high-minus-low deciles formed on Cm, Sim, and Cim earn average returns of 0.79%, 0.77%, and 0.78% per month ($t = 3.74, 3.37, \text{ and } 3.45$) with NYSE-VW, and 0.53%, 1.15%, and 1% ($t = 2.78, 5.24, \text{ and } 4.12$) with ABM-EW, respectively. The Carhart alphas are 0.76%, 0.51%, and 0.65% ($t = 2.98, 2.19, \text{ and } 2.98$) with NYSE-VW, and 0.43%, 0.99%, and 0.8% ($t = 1.94, 4.3, \text{ and } 3.5$) with ABM-EW, the q -factor alphas 0.72%, 0.54%, and 0.64% ($t = 2.75, 1.65, \text{ and } 2.29$) with NYSE-VW, and 0.38%, 0.95%, and 0.87% ($t = 1.48, 2.76, \text{ and } 2.53$) with ABM-EW, and the five-factor alphas 0.82%, 0.81%, and 0.76% ($t = 3.52, 2.76, \text{ and } 2.99$) with NYSE-VW, and 0.53%, 1.17%, and 1.06% ($t = 2.38, 3.98, \text{ and } 3.45$) with ABM-EW, respectively. However, the average returns are more than halved, once the horizon extends to 6-month, and are further weakened at the 12-month.

3.3.2 Value-versus-growth

Columns 38–68 in Table 3 report the results for the 31 significant value-versus-growth anomalies with NYSE-VW, and columns 51–88 in Table 4 for the 38 significant anomalies with ABM-EW.

The high-minus-low book-to-market (Bm) decile earns an average return of 0.59% per month ($t = 2.84$) with NYSE-VW and 0.74% ($t = 3.24$) with ABM-EW. The q -factor and five-factor alphas are 0.18% ($t = 1.15$) and 0.01% ($t = 0.12$) with NYSE-VW, as well as 0.08% ($t = 0.37$) and 0.01% ($t = 0.08$) with ABM-EW, respectively.

In addition to annual sorts commonly applied to the value-versus-growth anomalies, we also perform monthly sorts on quarterly variables, such as earnings-to-price, cash flow-to-price (Cp^q), (net) payout yield, enterprise multiple, and sales-to-price (Sp^q). The q -factor model underperforms

the five-factor model in explaining the Cp^q effect with NYSE-VW. At the 1-, 6-, and 12-month, the average returns of the high-minus-low decile are 0.69%, 0.55%, and 0.45% per month ($t = 3.25$, 2.77, and 2.44), the q -factor alphas 0.5%, 0.38%, and 0.22% ($t = 2.27$, 1.98, and 1.24), but the five-factor alphas only 0.17%, 0.07%, and -0.04% , respectively, all of which are within one standard error from zero. With ABM-EW, the average returns are 0.83%, 0.53%, and 0.54% ($t = 3.49$, 2.38, and 2.62), the q -factor alphas 0.43%, 0.14%, and 0.07% ($t = 1.54$, 0.56, 0.31), and the five-factor alphas 0.14%, -0.11% , and -0.1% ($t = 0.78$, -0.71 , and -0.82), respectively.

For most of the other value-minus-growth anomalies, the performance of the q -factor model is largely comparable with that of the five-factor model. For example, with NYSE-VW, the average returns of the high-minus-low Sp^q decile at the 1-, 6-, and 12-month horizons are 0.61%, 0.58%, and 0.55% per month ($t = 2.39$, 2.43, and 2.49), the q -factor alphas 0.21%, 0.15%, and 0.06% ($t = 0.7$, 0.59, and 0.28), and the five-factor alphas -0.2% , -0.23% , and -0.22% ($t = -0.98$, -1.33 , and -1.52), respectively. With ABM-EW, the average returns are 0.77%, 0.67%, and 0.64% ($t = 2.53$, 2.37, and 2.35), the q -factor alphas -0.02% , -0.16% , and -0.27% ($t = -0.05$, -0.53 , and -0.98), and the five-factor alphas -0.39% , -0.46% , and -0.48% ($t = -1.83$, -2.69 , and -3.2), respectively.

3.3.3 Investment

Columns 69–95 in Table 3 report the results for the 27 significant investment anomalies with NYSE-VW, and columns 89–124 in Table 4 for the 36 significant anomalies with ABM-EW.

The high-minus-low decile formed on abnormal corporate investment (Aci, Titman, Wei, and Xie 2004) earns on average -0.31% per month ($t = -2.2$) with NYSE-VW and -0.31% ($t = -3.64$) with ABM-EW. The q -factor and five-factor alphas are -0.17% per month ($t = -1.05$) and -0.31% ($t = -2.05$) with NYSE-VW, as well as -0.12% ($t = -1.27$) and -0.24% ($t = -2.72$) with ABM-EW, respectively. The high-minus-low decile on composite equity issuance (Cei, Daniel and Titman 2006) earns an average return of -0.56% ($t = -3.16$) with NYSE-VW and -0.67% ($t = -4.09$) with ABM-EW. The q -factor and five-factor alphas are -0.24% ($t = -1.85$) and -0.25% ($t = -2.4$) with

NYSE-VW, as well as -0.31% ($t = -2.4$) and -0.47% ($t = -4.35$) with ABM-EW, respectively.

Neither of the models explains the operating accruals anomaly (Oa, Sloan 1996). The high-minus-low average return is -0.27% per month ($t = -2.13$) with NYSE-VW and -0.28% ($t = -2.27$) with ABM-EW. The q -factor and five-factor alphas are -0.54% ($t = -3.77$) and -0.52% ($t = -4.06$) with NYSE-VW and -0.5% ($t = -3.82$) and -0.47% ($t = -4.36$) with ABM-EW, respectively. The models do better for percent operating accruals (Poa, Hafzalla, Lundholm, and Van Winkle 2011), in which accruals are scaled by absolute earnings. The high-minus-low Poa decile earns an average return of -0.4% ($t = -2.85$) with NYSE-VW and -0.41% ($t = -3.75$) with ABM-EW. The q -factor and five-factor alphas are -0.07% ($t = -0.57$) and -0.11% ($t = -0.95$) with NYSE-VW and -0.15% ($t = -1.54$) and -0.24% ($t = -2.8$) with ABM-EW, respectively.

3.3.4 Profitability

Columns 96–128 in Table 3 report factor regressions for the 33 significant profitability anomalies with NYSE-VW, and columns 125–171 in Table 4 for the 47 significant anomalies with ABM-EW.

Sorting on the change in return on equity (dRoe, current Roe minus four-quarter-lagged Roe) yields more precise average returns than sorting on the Roe level. At the 1-, 6-, and 12-month, the high-minus-low dRoe decile earns average returns of 0.76% , 0.39% , and 0.27% per month ($t = 5.43$, 3.28 , and 2.57) with NYSE-VW, and 0.87% , 0.44% , and 0.24% ($t = 6.6$, 4.03 , and 2.62) with ABM-EW, respectively. In contrast, across the three horizons, the high-minus-low Roe decile earns on average 0.69% , 0.42% , and 0.24% ($t = 3.07$, 1.95 , and 1.19) with NYSE-VW, and 0.97% , 0.66% , and 0.35% ($t = 4.53$, 3.39 , and 1.84), respectively. We interpret the evidence as indicating earnings seasonality. Sorting on the fourth-quarter Roe change controls for seasonality, and likely better captures the underlying economic profitability than the Roe level.

The high-minus-low decile on gross profits-to-current assets (Gpa) earns significant average returns of 0.38% per month ($t = 2.62$) with NYSE-VW and 0.62% ($t = 3.52$) with ABM-EW. Both the q -factor and five-factor models capture the average returns. However, Table A2 shows

that sorting on gross profits-to-lagged assets (Gla) yields insignificant average returns of 0.16% ($t = 1.04$) with NYSE-VW and 0.29% ($t = 1.85$) with ABM-EW. Intuitively, the gross profits-to-current assets ratio equals the gross profits-to-lagged assets ratio divided by asset growth (current assets-to-lagged assets). As such, the Gpa effect is mixed with a hidden investment effect, and once the hidden effect is purged, the remaining Gla effect is insignificant.

Which deflator should be used to scale economic profits, lagged or current assets? Economic logic would imply that profits should be scaled by one-period-lagged assets. Intuitively, profits are generated by one-period-lagged assets. Contemporaneous assets at the end of the period are accumulated via the investment process over the course of the current period. Under, for instance, one-period time-to-build, current assets can start to generate profits only at the end of the period.

The q -factor model outperforms the five-factor model for profitability anomalies. At the 1-, 6-, and 12-month, the high-minus-low quarterly F-score (F^q) decile earns average returns of 0.58%, 0.53%, and 0.42% per month ($t = 2.47, 2.52, \text{ and } 2.22$) with NYSE-VW and 0.93%, 0.7%, and 0.54% ($t = 3.82, 3.29, \text{ and } 2.75$) with ABM-EW, respectively. The q -factor alphas are 0.13%, 0.15%, and 0.07% ($t = 0.58, 0.86, \text{ and } 0.49$) with NYSE-VW and 0.41%, 0.16%, and 0.01% ($t = 1.98, 0.92, \text{ and } 0.04$) with ABM-EW. The five-factor alphas are 0.39%, 0.39%, and 0.3% ($t = 1.72, 2.25, \text{ and } 2.16$) with NYSE-VW and 0.7%, 0.47%, and 0.32% ($t = 3.69, 2.99, \text{ and } 2.33$) with ABM-EW, respectively.

3.3.5 Intangibles and Trading Frictions

Columns 129–154 in Table 3 report the results for the 26 significant intangibles anomalies with NYSE-VW, and columns 172–200 in Table 4 for the 29 significant anomalies in the same category for ABM-EW. The remaining columns in both tables report significant trading frictions anomalies.

The q -factor model underperforms the five-factor model in capturing the R&D-to-market (Rdm) anomaly. In annual sorts, the high-minus-low decile earns on average 0.68% per month ($t = 2.58$) with NYSE-VW and 1% ($t = 3.99$) in ABM-EW. The q -factor alpha is 0.7% ($t = 2.89$) with NYSE-VW, in contrast to the five-factor alpha of 0.46% ($t = 1.93$). The q -factor alpha is 0.9%

($t = 3.23$) with ABM-EW, which is still higher than 0.8% ($t = 3.28$) for the five-factor alpha. The underperformance is starker in monthly sorts on quarterly R&D-to-market (R_{dm}^q). At the 1-, 6-, and 12-month horizons, the high-minus-low decile earns average returns of 1.19%, 0.83%, and 0.83% ($t = 2.93, 2.12$, and 2.32) with NYSE-VW, respectively. The q -factor alphas are 1.47%, 0.97%, and 0.8% ($t = 2.97, 2.73$, and 2.8), whereas the five-factor alphas are 0.85%, 0.57%, and 0.5% ($t = 2.05, 1.67$, and 1.73), respectively. The results with ABM-EW are largely similar.

All the factor models fail to capture the Heston-Sadka (2008) seasonality anomalies. At the beginning of each month t , we split stocks into deciles based on various measures of past performance, including returns in month $t-12$ (R_a^1), average returns across months $t-24, t-36, t-48$, and $t-60$ ($R_a^{[2,5]}$), average returns across months $t-72, t-84, t-96, t-108$, and $t-120$ ($R_a^{[6,10]}$), average returns across months $t-132, t-144, t-156, t-168$, and $t-180$ ($R_a^{[11,15]}$), and average returns across months $t-192, t-204, t-216, t-228$, and $t-240$ ($R_a^{[16,20]}$). Monthly decile returns are calculated for the current month t , and the deciles are rebalanced at the beginning of month $t+1$.

With NYSE-VW, the average returns of the high-minus-low deciles formed on $R_a^1, R_a^{[2,5]}, R_a^{[6,10]}, R_a^{[11,15]}$, and $R_a^{[16,20]}$ are 0.65%, 0.69%, 0.83%, 0.67%, and 0.56% per month ($t = 3.23, 4, 4.91, 4.66$, and 3.29), the q -factor alphas 0.55%, 0.81%, 1.13%, 0.65%, and 0.64% ($t = 2.48, 3.9, 4.88, 3.6$, and 3.14), and the five-factor alphas 0.65%, 0.73%, 1.05%, 0.73%, and 0.61% ($t = 3.35, 3.93, 5.22, 4.07$, and 3.67), respectively. With ABM-EW, the average returns are 0.6%, 0.54%, 0.65%, 0.44%, and 0.49% ($t = 3.4, 4.04, 5.77, 4.09$, and 4.5), the q -factor alphas 0.51%, 0.73%, 0.82%, 0.37%, and 0.59% ($t = 2.89, 4.76, 5.05, 2.74$, and 4.6), and the five-factor alphas 0.64%, 0.64%, 0.74%, 0.43%, and 0.53% ($t = 3.75, 4.55, 5.53, 3.39$, and 4.66), respectively.

Finally, the q -factor model does a better job than the five-factor model in capturing several trading frictions anomalies with ABM-EW. For instance, at the 1-month horizon, the high-minus-low deciles formed on the idiosyncratic volatility per the CAPM (Ivc), the idiosyncratic volatility per the q -factor model (Ivq), and maximum daily return (Mdr) earn average returns of -0.69% ,

−0.63%, and −0.67% per month ($t = -2.1, -1.97$, and -2.22), respectively. The q -factor alphas are −0.14%, −0.08%, and −0.18% ($t = -0.69, -0.42$, and -0.86), whereas the five-factor alphas −0.34%, −0.27%, and −0.31% ($t = -2.28, -1.97$, and -2.32), respectively.

3.4 Betas

To shed light on the driving forces behind the model performance, we examine factor loadings (betas). Table 5 reports their factor loadings for the 161 significant anomalies with NYSE-VW, and Table 6 for the 216 significant anomalies with ABM-EW.

3.4.1 Momentum

Columns 1–37 in Table 5 report the factor loadings for the 37 significant momentum anomalies with NYSE-VW, and columns 1–50 in Table 6 for the 50 significant anomalies with ABM-EW.

The ROE factor is the main source of the q -factor model’s success in capturing momentum. With NYSE-VW, 35 out of 37 winner-minus-loser deciles have positive ROE-factor loadings, and the two negative loadings are tiny and insignificant. The average loading is 0.57. All but three of the positive loadings are significant, including 28 with t -statistics above three. In contrast, the investment-factor loadings are generally small, on average only −0.029, with mixed signs, and most (29) are insignificant. With ABM-EW, the ROE-factor loadings are all positive, with an average of 0.56. Most (45 out of 50) loadings are significant, including 41 with t -statistics above three. In contrast, the investment-factor loadings are again small, with an average of −0.015, and 39 are insignificant.

The RMW loadings of the winner-minus-loser deciles are generally small, and mostly insignificant, rendering the five-factor model ineffective in explaining momentum. With NYSE-VW, only eight out of 37 loadings are significantly positive, and 12 are negative. The average loading is only 0.1. With ABM-EW, 15 out of 50 loadings are significantly positive, and 15 are negative, albeit insignificant. The average is 0.15. Intuitively, formed monthly on the latest announced quarterly earnings, the ROE factor is more powerful in capturing the expected profitability differences between

winners and losers. In contrast, formed annually on earnings from the last fiscal year end, RMW is weak. The evidence echoes Table A2, which reports insignificant average returns for the high-minus-low decile formed on the sorting variable underlying RMW (operating profits-to-book equity, Ope).

Two specific examples are in order. First, with NYSE-VW, the high-minus-low Sue1 decile has a large ROE-factor loading of 0.86 ($t = 11.24$), in contrast to the RMW loading of 0.47 ($t = 3.9$). The investment-factor loading is only -0.09 ($t = -0.95$). With ABM-EW, the ROE-factor loading is 0.89 ($t = 14.01$), which is higher than the RMW loading of 0.48 ($t = 6.96$). The investment-factor loading is -0.05 ($t = -0.8$). Second, the high-minus-low decile on R^6 (prior 6-month returns, with the 6-month holding period) with NYSE-VW has an ROE-factor loading of 0.99 ($t = 5.33$), in contrast to the RMW loading of 0.09 ($t = 0.37$). The investment-factor loading is tiny, -0.01 ($t = -0.04$). With ABM-EW, the ROE-factor loading is 1.19 ($t = 4.98$), which is higher than the RMW loading of 0.23 ($t = 0.69$). The investment-factor loading is only 0.1 ($t = 0.33$).

3.4.2 Value-versus-growth

Columns 38–68 in Table 5 report the loadings for the 31 significant value-minus-growth anomalies with NYSE-VW, and columns 51–88 in Table 6 for the 38 significant variables with ABM-EW.

The investment factor is the main source of the q -factor model's explanatory power for the value-versus-growth anomalies. All 31 high-minus-low deciles with NYSE-VW and all 38 with ABM-EW have investment-factor loadings that go in the right direction in explaining average returns. In particular, all the value-minus-growth deciles have significantly positive loadings on the low-minus-high investment factor. Intuitively, value firms invest less than growth firms in the data. All the loadings have t -statistics with magnitudes above three. The average magnitude of the loadings is 1.01 with NYSE-VW, and 1.24 with ABM-EW. In contrast, the ROE-factor loadings often go in the wrong direction in capturing average returns, and many (18 with NYSE-VW and 12 with ABM-EW) are significant. Intuitively, value firms are less profitable than growth firms in the data. More important, however, the investment-factor loadings dominate the ROE-factor loadings

quantitatively, allowing the q -factor model to fit the value-minus-growth anomalies.

In particular, the high-minus-low book-to-market decile has an investment-factor loadings of 1.33 ($t = 3.09$), in contrast to an ROE-factor loading of -0.55 ($t = -6.64$) with NYSE-VW. With ABM-EW, the two loadings are 1.78 ($t = 8.62$) and -0.13 ($t = -0.76$), respectively. Also, the strong ROE-factor loadings of the high-minus-low quarterly cash flow-to-price deciles are the source of the q -factor model's underperformance with NYSE-VW. At the 1-, 6-, and 12-month, these loadings are -0.61 , -0.56 , and -0.45 ($t = -4.3$, -4.7 , and -4.16), despite their strong investment-factor loadings of 0.99, 0.97, and 1.01 ($t = 6.12$, 6.74 , and 7.57), respectively. With ABM-EW, the ROE-factor loadings are weaker, -0.17 , -0.13 , and -0.04 ($t = -0.87$, -0.72 , and -0.23), but the investment-factor loadings remain strong, 1.22, 1.22, and 1.25 ($t = 5.73$, 5.96 , and 6.71), respectively. As such, the q -factor model's underperformance largely vanishes.

Not surprisingly, the value factor, HML, is the main source of the five-factor model's power in fitting the value-versus-growth anomalies. All the value-minus-growth deciles have significantly positive HML loadings. All but two loadings with NYSE-VW and all the loadings with ABM-EW have t -statistics above three, and many are highly significant. The investment factor, CMA, also helps, but its loadings are often insignificant, and their signs can be opposite to the HML loadings.

3.4.3 Investment

Columns 69–95 in Table 5 report the loadings for the 27 significant investment anomalies with NYSE-VW, and columns 89–124 in Table 6 for the 36 significant variables with ABM-EW. This category includes not only investment, but also financing, inventory, and accruals anomalies.

The investment factor is the main source of the q -factor model's explanatory power for the investment anomalies. Except for abnormal corporate investment (Aci), net operating assets (Noa), operating accruals (Oa), change in net financial assets (dFin), discretionary accruals (Dac), and percent discretionary accruals (Pda), the remaining high-minus-low deciles all have significantly negative loadings on the low-minus-high investment factor. All the remaining 21 loadings with

NYSE-VW and 30 with ABM-EW have t -statistics above three. The average of the 21 loadings with NYSE-VW is -0.86 , and the average of the 30 loading with ABM-EW is -0.79 . Intuitively, high investment, financing, and accruals firms invest more than low investment, financing, and accruals firms. In contrast, the ROE-factor loadings have mixed signs. Even though the ROE-factor loadings can occasionally be significantly positive, going in the wrong direction in capturing average returns, their loadings are dominated by the strong investment-factor loadings.

The high-minus-low Noa decile has tiny investment-factor loadings of -0.07 ($t = -0.44$) with NYSE-VW and 0.01 ($t = 0.07$) with ABM-EW. The ROE-factor loadings are also small and insignificant. As a result, the q -factor alpha is -0.41% per month ($t = -2.24$) with NYSE-VW (Table 3), and -0.74% ($t = -3.45$) with ABM-EW (Table 4). Although often viewed as an accrual variable, Noa is a stock, not a flow variable as accruals. Taking the first difference transforms the stock into a flow variable, the change in Noa (dNoa). With NYSE-VW, the high-minus-low dNoa decile earns on average -0.53% ($t = -3.89$), the q -factor alpha is only -0.1% ($t = -0.66$), and the investment-factor loading -1.05 ($t = -9.49$). With ABM-EW, the average return is -0.74% ($t = -5.79$), the q -factor alpha -0.44% ($t = -3.55$), and the investment-factor loading -0.83 ($t = -10.06$). The five-factor model also fails to capture the Noa anomaly. The HML and CMA loadings, which are both large and significant, have opposite signs that work to offset each other. The model does better for the dNoa effect, as the CMA loading dominates the HML loading.

The high-minus-low Oa decile has only small investment-factor loadings of -0.02 ($t = -0.23$) with NYSE-VW and -0.19 ($t = -1.75$) with ABM-EW. In contrast, the ROE-factor loadings are both large and significant, 0.26 ($t = 4.13$) and 0.42 ($t = 4.88$), which go in the wrong direction in capturing average returns. The same problem also plagues the five-factor model. The high-minus-low Oa decile has small and insignificant CMA loadings of 0.04 and -0.11 , but large and significant RMW loadings of 0.41 and 0.62 , with NYSE-VW and ABM-EW, respectively.

The problem deepens with discretionary accruals (Dac, Xie 2001), which purges information

on the sales change and property, plant, and equipment from Oa. The high-minus-low Dac decile earns average returns of -0.36% per month ($t = -2.73$) with NYSE-VW and -0.32% ($t = -3.32$) with ABM-EW. The q -factor alphas are -0.64% ($t = -4.37$) and -0.46% ($t = -4.23$), and the five-factor alphas -0.6% ($t = -4.3$) and -0.45% ($t = -5.03$), respectively. With NYSE-VW, both investment- and ROE-factor loadings go in the wrong direction, 0.23 and 0.19, respectively, both of which are significant. With ABM-EW, the investment-factor loading is close to zero, but the ROE-factor loading is still 0.22 ($t = 2.5$). The five-factor loadings paint a similar picture.

3.4.4 Profitability

Columns 96–128 in Table 5 report the loadings for the 33 significant profitability anomalies with NYSE-VW, and columns 125–171 in Table 6 for the 47 significant variables with ABM-EW.

Naturally, the ROE factor is the main source of the q -factor model's ability to fit the profitability anomalies. All but one ROE-factor loadings with NYSE-VW are highly significant, with the t -value magnitudes above five.² The average magnitude of the loadings is 0.73. In addition, all the ROE-factor loadings with ABM-EW are significant, with the t -value magnitudes all above four. The average magnitude of the loadings is 0.88. In the five-factor model, RMW is the main source of explanatory power. The magnitude of the RMW loadings is largely similar to that of the ROE-factor loadings. However, six RMW loadings with NYSE VW and seven with ABM-EW are insignificant.

In particular, at the 1-, 6-, and 12-month, the high-minus-low decile formed on the change in return on equity (dRoe) have ROE-factor loadings of 0.58, 0.56, and 0.52 ($t = 6.76$, 6.02, and 8.01) with NYSE-VW, and 0.54, 0.49, and 0.46 ($t = 4.79$, 6, and 9.13) with ABM-EW, respectively. In contrast, the RMW loadings are 0.02, 0.06, and 0.15 ($t = 0.19$, 0.61, and 1.92) with NYSE-VW, and 0.01, 0.02, and 0.07 ($t = 0.08$, 0.24, and 1.09) with ABM-EW, respectively. As a result, the q -factor

²The only exception is quarterly taxable income-to-book income (Tbi^q12), with the 12-month holding period. With NYSE-VW, the high-minus-low decile has an ROE-factor loading of 0.05 ($t = 0.66$) and an investment-factor loading of -0.14 ($t = -2.07$). The average return is marginally significant, 0.22% per month ($t = 1.96$), the q -factor alpha 0.34% ($t = 2.93$), and the five-factor alpha 0.26% ($t = 2.33$). However, the average returns at the 1- and 6-month horizons are both insignificant. With ABM-EW, the average returns are insignificant at all horizons.

alphas are mostly insignificant, whereas the five-factor alphas are all significant (Tables 3 and 4).

3.4.5 Intangibles and Trading Frictions

Columns 129–154 in Table 5 report the loadings for the 26 significant intangibles anomalies with NYSE-VW, and columns 172–200 in Table 6 for the 29 significant anomalies in the same category for ABM-EW. The remaining columns in both tables report significant trading frictions anomalies.

The q -factor model fails to capture the Heston-Sadka (2008) seasonality deciles because the loadings are mostly small and insignificant. With NYSE-VW, the high-minus-low deciles formed on R_a^1 , $R_a^{[2,5]}$, $R_a^{[6,10]}$, $R_a^{[11,15]}$, and $R_a^{[16,20]}$ have investment-factor loadings of -0.15 , -0.28 , -0.37 , -0.03 , and -0.04 ($t = -0.97, -2.46, -2.22, -0.23$, and -0.34), as well as ROE-factor loadings of 0.18 , 0.05 , -0.23 , 0.1 , and -0.001 ($t = 1.25, 0.47, -1.97, 1.09$, and -0.01), respectively. The loadings with ABM-EW, as well as the results for the five-factor model, are largely similar. Finally, both investment- and ROE-factor loadings help explain the average returns of the high-minus-low deciles formed on Ivc , Ivq , and Mdr at the 1-month horizon with ABM-EW. Their investment-factor loadings are -1.36 , -1.34 , and -1.2 ($t = -7.46, -7.62$, and -6.4), and the ROE-factor loadings are -0.89 , -0.85 , and -0.78 ($t = -5.34, -5.38$, and -4.6), respectively.

4 Conclusion

Using the hundreds of significant anomalies in the broad cross section, we compare the performance of a large array of empirical asset pricing models, including the CAPM, the Fama-French (1993) three-factor model, the Carhart (1997) four-factor model, the Pastor-Stambaugh (2003) four-factor model, the Hou-Xue-Zhang (2015) q -factor model, and the Fama-French (2015) five-factor model. The q -factor model and the five-factor model are the two best performing models. The q -factor model outperforms the five-factor model in factor spanning tests and in explaining momentum and profitability anomalies, but the five-factor model has an edge in explaining value-versus-growth anomalies. In all, our evidence suggests that fundamentals, such as investment and profitability,

not liquidity, are the dominating driving forces in the broad cross section of expected returns.

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Table 1 : Factor Spanning Tests, January 1967 to December 2014, 576 Months

r_{ME} , $r_{I/A}$, and r_{ROE} are the size, investment, ROE factors in the q -factor model, respectively. MKT is the value-weighted market return minus the one-month Treasury bill rate from CRSP. SMB, HML, RMW, and CMA are the size, value, profitability, and investment factors from the five-factor model, respectively. The data for SMB and HML in the three-factor model, SMB, HML, RMW, and CMA in the five-factor model (b, s, h, r , and c are the loadings, respectively), as well as the momentum factor UMD are from Kenneth French's Web site. LIQ is the liquidity factor from Robert Stambaugh's Web site. m is the average return, α_C is the Carhart alpha, α_q the q -model alpha, a is the five-factor alpha, and α_{PS} is the alpha from the four-factor model with the Fama-French three factors and LIQ. The numbers in parentheses in Panels A–D are heteroscedasticity-and-autocorrelation-adjusted t -statistics. In Panel E, the numbers in parentheses are p -values testing that a given correlation is zero. The sample when LIQ is used starts in January 1968. For all the other tests, the sample starts in January 1967.

Panel A: The Hou-Xue-Zhang q -factors							
	m	α_C	β_{MKT}	β_{SMB}	β_{HML}	β_{UMD}	R^2
r_{ME}	0.32 (2.42)	0.01 (0.25)	0.01 (1.08)	0.97 (67.08)	0.17 (7.21)	0.03 (1.87)	0.94
$r_{I/A}$	0.43 (5.08)	0.29 (4.57)	−0.06 (−4.51)	−0.04 (−1.88)	0.41 (13.36)	0.05 (1.93)	0.53
r_{ROE}	0.56 (5.24)	0.51 (5.58)	−0.04 (−1.39)	−0.30 (−4.31)	−0.12 (−1.79)	0.27 (6.19)	0.40
		α_{PS}	β_{MKT}	β_{SMB}	β_{HML}	β_{LIQ}	R^2
r_{ME}		0.05 (1.51)	0.01 (0.58)	0.98 (62.58)	0.17 (7.05)	−0.01 (−1.24)	0.93
$r_{I/A}$		0.35 (5.73)	−0.07 (−4.63)	−0.04 (−1.55)	0.40 (13.30)	0.00 (−0.22)	0.52
r_{ROE}		0.75 (7.61)	−0.09 (−2.45)	−0.33 (−6.23)	−0.21 (−2.71)	−0.05 (−1.47)	0.22
	a	b	s	h	r	c	R^2
r_{ME}	0.05 (1.39)	0.00 (0.39)	0.98 (68.34)	0.02 (1.14)	−0.01 (−0.21)	0.04 (1.19)	0.95
$r_{I/A}$	0.12 (3.35)	0.01 (0.73)	−0.05 (−2.86)	0.04 (1.60)	0.07 (2.77)	0.82 (26.52)	0.84
r_{ROE}	0.45 (5.60)	−0.04 (−1.45)	−0.11 (−2.69)	−0.24 (−3.54)	0.75 (13.46)	0.13 (1.34)	0.52
Panel B: The Fama-French five factors							
	m	α_C	β_{MKT}	β_{SMB}	β_{HML}	β_{UMD}	R^2
SMB	0.26 (1.92)	−0.02 (−1.24)	0.00 (0.96)	1.00 (89.87)	0.13 (8.07)	0.00 (0.11)	0.99
HML	0.36 (2.57)	−0.00 (−1.79)	0.00 (1.79)	−0.00 (−1.69)	1.00 (13282.85)	−0.00 (−0.87)	1.00
RMW	0.27 (2.58)	0.33 (3.31)	−0.04 (−1.32)	−0.28 (−3.20)	−0.00 (−0.03)	0.04 (0.81)	0.19
CMA	0.34 (3.63)	0.19 (2.83)	−0.09 (−4.42)	0.03 (0.86)	0.46 (13.52)	0.04 (1.51)	0.55
		α_{PS}	β_{MKT}	β_{SMB}	β_{HML}	β_{LIQ}	R^2
SMB		−0.02 (−1.01)	0.00 (0.77)	1.00 (89.07)	0.12 (8.11)	−0.01 (−1.08)	0.99
HML		0.00 (−2.04)	0.00 (1.89)	0.00 (−1.70)	1.00 (12795.89)	0.00 (1.11)	1.00
RMW		0.34 (3.19)	−0.05 (−1.38)	−0.28 (−3.38)	−0.01 (−0.17)	0.01 (0.34)	0.19
CMA		0.24 (3.71)	−0.09 (−4.13)	0.04 (0.98)	0.45 (12.77)	0.00 (−0.04)	0.54
		α_q	β_{MKT}	β_{ME}	$\beta_{I/A}$	β_{ROE}	R^2
SMB		0.05 (1.48)	−0.00 (−0.17)	0.94 (62.40)	−0.09 (−4.91)	−0.10 (−5.94)	0.96
HML		0.03 (0.28)	−0.05 (−1.33)	0.00 (0.03)	1.03 (11.72)	−0.17 (−2.17)	0.50
RMW		0.04 (0.42)	−0.03 (−0.99)	−0.12 (−1.78)	−0.03 (−0.35)	0.53 (8.59)	0.49
CMA		0.01 (0.32)	−0.05 (−3.63)	0.04 (1.68)	0.94 (35.26)	−0.11 (−3.95)	0.85

Panel C: The Carhart momentum factor, UMD									
UMD	m	α_q	β_{MKT}	β_{ME}	$\beta_{\text{I/A}}$	β_{ROE}	R^2		
	0.67 (3.66)	0.11 (0.43)	-0.07 (-1.09)	0.24 (1.75)	0.03 (0.17)	0.91 (5.59)	0.27		
		α_{PS}	β_{MKT}	β_{SMB}	β_{HML}	β_{LIQ}	R^2		
		0.89 (5.25)	-0.19 (-2.52)	-0.02 (-0.20)	-0.32 (-2.21)	-0.04 (-0.57)	0.06		
	a	b	s	h	r	c	R^2		
	0.69 (3.11)	-0.14 (-1.82)	0.03 (0.29)	-0.54 (-2.98)	0.25 (1.23)	0.47 (1.88)	0.09		
Panel D: The Pastor-Stambaugh liquidity factor, LIQ									
LIQ	m	α	β_{MKT}	β_{SMB}	β_{HML}	β_{UMD}	R^2		
	0.42 (2.81)	0.46 (2.61)	-0.04 (-0.66)	-0.01 (-0.18)	0.00 (0.02)	-0.03 (-0.56)	0.00		
		α_q	β_{MKT}	β_{ME}	$\beta_{\text{I/A}}$	β_{ROE}	R^2		
		0.53 (2.99)	-0.05 (-0.79)	-0.06 (-0.93)	-0.01 (-0.13)	-0.12 (-1.48)	0.01		
	a	b	s	h	r	c	R^2		
	0.43 (2.66)	-0.03 (-0.57)	-0.01 (-0.18)	0.01 (0.06)	0.02 (0.30)	0.00 (0.03)	0.00		
Panel E: Correlation matrix									
	$r_{\text{I/A}}$	r_{ROE}	MKT	SMB	HML	UMD	RMW	CMA	LIQ
r_{ME}	-0.15 (0.00)	-0.31 (0.00)	0.27 (0.00)	0.95 (0.00)	-0.07 (0.08)	-0.01 (0.72)	-0.37 (0.00)	-0.06 (0.16)	-0.04 (0.36)
$r_{\text{I/A}}$		0.04 (0.35)	-0.39 (0.00)	-0.27 (0.00)	0.69 (0.00)	0.04 (0.37)	0.05 (0.24)	0.91 (0.00)	0.02 (0.65)
r_{ROE}			-0.20 (0.00)	-0.38 (0.00)	-0.11 (0.01)	0.49 (0.00)	0.68 (0.00)	-0.10 (0.01)	-0.06 (0.18)
MKT				0.32 (0.00)	-0.31 (0.00)	-0.14 (0.00)	-0.22 (0.00)	-0.40 (0.00)	-0.05 (0.21)
SMB					-0.23 (0.00)	-0.03 (0.50)	-0.42 (0.00)	-0.17 (0.00)	-0.03 (0.53)
HML						-0.16 (0.00)	0.10 (0.01)	0.71 (0.00)	0.03 (0.54)
UMD							0.10 (0.02)	0.01 (0.88)	-0.03 (0.55)
RMW								-0.08 (0.06)	0.03 (0.48)
CMA									0.03 (0.55)

Table 2 : Overall Performance of Factor Models, January 1967 to December 2014, 576 Months

“Mom,” “V–G,” “Inv,” “Prof,” “Intan,” and “Fric” denote momentum, value-versus-growth, investment, profitability, intangibles, and frictions categories of anomalies, respectively, and “All” is all the significant anomalies combined. The number in the parenthesis beside a given category is the number of significant anomalies in the category. $\overline{|\alpha_{H-L}|}$ is the average magnitude of the high-minus-low alphas, $\#_{H-L}^*$ is the number of significant high-minus-low alphas, $\overline{|\alpha|}$ is the mean absolute alpha across the significant anomalies in each category, and $\#_{GRS}^*$ is the number of the sets of anomaly deciles across which a given factor model is rejected by the GRS test. All the significance is at the 5% level. CAPM is the CAPM, FF3 the Fama-French three-factor model, PS4 the four-factor model in Pastor and Stambaugh (2003) that augments the three-factor model with their liquidity factor, CARH the Carhart four-factor model, HXZ- q the Hou-Xue-Zhang q -factor model, and FF5 the Fama-French five-factor model.

Panel A: NYSE breakpoints with value-weighted returns														
	All (161)		Mom (37)		V–G (31)		Inv (27)		Prof (33)		Intan (26)		Fric (7)	
	$\overline{ \alpha_{H-L} }$	$\#_{H-L}^*$	$\overline{ \alpha_{H-L} }$	$\#_{H-L}^*$	$\overline{ \alpha_{H-L} }$	$\#_{H-L}^*$	$\overline{ \alpha_{H-L} }$	$\#_{H-L}^*$	$\overline{ \alpha_{H-L} }$	$\#_{H-L}^*$	$\overline{ \alpha_{H-L} }$	$\#_{H-L}^*$	$\overline{ \alpha_{H-L} }$	$\#_{H-L}^*$
CAPM	0.56	152	0.61	37	0.61	31	0.48	26	0.57	30	0.58	24	0.36	4
FF3	0.49	116	0.73	37	0.18	4	0.34	21	0.72	33	0.46	17	0.21	4
PS4	0.49	113	0.74	37	0.16	3	0.35	21	0.72	32	0.46	17	0.22	3
CARH	0.36	94	0.30	18	0.25	10	0.28	17	0.52	29	0.49	16	0.21	4
HXZ- q	0.26	46	0.26	9	0.23	6	0.19	7	0.23	9	0.41	11	0.24	4
FF5	0.37	84	0.65	35	0.13	2	0.22	11	0.39	23	0.39	10	0.20	3
	$\overline{ \alpha }$	$\#_{GRS}^*$	$\overline{ \alpha }$	$\#_{GRS}^*$	$\overline{ \alpha }$	$\#_{GRS}^*$	$\overline{ \alpha }$	$\#_{GRS}^*$	$\overline{ \alpha }$	$\#_{GRS}^*$	$\overline{ \alpha }$	$\#_{GRS}^*$	$\overline{ \alpha }$	$\#_{GRS}^*$
CAPM	0.159	129	0.152	33	0.210	26	0.143	27	0.140	27	0.162	15	0.118	2
FF3	0.144	128	0.170	34	0.098	14	0.125	24	0.179	32	0.153	17	0.086	7
PS4	0.144	126	0.173	35	0.095	12	0.125	24	0.179	32	0.151	17	0.086	6
CARH	0.126	119	0.109	27	0.118	15	0.115	24	0.139	30	0.166	17	0.083	6
HXZ- q	0.122	107	0.110	25	0.121	18	0.099	17	0.121	20	0.174	20	0.102	7
FF5	0.130	108	0.160	35	0.093	10	0.090	16	0.161	26	0.148	15	0.081	6
Panel B: All-but-micro breakpoints with equal-weighted returns														
	All (216)		Mom (50)		V–G (38)		Inv (36)		Prof (47)		Intan (29)		Fric (16)	
	$\overline{ \alpha_{H-L} }$	$\#_{H-L}^*$	$\overline{ \alpha_{H-L} }$	$\#_{H-L}^*$	$\overline{ \alpha_{H-L} }$	$\#_{H-L}^*$	$\overline{ \alpha_{H-L} }$	$\#_{H-L}^*$	$\overline{ \alpha_{H-L} }$	$\#_{H-L}^*$	$\overline{ \alpha_{H-L} }$	$\#_{H-L}^*$	$\overline{ \alpha_{H-L} }$	$\#_{H-L}^*$
CAPM	0.67	212	0.62	48	0.76	38	0.61	36	0.71	46	0.65	29	0.74	15
FF3	0.55	185	0.70	50	0.28	19	0.48	36	0.74	44	0.52	24	0.49	12
PS4	0.56	184	0.72	50	0.26	18	0.49	36	0.75	45	0.51	24	0.47	11
CARH	0.42	154	0.31	27	0.36	22	0.39	34	0.55	39	0.52	21	0.40	11
HXZ- q	0.26	66	0.28	14	0.19	2	0.28	24	0.22	11	0.41	14	0.12	1
FF5	0.38	128	0.61	44	0.18	7	0.35	31	0.37	27	0.41	16	0.18	3
	$\overline{ \alpha }$	$\#_{GRS}^*$	$\overline{ \alpha }$	$\#_{GRS}^*$	$\overline{ \alpha }$	$\#_{GRS}^*$	$\overline{ \alpha }$	$\#_{GRS}^*$	$\overline{ \alpha }$	$\#_{GRS}^*$	$\overline{ \alpha }$	$\#_{GRS}^*$	$\overline{ \alpha }$	$\#_{GRS}^*$
CAPM	0.224	202	0.204	46	0.260	34	0.236	36	0.221	46	0.209	24	0.203	16
FF3	0.142	173	0.168	45	0.099	16	0.127	36	0.177	44	0.131	20	0.120	12
PS4	0.142	172	0.172	45	0.093	14	0.125	36	0.180	45	0.131	20	0.117	12
CARH	0.171	183	0.140	34	0.169	28	0.183	36	0.180	45	0.201	26	0.150	14
HXZ- q	0.145	172	0.133	37	0.116	25	0.136	30	0.141	42	0.208	25	0.152	13
FF5	0.115	151	0.155	43	0.072	15	0.094	30	0.116	38	0.141	19	0.080	6

Table 3 : Significant Anomalies, NYSE-VW, January 1967 to December 2014, 576 Months

For each high-minus-low decile, $m, \alpha, \alpha_{FF}, \alpha_{PS}, \alpha_C, \alpha_q$, and a are the average return, the Fama-French three-factor alpha, the Pastor-Stambaugh alpha, the Carhart alpha, the q -model alpha, and the five-factor alpha. and $t_m, t_\alpha, t_{FF}, t_{PS}, t_C, t_q$, and t_a are their t -statistics adjusted for heteroscedasticity and autocorrelations, respectively. $|\alpha|, |\alpha_{FF}|, |\alpha_{PS}|, |\alpha_C|, |\alpha_q|$, and $|a|$ are the mean absolute alpha across a given set of deciles, and $p, p_{FF}, p_{PS}, p_C, p_q$, and p_a are the p -value of the GRS test on the null that the alphas across the deciles are jointly zero. Table A1 describes the symbols, and Appendix ?? details variable definition and portfolio construction.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Sue1	Abr1	Abr6	Abr12	Re1	Re6	R^6_1	R^6_6	R^6_{12}	R^{11}_1	R^{11}_6	Im1	Im6	Im12	Rs1	dEf1	dEf6	dEf12
m	0.47	0.74	0.30	0.22	0.81	0.54	0.60	0.82	0.55	1.19	0.81	0.67	0.60	0.64	0.31	1.03	0.58	0.35
α	0.53	0.77	0.31	0.21	0.96	0.66	0.76	0.90	0.58	1.31	0.88	0.80	0.67	0.67	0.35	1.07	0.58	0.36
α_{FF}	0.72	0.84	0.38	0.32	1.13	0.86	0.92	1.08	0.82	1.52	1.16	0.93	0.77	0.83	0.63	1.25	0.76	0.56
α_{PS}	0.78	0.85	0.38	0.31	1.15	0.85	0.93	1.07	0.83	1.55	1.18	0.92	0.72	0.81	0.68	1.27	0.76	0.55
α_C	0.43	0.63	0.19	0.16	0.52	0.31	-0.26	0.08	0.09	0.19	0.10	0.09	-0.06	0.19	0.48	0.76	0.32	0.23
α_q	0.05	0.66	0.27	0.23	0.11	0.02	-0.04	0.24	0.16	0.31	0.12	0.26	0.06	0.32	0.22	0.64	0.20	0.09
a	0.51	0.85	0.44	0.40	0.88	0.68	0.73	0.97	0.77	1.26	1.02	0.73	0.64	0.82	0.53	1.22	0.78	0.54
t_m	3.42	5.85	3.24	2.84	3.28	2.49	2.04	3.49	2.90	4.06	3.14	2.74	3.08	3.71	2.21	4.65	3.23	2.45
t_α	4.11	6.21	3.48	2.76	4.18	3.29	2.85	4.09	3.17	4.91	3.65	3.34	3.49	3.88	2.50	4.89	3.29	2.54
t_{FF}	6.08	6.24	4.08	4.47	4.99	4.48	3.53	4.98	4.78	5.83	5.14	4.01	4.06	5.00	4.80	5.88	4.56	4.37
t_{PS}	6.35	6.13	4.00	4.41	5.34	4.64	3.46	4.81	4.75	5.79	5.12	3.84	3.64	4.61	5.33	5.99	4.76	4.44
t_C	3.61	4.62	2.21	2.53	2.61	1.88	-1.31	0.79	0.90	1.58	0.89	0.45	-0.43	1.37	3.45	3.85	2.34	2.16
t_q	0.40	4.49	2.41	2.65	0.45	0.11	-0.10	0.78	0.75	0.77	0.41	0.80	0.23	1.44	1.52	2.81	1.15	0.70
t_a	3.69	6.12	4.43	5.24	3.46	3.03	2.11	3.50	3.93	3.59	3.69	2.53	2.68	4.16	3.85	5.23	4.37	4.07
$ \alpha $	0.16	0.14	0.09	0.08	0.18	0.13	0.15	0.16	0.13	0.21	0.16	0.23	0.17	0.17	0.10	0.27	0.16	0.12
$ \alpha_{FF} $	0.23	0.16	0.11	0.11	0.24	0.21	0.16	0.18	0.15	0.25	0.21	0.22	0.16	0.17	0.15	0.32	0.21	0.18
$ \alpha_{PS} $	0.23	0.16	0.11	0.10	0.25	0.22	0.16	0.18	0.15	0.25	0.21	0.21	0.15	0.16	0.16	0.32	0.21	0.18
$ \alpha_C $	0.14	0.13	0.09	0.09	0.11	0.09	0.22	0.10	0.07	0.13	0.08	0.06	0.04	0.05	0.12	0.19	0.12	0.10
$ \alpha_q $	0.11	0.13	0.08	0.07	0.11	0.12	0.18	0.09	0.07	0.13	0.11	0.13	0.11	0.13	0.08	0.17	0.12	0.12
$ a $	0.20	0.16	0.09	0.09	0.19	0.17	0.13	0.16	0.15	0.23	0.20	0.23	0.21	0.22	0.14	0.27	0.17	0.15
p	0.00	0.00	0.00	0.01	0.01	0.13	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.07	0.00	0.01	0.03
p_{FF}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.01	0.00	0.00	0.00	0.00	0.00
p_{PS}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.01	0.00	0.00	0.00	0.00	0.00
p_C	0.00	0.00	0.00	0.00	0.02	0.04	0.00	0.00	0.00	0.00	0.01	0.90	0.30	0.19	0.00	0.00	0.00	0.01
p_q	0.00	0.00	0.00	0.00	0.08	0.01	0.00	0.00	0.03	0.00	0.01	0.51	0.03	0.11	0.03	0.00	0.00	0.03
p_a	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
	Ne1	Ne6	52w6	ϵ^6_6	ϵ^6_{12}	ϵ^{11}_1	ϵ^{11}_6	ϵ^{11}_{12}	Sm1	Ilr1	Ilr6	Ilr12	Ile1	Cm1	Cm12	Sim1	Cim1	Cim6
m	0.37	0.22	0.57	0.49	0.39	0.67	0.55	0.36	0.59	0.74	0.33	0.35	0.62	0.79	0.16	0.77	0.78	0.30
α	0.39	0.24	0.94	0.51	0.41	0.70	0.56	0.37	0.64	0.86	0.41	0.38	0.66	0.78	0.15	0.78	0.82	0.34
α_{FF}	0.58	0.45	1.10	0.53	0.49	0.73	0.63	0.47	0.64	0.91	0.45	0.44	0.89	0.79	0.18	0.76	0.82	0.34
α_{PS}	0.61	0.47	1.09	0.51	0.48	0.75	0.62	0.46	0.64	0.99	0.44	0.42	0.91	0.82	0.16	0.83	0.88	0.37
α_C	0.39	0.30	0.17	0.18	0.17	0.25	0.19	0.13	0.53	0.73	0.10	0.10	0.62	0.76	0.02	0.51	0.65	0.02
α_q	0.16	0.10	-0.01	0.30	0.22	0.32	0.25	0.15	0.61	0.79	0.17	0.18	0.37	0.72	0.05	0.54	0.64	0.05
a	0.44	0.33	0.73	0.52	0.47	0.63	0.61	0.46	0.76	0.89	0.37	0.38	0.71	0.82	0.13	0.81	0.76	0.26
t_m	3.31	2.03	2.02	3.86	3.92	3.91	3.94	2.96	2.57	3.61	3.18	4.18	3.70	3.74	2.30	3.37	3.45	2.83
t_α	3.52	2.20	4.33	4.05	4.14	4.10	4.06	3.09	2.73	4.14	4.01	4.82	3.93	3.58	2.14	3.30	3.51	3.20
t_{FF}	6.00	4.87	5.71	4.10	4.67	4.12	4.32	3.82	2.71	4.14	4.46	5.49	5.32	3.67	2.50	3.07	3.60	3.07
t_{PS}	6.15	5.06	5.62	3.90	4.54	4.15	4.23	3.68	2.73	4.35	4.19	5.15	5.18	3.80	2.28	3.33	3.72	3.24
t_C	3.75	3.02	1.44	1.64	1.98	1.63	1.64	1.24	2.28	3.41	1.13	1.72	3.74	2.98	0.24	2.19	2.98	0.24
t_q	1.60	1.07	-0.04	1.79	1.66	1.46	1.39	0.94	2.18	3.15	1.22	1.59	2.13	2.75	0.49	1.65	2.29	0.27
t_a	4.55	3.68	2.93	3.56	3.95	3.17	3.79	3.37	3.16	3.71	2.97	3.67	4.17	3.52	1.45	2.76	2.99	1.73
$ \alpha $	0.17	0.13	0.20	0.11	0.10	0.18	0.14	0.10	0.16	0.23	0.12	0.13	0.15	0.20	0.06	0.22	0.22	0.10
$ \alpha_{FF} $	0.21	0.17	0.23	0.11	0.11	0.18	0.15	0.12	0.16	0.21	0.08	0.09	0.18	0.22	0.08	0.21	0.21	0.08
$ \alpha_{PS} $	0.21	0.17	0.23	0.11	0.11	0.19	0.16	0.13	0.16	0.23	0.08	0.09	0.18	0.24	0.08	0.22	0.23	0.09
$ \alpha_C $	0.13	0.10	0.12	0.08	0.08	0.10	0.09	0.08	0.14	0.19	0.05	0.05	0.11	0.22	0.07	0.17	0.18	0.08
$ \alpha_q $	0.09	0.08	0.05	0.06	0.06	0.10	0.06	0.06	0.13	0.21	0.10	0.10	0.13	0.24	0.13	0.15	0.16	0.06
$ \alpha_a $	0.16	0.13	0.14	0.09	0.08	0.17	0.13	0.09	0.16	0.24	0.15	0.15	0.20	0.25	0.11	0.19	0.19	0.06
p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.15	0.00	0.01	0.00	0.00	0.03	0.12	0.01	0.00	0.02
p_{FF}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.23	0.00	0.01	0.00	0.00	0.03	0.05	0.02	0.00	0.07
p_{PS}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.17	0.00	0.01	0.00	0.00	0.02	0.05	0.01	0.00	0.04
p_C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.33	0.01	0.46	0.22	0.01	0.03	0.37	0.09	0.00	0.22
p_q	0.02	0.01	0.25	0.00	0.00	0.00	0.00	0.01	0.26	0.02	0.20	0.10	0.06	0.07	0.02	0.27	0.02	0.26
p_a	0.00	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.03	0.00	0.01	0.00	0.00	0.03	0.02	0.03	0.01	0.37

	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
	Cim12	Bm	Bmj	Bm ^q 12	Rev1	Rev6	Rev12	Ep	Ep ^q 1	Ep ^q 6	Ep ^q 12	Efp1	Cp	Cp ^q 1	Cp ^q 6	Cp ^q 12	Nop	Em
m	0.26	0.59	0.49	0.51	-0.45	-0.44	-0.41	0.47	0.98	0.65	0.49	0.48	0.49	0.69	0.55	0.45	0.65	-0.59
α	0.28	0.62	0.55	0.51	-0.47	-0.47	-0.44	0.57	1.03	0.72	0.57	0.68	0.55	0.68	0.58	0.51	0.85	-0.71
α_{FF}	0.31	-0.05	-0.12	-0.16	0.04	-0.02	-0.04	-0.02	0.56	0.28	0.13	0.25	-0.09	0.12	0.03	-0.06	0.54	-0.24
α_{PS}	0.32	-0.03	-0.11	-0.15	0.04	-0.03	-0.05	-0.06	0.52	0.24	0.09	0.18	-0.12	0.03	-0.06	-0.13	0.51	-0.20
α_C	0.02	-0.04	0.11	0.18	-0.08	-0.11	-0.07	-0.06	0.72	0.37	0.17	0.49	-0.06	0.56	0.36	0.14	0.52	-0.16
α_q	0.06	0.18	0.30	0.39	-0.16	-0.20	-0.13	0.03	0.46	0.13	0.01	0.22	0.09	0.50	0.38	0.22	0.36	-0.27
a	0.28	0.01	-0.02	-0.01	0.06	-0.02	-0.03	-0.03	0.50	0.17	0.02	0.18	-0.09	0.17	0.07	-0.04	0.22	-0.12
t_m	3.38	2.84	2.27	2.35	-1.98	-2.04	-2.04	2.34	5.08	3.69	2.93	1.99	2.47	3.25	2.77	2.44	3.36	-3.12
t_α	3.64	2.95	2.50	2.39	-2.08	-2.19	-2.20	2.84	5.24	4.04	3.31	2.84	2.71	3.20	2.94	2.68	4.70	-3.75
t_{FF}	3.73	-0.40	-0.96	-1.23	0.24	-0.10	-0.22	-0.15	3.53	1.94	1.01	1.44	-0.77	0.76	0.21	-0.50	3.71	-1.77
t_{PS}	3.63	-0.28	-0.88	-1.24	0.21	-0.18	-0.30	-0.46	3.27	1.64	0.68	1.04	-1.01	0.21	-0.48	-1.17	3.49	-1.45
t_C	0.28	-0.34	0.80	1.42	-0.45	-0.59	-0.40	-0.47	4.40	2.66	1.39	2.94	-0.54	4.14	3.00	1.37	3.60	-1.18
t_q	0.49	1.15	1.70	2.25	-0.91	-1.15	-0.76	0.14	1.86	0.68	0.06	1.22	0.49	2.27	1.98	1.24	2.41	-1.56
t_a	2.44	0.12	-0.15	-0.07	0.37	-0.12	-0.17	-0.23	2.86	1.17	0.15	0.70	-0.76	0.90	0.49	-0.33	1.64	-0.89
$ \alpha $	0.09	0.19	0.17	0.16	0.21	0.18	0.18	0.20	0.27	0.21	0.19	0.20	0.20	0.23	0.19	0.19	0.24	0.20
$ \alpha_{FF} $	0.08	0.06	0.07	0.05	0.10	0.09	0.10	0.09	0.17	0.12	0.10	0.10	0.08	0.09	0.06	0.06	0.16	0.10
$ \alpha_{PS} $	0.08	0.07	0.07	0.06	0.10	0.09	0.10	0.08	0.16	0.12	0.10	0.09	0.09	0.07	0.05	0.07	0.15	0.09
$ \alpha_C $	0.06	0.06	0.09	0.09	0.10	0.08	0.08	0.09	0.20	0.15	0.11	0.16	0.08	0.18	0.11	0.08	0.15	0.09
$ \alpha_q $	0.06	0.09	0.12	0.13	0.08	0.07	0.06	0.10	0.17	0.14	0.11	0.18	0.12	0.20	0.15	0.12	0.12	0.12
$ \alpha_a $	0.07	0.06	0.09	0.09	0.05	0.04	0.03	0.08	0.16	0.11	0.09	0.10	0.10	0.13	0.09	0.07	0.10	0.11
p	0.01	0.02	0.05	0.03	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.02	0.04	0.00	0.00
p_{FF}	0.04	0.11	0.07	0.02	0.08	0.08	0.11	0.04	0.00	0.01	0.02	0.23	0.00	0.32	0.44	0.53	0.00	0.03
p_{PS}	0.04	0.05	0.11	0.03	0.13	0.13	0.16	0.07	0.02	0.00	0.01	0.43	0.01	0.53	0.52	0.34	0.00	0.07
p_C	0.81	0.11	0.07	0.01	0.30	0.13	0.23	0.14	0.00	0.00	0.01	0.00	0.03	0.00	0.01	0.38	0.00	0.09
p_q	0.22	0.11	0.01	0.00	0.32	0.09	0.27	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.00
p_a	0.13	0.55	0.13	0.07	0.57	0.54	0.77	0.27	0.00	0.00	0.02	0.17	0.01	0.03	0.08	0.25	0.01	0.02
	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72
	Em ^q 1	Em ^q 6	Em ^q 12	Sp	Sp ^q 1	Sp ^q 6	Sp ^q 12	Ocp	Ocp ^q 1	Ir	Vhp	Vfp	Ebp	Dur	Ac	I/A	Ia ^q 6	Ia ^q 12
m	-0.81	-0.53	-0.53	0.53	0.61	0.58	0.55	0.77	0.66	-0.51	0.38	0.53	0.47	-0.47	-0.31	-0.46	-0.52	-0.50
α	-0.93	-0.65	-0.66	0.50	0.54	0.54	0.52	0.84	0.63	-0.50	0.45	0.61	0.46	-0.55	-0.29	-0.56	-0.63	-0.60
α_{FF}	-0.60	-0.32	-0.31	-0.13	-0.12	-0.12	-0.12	0.16	0.21	0.13	-0.09	0.25	-0.17	0.01	-0.32	-0.23	-0.22	-0.23
α_{PS}	-0.52	-0.24	-0.23	-0.15	-0.16	-0.15	-0.14	0.12	0.14	0.12	-0.14	0.19	-0.16	0.04	-0.31	-0.27	-0.23	-0.24
α_C	-0.74	-0.43	-0.33	-0.02	0.31	0.23	0.12	0.22	0.61	0.05	-0.10	0.23	-0.09	0.02	-0.20	-0.18	-0.26	-0.20
α_q	-0.63	-0.34	-0.30	-0.04	0.21	0.15	0.06	0.41	0.46	-0.18	-0.01	0.22	0.09	-0.10	-0.17	0.07	-0.11	0.00
a	-0.44	-0.14	-0.12	-0.23	-0.20	-0.23	-0.22	0.12	0.17	0.04	-0.11	0.19	-0.09	-0.01	-0.31	0.02	0.01	0.03
t_m	-3.67	-2.57	-2.62	2.44	2.39	2.43	2.49	3.50	2.24	-2.41	2.03	2.42	2.36	-2.39	-2.20	-2.92	-3.04	-3.19
t_α	-4.12	-3.16	-3.28	2.27	2.18	2.27	2.32	3.72	2.15	-2.30	2.33	2.65	2.23	-2.76	-1.93	-3.57	-3.63	-3.72
t_{FF}	-3.20	-1.87	-2.00	-0.96	-0.66	-0.74	-0.79	1.18	0.95	0.92	-0.61	1.33	-1.48	0.11	-2.13	-1.79	-1.84	-2.01
t_{PS}	-2.83	-1.44	-1.50	-1.11	-0.91	-0.97	-1.02	0.89	0.65	0.88	-0.91	0.98	-1.33	0.34	-1.90	-2.06	-1.93	-2.13
t_C	-4.16	-2.68	-2.25	-0.14	1.62	1.42	0.85	1.68	3.26	0.38	-0.68	1.18	-0.78	0.19	-1.32	-1.33	-1.95	-1.56
t_q	-2.55	-1.59	-1.55	-0.19	0.70	0.59	0.28	2.25	1.47	-1.13	-0.05	0.95	0.66	-0.53	-1.05	0.61	-0.96	0.04
t_a	-2.29	-0.82	-0.82	-1.67	-0.98	-1.33	-1.52	0.89	0.74	0.31	-0.73	1.02	-0.83	-0.04	-2.05	0.20	0.13	0.34
$ \alpha $	0.31	0.23	0.23	0.21	0.22	0.20	0.20	0.26	0.29	0.20	0.19	0.16	0.18	0.21	0.10	0.16	0.18	0.20
$ \alpha_{FF} $	0.21	0.13	0.13	0.06	0.05	0.05	0.05	0.12	0.21	0.08	0.08	0.10	0.08	0.09	0.12	0.12	0.10	0.12
$ \alpha_{PS} $	0.19	0.11	0.11	0.06	0.05	0.05	0.06	0.11	0.20	0.07	0.09	0.10	0.07	0.09	0.12	0.13	0.09	0.12
$ \alpha_C $	0.25	0.16	0.13	0.06	0.15	0.12	0.10	0.11	0.30	0.07	0.08	0.08	0.08	0.07	0.11	0.11	0.12	0.12
$ \alpha_q $	0.23	0.15	0.13	0.06	0.08	0.07	0.07	0.11	0.19	0.07	0.14	0.15	0.12	0.08	0.13	0.09	0.07	0.06
$ \alpha_a $	0.19	0.11	0.10	0.09	0.08	0.09	0.09	0.07	0.12	0.06	0.11	0.14	0.08	0.05	0.14	0.10	0.05	0.05
p	0.00	0.00	0.00	0.09	0.21	0.17	0.08	0.00	0.00	0.01	0.00	0.10	0.01	0.00	0.01	0.00	0.00	0.00
p_{FF}	0.00	0.07	0.03	0.14	0.47	0.26	0.07	0.02	0.02	0.15	0.01	0.30	0.01	0.14	0.00	0.00	0.15	0.07
p_{PS}	0.00	0.11	0.07	0.15	0.54	0.37	0.06	0.03	0.03	0.20	0.02	0.35	0.01	0.10	0.00	0.00	0.26	0.12
p_C	0.00	0.01	0.02	0.24	0.10	0.12	0.05	0.03	0.00	0.44	0.06	0.29	0.02	0.31	0.00	0.00	0.03	0.03
p_q	0.00	0.00	0.00	0.27	0.32	0.41	0.21	0.03	0.28	0.44	0.01	0.06	0.00	0.42	0.00	0.00	0.05	0.14
p_a	0.00	0.01	0.00	0.10	0.54	0.44	0.17	0.38	0.44	0.18	0.06	0.09	0.10	0.69	0.00	0.00	0.19	0.24

	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
	dPia	Noa	dNoa	dLno	Ig	2Ig	Nsi	dli	Cei	Ivg	Ivc	Oa	dWc	dCoa	dNco	dNca	dFin	dFnl
m	-0.51	-0.40	-0.53	-0.40	-0.45	-0.37	-0.66	-0.30	-0.56	-0.36	-0.45	-0.27	-0.41	-0.29	-0.40	-0.41	0.28	-0.34
α	-0.57	-0.41	-0.62	-0.42	-0.49	-0.44	-0.77	-0.37	-0.79	-0.43	-0.52	-0.32	-0.48	-0.39	-0.45	-0.45	0.28	-0.39
α_{FF}	-0.40	-0.54	-0.41	-0.24	-0.30	-0.25	-0.65	-0.17	-0.54	-0.25	-0.39	-0.37	-0.47	-0.10	-0.31	-0.29	0.43	-0.35
α_{PS}	-0.44	-0.56	-0.42	-0.27	-0.29	-0.27	-0.65	-0.17	-0.51	-0.23	-0.45	-0.32	-0.47	-0.10	-0.35	-0.32	0.47	-0.39
α_{C}	-0.35	-0.44	-0.34	-0.17	-0.25	-0.12	-0.58	-0.04	-0.46	-0.16	-0.30	-0.32	-0.41	-0.06	-0.25	-0.26	0.41	-0.31
α_q	-0.22	-0.41	-0.10	0.03	-0.03	0.05	-0.29	0.12	-0.24	0.01	-0.30	-0.54	-0.48	0.12	-0.03	0.00	0.44	-0.08
a	-0.31	-0.45	-0.23	-0.07	-0.14	-0.08	-0.26	-0.02	-0.25	-0.10	-0.38	-0.52	-0.50	0.08	-0.15	-0.12	0.49	-0.16
t_m	-3.76	-2.94	-3.89	-3.03	-3.56	-2.74	-4.45	-2.70	-3.16	-2.57	-3.32	-2.13	-3.13	-2.08	-3.33	-3.32	2.31	-3.21
t_α	-4.31	-3.04	-4.60	-3.00	-3.79	-3.23	-5.34	-3.19	-5.29	-3.17	-3.74	-2.49	-3.62	-2.85	-3.72	-3.57	2.40	-3.66
t_{FF}	-3.13	-3.82	-3.16	-1.74	-2.61	-2.03	-4.78	-1.78	-4.56	-1.92	-2.95	-2.95	-3.73	-0.92	-2.50	-2.37	3.80	-3.35
t_{PS}	-3.28	-3.77	-3.21	-1.83	-2.43	-2.10	-4.59	-1.71	-4.16	-1.77	-3.45	-2.50	-3.64	-0.83	-2.82	-2.57	3.93	-3.52
t_{C}	-2.65	-3.21	-2.48	-1.10	-2.15	-1.06	-4.28	-0.39	-3.77	-1.19	-2.26	-2.30	-3.08	-0.54	-1.99	-2.03	3.45	-2.76
t_q	-1.77	-2.24	-0.66	0.16	-0.27	0.42	-2.19	1.14	-1.85	0.11	-2.11	-3.77	-3.43	1.16	-0.23	0.03	2.94	-0.73
t_a	-2.64	-2.76	-1.51	-0.47	-1.24	-0.64	-2.20	-0.23	-2.40	-0.85	-3.00	-4.06	-3.79	0.77	-1.19	-0.94	3.89	-1.54
$ \alpha $	0.15	0.15	0.18	0.13	0.12	0.13	0.20	0.13	0.17	0.14	0.15	0.15	0.14	0.14	0.15	0.16	0.11	0.13
$ \alpha_{\text{FF}} $	0.13	0.17	0.14	0.11	0.12	0.12	0.18	0.10	0.15	0.10	0.11	0.13	0.13	0.10	0.13	0.13	0.12	0.13
$ \alpha_{\text{PS}} $	0.14	0.17	0.14	0.11	0.12	0.12	0.19	0.10	0.15	0.10	0.12	0.12	0.13	0.09	0.14	0.14	0.12	0.13
$ \alpha_{\text{C}} $	0.12	0.15	0.13	0.11	0.10	0.11	0.16	0.07	0.14	0.09	0.10	0.12	0.12	0.09	0.12	0.12	0.11	0.12
$ \alpha_q $	0.12	0.11	0.07	0.05	0.09	0.08	0.11	0.07	0.12	0.10	0.08	0.13	0.13	0.08	0.10	0.10	0.08	0.09
$ \alpha_a $	0.10	0.10	0.08	0.06	0.07	0.06	0.11	0.05	0.10	0.10	0.08	0.12	0.12	0.06	0.08	0.09	0.09	0.08
p	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_{FF}	0.00	0.00	0.00	0.04	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_{PS}	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.05	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.00
p_{C}	0.00	0.00	0.00	0.05	0.00	0.00	0.00	0.03	0.00	0.07	0.02	0.00	0.00	0.02	0.00	0.00	0.00	0.00
p_q	0.00	0.00	0.21	0.62	0.01	0.08	0.00	0.42	0.00	0.07	0.27	0.00	0.00	0.06	0.00	0.01	0.02	0.05
p_a	0.01	0.00	0.09	0.82	0.15	0.28	0.00	0.49	0.01	0.05	0.14	0.00	0.00	0.29	0.05	0.03	0.00	0.04
	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108
	Dac	Poa	Pta	Pda	Ndf	Roe1	dRoe1	dRoe6	dRoe12	Roal	dRoal	dRoal6	Rna ^{q1}	Ato ^{q1}	Ato ^{q6}	Ato ^{q12}	Cto ^{q1}	Cto ^{q6}
m	-0.36	-0.40	-0.42	-0.37	-0.31	0.69	0.76	0.39	0.27	0.57	0.58	0.31	0.64	0.58	0.50	0.40	0.44	0.41
α	-0.36	-0.48	-0.53	-0.41	-0.38	0.86	0.80	0.42	0.30	0.75	0.58	0.30	0.86	0.48	0.41	0.30	0.39	0.35
α_{FF}	-0.46	-0.28	-0.33	-0.44	-0.29	1.07	0.89	0.51	0.38	0.95	0.66	0.39	1.10	0.69	0.65	0.56	0.45	0.43
α_{PS}	-0.41	-0.25	-0.32	-0.40	-0.30	1.11	0.93	0.53	0.39	0.99	0.70	0.43	1.14	0.70	0.65	0.56	0.46	0.44
α_{C}	-0.47	-0.21	-0.31	-0.36	-0.25	0.78	0.59	0.26	0.18	0.62	0.36	0.13	0.83	0.55	0.52	0.45	0.37	0.34
α_q	-0.64	-0.07	-0.15	-0.28	0.03	-0.03	0.34	-0.02	-0.09	0.04	0.06	-0.18	0.18	0.31	0.32	0.30	-0.11	-0.08
a	-0.60	-0.11	-0.13	-0.32	-0.04	0.51	0.79	0.41	0.28	0.49	0.53	0.26	0.50	0.37	0.38	0.33	-0.05	-0.03
t_m	-2.73	-2.85	-3.00	-3.19	-2.44	3.07	5.43	3.28	2.57	2.59	3.77	2.19	2.68	3.17	2.87	2.37	2.37	2.30
t_α	-2.66	-3.60	-4.14	-3.64	-2.94	4.09	6.14	3.83	3.17	3.63	3.84	2.25	3.99	2.56	2.25	1.76	1.97	1.85
t_{FF}	-3.56	-2.36	-2.66	-3.95	-2.37	5.55	6.29	4.17	3.68	5.23	4.28	2.76	5.70	4.22	4.36	3.91	2.44	2.44
t_{PS}	-3.13	-2.08	-2.50	-3.60	-2.46	5.55	6.17	4.10	3.61	5.46	4.50	2.97	6.12	4.06	4.22	3.76	2.42	2.43
t_{C}	-3.44	-1.78	-2.34	-3.02	-2.02	4.19	4.48	2.50	2.07	3.41	2.48	0.97	4.38	3.34	3.38	3.02	1.95	1.88
t_q	-4.37	-0.57	-1.07	-1.88	0.25	-0.27	2.29	-0.20	-0.96	0.31	0.36	-1.23	1.32	1.75	1.88	1.85	-0.65	-0.48
t_a	-4.30	-0.95	-1.03	-2.50	-0.36	3.64	5.39	3.29	2.62	3.46	3.24	1.83	3.55	2.39	2.69	2.46	-0.32	-0.23
$ \alpha $	0.11	0.12	0.13	0.12	0.12	0.16	0.20	0.13	0.10	0.15	0.19	0.12	0.16	0.13	0.11	0.08	0.12	0.11
$ \alpha_{\text{FF}} $	0.12	0.12	0.11	0.15	0.10	0.23	0.21	0.15	0.12	0.22	0.19	0.14	0.23	0.16	0.14	0.12	0.13	0.13
$ \alpha_{\text{PS}} $	0.12	0.11	0.11	0.14	0.10	0.23	0.22	0.16	0.13	0.23	0.20	0.14	0.23	0.16	0.14	0.12	0.12	0.13
$ \alpha_{\text{C}} $	0.12	0.11	0.11	0.14	0.09	0.15	0.13	0.10	0.09	0.13	0.12	0.07	0.16	0.13	0.12	0.12	0.10	0.11
$ \alpha_q $	0.15	0.12	0.08	0.17	0.08	0.10	0.09	0.07	0.08	0.06	0.10	0.08	0.07	0.11	0.07	0.08	0.09	0.09
$ \alpha_a $	0.13	0.12	0.07	0.16	0.05	0.11	0.15	0.09	0.06	0.14	0.16	0.10	0.14	0.15	0.11	0.10	0.08	0.09
p	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.01	0.04	0.09	0.00	0.00	0.04	0.01	0.03	0.19	0.17	0.02
p_{FF}	0.00	0.00	0.00	0.00	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.11	0.00
p_{PS}	0.00	0.00	0.00	0.00	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.13	0.00
p_{C}	0.00	0.00	0.00	0.00	0.11	0.00	0.00	0.00	0.00	0.06	0.07	0.05	0.00	0.01	0.02	0.03	0.22	0.01
p_q	0.00	0.00	0.04	0.00	0.36	0.01	0.04	0.05	0.01	0.85	0.42	0.04	0.23	0.03	0.14	0.09	0.25	0.01
p_a	0.00	0.00	0.09	0.00	0.87	0.01	0.00	0.02	0.04	0.07	0.01	0.03	0.01	0.00	0.01	0.01	0.38	0.03

	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126
	Cto ^q 12	Gpa	Gla ^q 1	Gla ^q 6	Gla ^q 12	Ole ^q 1	Ole ^q 6	Ola ^q 1	Ola ^q 6	Ola ^q 12	Cop	Cla	Cla ^q 1	Cla ^q 6	Cla ^q 12	F ^q 1	F ^q 6	F ^q 12
m	0.37	0.38	0.51	0.34	0.29	0.67	0.45	0.72	0.51	0.47	0.63	0.53	0.49	0.48	0.47	0.58	0.53	0.42
α	0.31	0.37	0.53	0.33	0.28	0.82	0.60	0.88	0.66	0.63	0.82	0.69	0.60	0.56	0.56	0.81	0.72	0.62
α_{FF}	0.41	0.55	0.71	0.53	0.47	0.80	0.59	1.16	0.94	0.90	1.08	0.99	0.80	0.74	0.75	0.73	0.66	0.54
α_{PS}	0.41	0.50	0.72	0.53	0.47	0.82	0.59	1.19	0.96	0.92	1.06	0.96	0.82	0.74	0.76	0.67	0.59	0.49
α_{C}	0.34	0.49	0.56	0.41	0.39	0.56	0.41	0.88	0.70	0.70	0.94	0.88	0.64	0.57	0.60	0.48	0.46	0.39
α_q	-0.05	0.18	0.20	0.10	0.13	-0.04	-0.16	0.37	0.25	0.33	0.69	0.74	0.43	0.40	0.46	0.13	0.15	0.07
a	-0.01	0.19	0.30	0.18	0.17	0.22	0.04	0.64	0.46	0.49	0.76	0.80	0.56	0.54	0.59	0.39	0.39	0.30
t_m	2.13	2.62	3.40	2.43	2.12	3.14	2.22	3.35	2.51	2.46	3.44	3.02	3.02	3.45	3.57	2.47	2.52	2.22
t_α	1.70	2.44	3.42	2.32	2.03	3.89	2.97	4.26	3.38	3.43	5.17	4.33	3.74	4.21	4.54	3.57	3.56	3.50
t_{FF}	2.36	3.84	4.65	3.80	3.60	4.20	3.24	6.43	5.58	5.91	8.12	7.90	5.59	6.35	7.19	3.59	3.72	3.66
t_{PS}	2.35	3.49	4.64	3.74	3.56	4.23	3.25	6.62	5.76	6.04	7.98	7.70	5.63	6.27	7.21	3.39	3.40	3.32
t_{C}	1.88	3.39	3.81	3.10	2.97	3.08	2.31	4.97	4.24	4.56	7.00	6.83	4.50	4.97	5.82	2.45	2.64	2.66
t_q	-0.31	1.24	1.41	0.79	1.01	-0.25	-1.06	2.34	1.78	2.48	4.77	4.89	2.69	2.82	3.56	0.58	0.86	0.49
t_a	-0.10	1.46	2.14	1.46	1.42	1.60	0.32	3.85	3.28	3.81	5.95	6.21	3.66	3.92	4.76	1.72	2.25	2.16
$ \alpha $	0.10	0.08	0.11	0.07	0.06	0.18	0.12	0.20	0.16	0.14	0.16	0.16	0.22	0.15	0.15	0.21	0.19	0.15
$ \alpha_{\text{FF}} $	0.12	0.15	0.18	0.16	0.14	0.21	0.15	0.27	0.21	0.19	0.22	0.20	0.26	0.19	0.20	0.21	0.19	0.16
$ \alpha_{\text{PS}} $	0.12	0.14	0.19	0.17	0.15	0.20	0.15	0.27	0.21	0.18	0.22	0.20	0.26	0.19	0.20	0.21	0.19	0.15
$ \alpha_{\text{C}} $	0.11	0.15	0.17	0.15	0.14	0.14	0.12	0.20	0.16	0.14	0.19	0.17	0.24	0.15	0.15	0.16	0.15	0.13
$ \alpha_q $	0.09	0.12	0.11	0.11	0.10	0.07	0.09	0.12	0.08	0.08	0.17	0.14	0.18	0.10	0.11	0.10	0.15	0.11
$ \alpha_a $	0.08	0.10	0.12	0.11	0.10	0.08	0.05	0.21	0.15	0.14	0.19	0.17	0.20	0.13	0.15	0.13	0.12	0.08
p	0.07	0.04	0.01	0.36	0.46	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_{FF}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_{PS}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_{C}	0.02	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_q	0.01	0.11	0.12	0.20	0.41	0.09	0.01	0.01	0.03	0.04	0.00	0.00	0.00	0.05	0.00	0.10	0.00	0.00
p_a	0.02	0.04	0.02	0.09	0.26	0.20	0.31	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.06	0.00	0.00
	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144
	Fp6	Tbi ^q 12	Oca	loca	Adm	Rdm	Rdm ^q 1	Rdm ^q 6	Rdm ^q 12	Ol	Ol ^q 1	Ol ^q 6	Ol ^q 12	Hs	Etr	Rer	Eprd	Etl
m	-0.63	0.22	0.54	0.55	0.70	0.68	1.19	0.83	0.83	0.46	0.49	0.49	0.49	-0.31	0.25	0.32	-0.49	0.36
α	-1.04	0.26	0.66	0.62	0.72	0.50	1.06	0.74	0.78	0.49	0.56	0.57	0.57	-0.20	0.25	0.29	-0.58	0.32
α_{FF}	-1.36	0.31	0.69	0.57	0.09	0.29	0.86	0.49	0.52	0.43	0.55	0.53	0.53	-0.35	0.25	0.35	-0.94	0.36
α_{PS}	-1.37	0.21	0.71	0.56	0.06	0.29	0.78	0.45	0.49	0.39	0.53	0.50	0.52	-0.34	0.27	0.39	-1.02	0.43
α_{C}	-0.61	0.27	0.64	0.37	0.26	0.39	1.47	0.89	0.77	0.41	0.53	0.51	0.50	-0.27	0.21	0.33	-0.80	0.24
α_q	-0.17	0.34	0.13	0.07	0.08	0.70	1.47	0.97	0.80	0.03	0.07	0.09	0.12	-0.31	0.09	0.39	-0.49	0.28
a	-0.78	0.26	0.27	0.30	-0.09	0.46	0.85	0.57	0.50	0.11	0.19	0.18	0.22	-0.43	0.21	0.31	-0.80	0.38
t_m	-2.03	1.96	2.64	4.34	2.73	2.58	2.93	2.12	2.32	2.70	2.52	2.58	2.73	-2.08	2.35	2.25	-2.75	2.85
t_α	-4.04	2.42	3.09	4.65	2.80	2.00	2.62	1.87	2.08	2.75	2.77	2.93	3.11	-1.39	2.26	2.05	-3.30	2.61
t_{FF}	-6.43	2.78	3.35	4.44	0.50	1.27	2.24	1.47	1.75	2.52	2.78	2.81	2.99	-2.34	2.38	2.38	-6.02	2.87
t_{PS}	-6.63	1.88	3.34	4.23	0.33	1.27	2.04	1.38	1.70	2.21	2.67	2.64	2.85	-2.15	2.50	2.60	-6.51	3.19
t_{C}	-3.70	2.36	3.14	2.94	1.18	1.84	3.64	2.89	2.74	2.38	2.69	2.69	2.74	-1.61	1.92	2.10	-5.01	1.92
t_q	-0.63	2.93	0.65	0.53	0.31	2.89	2.97	2.73	2.80	0.19	0.37	0.54	0.69	-1.51	0.69	2.20	-2.77	1.55
t_a	-2.92	2.33	1.34	2.35	-0.44	1.93	2.05	1.67	1.73	0.68	1.05	1.08	1.34	-2.54	1.90	1.91	-5.01	2.65
$ \alpha $	0.15	0.08	0.15	0.13	0.23	0.14	0.26	0.22	0.26	0.11	0.12	0.11	0.12	0.09	0.10	0.07	0.13	0.10
$ \alpha_{\text{FF}} $	0.22	0.10	0.20	0.14	0.14	0.18	0.30	0.27	0.30	0.12	0.10	0.10	0.10	0.09	0.11	0.12	0.20	0.10
$ \alpha_{\text{PS}} $	0.22	0.09	0.20	0.14	0.14	0.19	0.29	0.28	0.31	0.10	0.10	0.09	0.09	0.09	0.12	0.13	0.22	0.11
$ \alpha_{\text{C}} $	0.12	0.10	0.20	0.11	0.20	0.21	0.45	0.37	0.37	0.13	0.12	0.12	0.12	0.09	0.11	0.13	0.17	0.09
$ \alpha_q $	0.12	0.10	0.12	0.10	0.07	0.27	0.55	0.49	0.47	0.11	0.09	0.09	0.09	0.14	0.10	0.15	0.17	0.08
$ \alpha_a $	0.10	0.10	0.11	0.10	0.07	0.21	0.38	0.34	0.34	0.08	0.08	0.07	0.08	0.16	0.08	0.12	0.22	0.09
p	0.00	0.02	0.03	0.00	0.08	0.24	0.10	0.45	0.19	0.23	0.13	0.01	0.02	0.27	0.01	0.69	0.01	0.04
p_{FF}	0.00	0.00	0.00	0.00	0.32	0.01	0.06	0.11	0.03	0.01	0.08	0.01	0.01	0.06	0.00	0.13	0.00	0.02
p_{PS}	0.00	0.00	0.00	0.00	0.25	0.01	0.08	0.09	0.03	0.02	0.08	0.02	0.02	0.09	0.00	0.08	0.00	0.00
p_{C}	0.00	0.00	0.00	0.00	0.17	0.00	0.00	0.01	0.00	0.01	0.08	0.01	0.02	0.07	0.01	0.05	0.00	0.19
p_q	0.00	0.00	0.05	0.01	0.69	0.00	0.00	0.00	0.00	0.03	0.15	0.03	0.01	0.03	0.02	0.02	0.01	0.23
p_a	0.00	0.00	0.10	0.01	0.77	0.01	0.01	0.02	0.01	0.06	0.24	0.10	0.04	0.00	0.05	0.06	0.00	0.06

	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161
	Alm ^q 1	Alm ^q 6	Alm ^q 12	R_a^1	$R_a^{[2,5]}$	$R_n^{[2,5]}$	$R_a^{[6,10]}$	$R_n^{[6,10]}$	$R_a^{[11,15]}$	$R_a^{[16,20]}$	Sv1	Dtv6	Dtv12	Ami12	Ts1	Isff1	Isq1
m	0.62	0.63	0.57	0.65	0.69	-0.51	0.83	-0.45	0.67	0.56	-0.53	-0.37	-0.42	0.42	0.23	0.34	0.27
α	0.55	0.56	0.50	0.55	0.66	-0.67	0.80	-0.59	0.68	0.60	-0.67	-0.35	-0.40	0.31	0.19	0.33	0.25
α_{FF}	0.03	0.08	0.05	0.65	0.71	-0.14	0.87	-0.25	0.70	0.63	-0.63	0.03	-0.03	-0.03	0.22	0.31	0.24
α_{PS}	-0.02	0.03	0.00	0.69	0.68	-0.12	0.88	-0.27	0.75	0.68	-0.67	0.05	-0.02	-0.05	0.21	0.32	0.24
α_{C}	0.12	0.10	0.02	0.42	0.74	-0.23	0.97	-0.14	0.68	0.64	-0.60	-0.05	-0.04	-0.04	0.26	0.26	0.20
α_q	0.28	0.25	0.15	0.55	0.81	-0.16	1.13	0.07	0.65	0.64	-0.35	-0.11	-0.13	0.15	0.31	0.31	0.31
a	0.10	0.14	0.11	0.65	0.73	0.03	1.05	-0.06	0.73	0.61	-0.34	0.00	-0.07	0.08	0.29	0.34	0.30
t_m	2.87	3.13	2.94	3.23	4.00	-2.22	4.91	-2.24	4.66	3.29	-2.47	-1.99	-2.28	1.99	2.11	3.50	2.88
t_α	2.48	2.74	2.55	2.75	3.70	-3.07	4.86	-2.95	4.87	3.48	-3.13	-1.89	-2.14	1.54	1.70	3.28	2.69
t_{FF}	0.22	0.66	0.39	3.57	3.95	-0.82	4.92	-1.54	4.75	3.74	-2.89	0.39	-0.40	-0.38	2.06	3.24	2.65
t_{PS}	-0.18	0.28	-0.02	3.68	3.65	-0.68	4.87	-1.57	4.85	3.85	-3.07	0.53	-0.29	-0.54	1.94	3.22	2.65
t_{C}	0.85	0.82	0.13	2.42	3.69	-1.31	5.28	-0.75	4.83	3.48	-2.63	-0.52	-0.43	-0.46	2.38	2.61	2.19
t_q	1.77	1.78	1.08	2.48	3.90	-0.86	4.88	0.35	3.60	3.14	-1.42	-1.21	-1.65	2.03	2.75	2.64	3.01
t_a	0.70	1.24	0.94	3.35	3.93	0.18	5.22	-0.34	4.07	3.67	-1.43	0.04	-0.87	1.12	2.56	3.05	2.91
$ \alpha $	0.18	0.21	0.19	0.15	0.14	0.26	0.19	0.19	0.17	0.18	0.19	0.15	0.17	0.12	0.05	0.07	0.07
$ \alpha_{\text{FF}} $	0.07	0.09	0.09	0.17	0.15	0.15	0.20	0.13	0.18	0.18	0.20	0.04	0.05	0.04	0.08	0.09	0.09
$ \alpha_{\text{PS}} $	0.06	0.08	0.07	0.18	0.14	0.15	0.20	0.13	0.18	0.18	0.21	0.04	0.05	0.04	0.08	0.09	0.09
$ \alpha_{\text{C}} $	0.07	0.08	0.07	0.12	0.15	0.15	0.23	0.11	0.17	0.17	0.18	0.06	0.06	0.04	0.08	0.08	0.09
$ \alpha_q $	0.09	0.10	0.07	0.15	0.17	0.13	0.24	0.15	0.18	0.17	0.12	0.08	0.09	0.12	0.10	0.10	0.11
$ \alpha_a $	0.06	0.07	0.06	0.17	0.16	0.09	0.22	0.15	0.20	0.16	0.12	0.04	0.05	0.08	0.08	0.10	0.09
p	0.09	0.03	0.05	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.08	0.04	0.08	0.51	0.08	0.06
p_{FF}	0.30	0.22	0.25	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.02	0.01	0.01	0.02	0.00	0.00
p_{PS}	0.29	0.32	0.29	0.02	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.06	0.03	0.04	0.02	0.00	0.00
p_{C}	0.16	0.25	0.30	0.16	0.00	0.00	0.00	0.02	0.00	0.00	0.01	0.04	0.05	0.03	0.05	0.01	0.00
p_q	0.07	0.04	0.22	0.09	0.00	0.00	0.00	0.00	0.00	0.01	0.05	0.00	0.00	0.00	0.01	0.00	0.00
p_a	0.34	0.18	0.33	0.06	0.00	0.00	0.00	0.00	0.00	0.01	0.10	0.04	0.03	0.01	0.02	0.00	0.00

Table 4 : Significant Anomalies, ABM-EW, January 1967 to December 2014, 576 Months

For each high-minus-low decile, $m, \alpha, \alpha_{FF}, \alpha_{PS}, \alpha_C, \alpha_q$, and a are the average return, the Fama-French three-factor alpha, the Pastor-Stambaugh alpha, the Carhart alpha, the q -model alpha, and the five-factor alpha. and $t_m, t_\alpha, t_{FF}, t_{PS}, t_C, t_q$, and t_a are their t -statistics adjusted for heteroscedasticity and autocorrelations, respectively. $|\alpha|, |\alpha_{FF}|, |\alpha_{PS}|, |\alpha_C|, |\alpha_q|$, and $|a|$ are the mean absolute alpha across a given set of deciles, and $p, p_{FF}, p_{PS}, p_C, p_q$, and p_a are the p -value of the GRS test on the null that the alphas across the deciles are jointly zero. Table A1 describes the symbols, and Appendix ?? details variable definition and portfolio construction.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Sue1	Sue6	Abr1	Abr6	Abr12	Re1	Re6	Re12	R^6_1	R^6_6	R^6_{12}	R^{11}_1	R^{11}_6	Im1	Im6	Im12	Rs1	Rs6
m	0.84	0.40	0.96	0.45	0.31	0.76	0.43	0.26	1.06	0.91	0.56	1.22	0.76	0.97	0.69	0.67	0.57	0.28
α	0.88	0.43	1.00	0.46	0.30	0.80	0.47	0.31	1.19	0.96	0.58	1.29	0.78	1.08	0.74	0.69	0.63	0.32
α_{FF}	1.07	0.63	1.05	0.52	0.37	0.91	0.59	0.43	1.33	1.11	0.79	1.54	1.08	1.14	0.78	0.79	0.83	0.54
α_{PS}	1.10	0.66	1.05	0.52	0.38	0.90	0.59	0.42	1.35	1.13	0.82	1.58	1.12	1.13	0.74	0.78	0.88	0.58
α_C	0.74	0.36	0.87	0.31	0.21	0.45	0.19	0.15	0.18	0.04	0.01	0.24	0.03	0.39	0.02	0.20	0.67	0.38
α_q	0.37	0.00	0.85	0.31	0.22	0.24	-0.05	-0.08	0.38	0.08	0.01	0.39	0.07	0.54	0.15	0.33	0.27	0.04
a	0.84	0.43	1.01	0.52	0.38	0.82	0.48	0.31	1.13	0.91	0.68	1.34	0.96	0.94	0.64	0.75	0.60	0.33
t_m	6.31	3.59	8.67	5.65	5.13	4.01	2.69	2.04	3.87	3.86	2.87	4.28	2.89	4.15	3.63	3.98	4.56	2.44
t_α	7.25	4.16	9.39	6.05	5.37	4.51	3.11	2.53	4.56	4.35	3.15	4.81	3.14	4.53	3.88	4.07	5.20	2.85
t_{FF}	8.71	6.03	9.12	6.24	6.24	5.19	4.07	3.72	5.30	4.82	4.26	5.73	4.43	4.95	4.16	4.70	7.55	5.58
t_{PS}	8.54	6.15	8.85	6.41	6.59	5.05	3.99	3.65	5.15	4.75	4.35	5.76	4.51	4.71	3.78	4.41	8.11	5.98
t_C	6.21	3.60	8.60	3.53	3.39	2.66	1.37	1.31	0.95	0.28	0.06	1.86	0.17	1.97	0.15	1.49	5.67	3.60
t_q	3.50	0.03	5.66	2.19	2.33	1.43	-0.33	-0.65	0.96	0.21	0.03	0.97	0.21	1.66	0.59	1.47	2.56	0.37
t_a	6.73	4.06	8.13	4.82	5.30	4.53	2.90	2.37	3.26	2.80	2.91	3.76	3.25	3.18	2.60	3.68	5.26	3.14
$ \alpha $	0.25	0.18	0.24	0.17	0.15	0.19	0.13	0.11	0.25	0.24	0.19	0.30	0.23	0.30	0.22	0.20	0.19	0.16
$ \alpha_{FF} $	0.27	0.14	0.20	0.11	0.09	0.24	0.16	0.12	0.20	0.21	0.16	0.31	0.22	0.26	0.18	0.16	0.18	0.11
$ \alpha_{PS} $	0.28	0.15	0.19	0.11	0.09	0.24	0.17	0.12	0.20	0.21	0.16	0.32	0.23	0.25	0.17	0.16	0.19	0.12
$ \alpha_C $	0.20	0.13	0.19	0.15	0.17	0.15	0.13	0.14	0.13	0.11	0.13	0.10	0.12	0.13	0.04	0.06	0.16	0.14
$ \alpha_q $	0.10	0.10	0.19	0.17	0.18	0.13	0.15	0.16	0.16	0.14	0.15	0.11	0.13	0.16	0.09	0.11	0.09	0.10
$ \alpha_a $	0.22	0.09	0.20	0.11	0.09	0.22	0.14	0.10	0.18	0.16	0.13	0.27	0.18	0.26	0.20	0.20	0.12	0.07
p	0.00	0.00	0.00	0.00	0.00	0.00	0.09	0.17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
p_{FF}	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_{PS}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_C	0.00	0.00	0.00	0.00	0.00	0.01	0.13	0.09	0.00	0.00	0.00	0.01	0.01	0.10	0.22	0.25	0.00	0.00
p_q	0.00	0.02	0.00	0.00	0.00	0.00	0.02	0.01	0.00	0.00	0.00	0.00	0.00	0.29	0.18	0.21	0.01	0.04
p_a	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.06
	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
	Tes1	Tes6	dEf1	dEf6	dEf12	Nei1	Nei6	52w6	52w12	ϵ^6_1	ϵ^6_6	ϵ^6_{12}	ϵ^{11}_1	ϵ^{11}_6	ϵ^{11}_{12}	Sm1	Sm6	Sm12
m	0.31	0.25	1.20	0.58	0.37	0.46	0.25	0.73	0.60	0.51	0.61	0.40	0.93	0.60	0.33	0.90	0.27	0.26
α	0.23	0.18	1.22	0.58	0.36	0.49	0.28	1.10	0.92	0.57	0.65	0.43	0.95	0.61	0.34	0.95	0.31	0.27
α_{FF}	0.39	0.33	1.38	0.71	0.50	0.61	0.39	1.14	1.00	0.61	0.68	0.50	1.01	0.70	0.45	0.92	0.27	0.26
α_{PS}	0.41	0.36	1.38	0.71	0.50	0.64	0.42	1.14	1.01	0.61	0.68	0.50	1.03	0.70	0.44	0.95	0.26	0.25
α_C	0.24	0.22	0.98	0.36	0.24	0.42	0.22	0.14	0.15	0.17	0.28	0.18	0.50	0.28	0.13	0.84	0.02	0.04
α_q	0.02	-0.01	0.95	0.27	0.14	0.04	-0.15	-0.17	-0.16	0.30	0.35	0.20	0.57	0.31	0.14	0.89	0.04	0.03
a	0.29	0.25	1.37	0.72	0.49	0.35	0.13	0.65	0.59	0.54	0.66	0.49	0.94	0.69	0.46	1.00	0.21	0.21
t_m	2.54	2.15	6.23	3.97	3.20	4.32	2.36	2.62	2.41	3.55	4.86	3.87	5.87	4.22	2.83	4.47	2.64	3.59
t_α	1.95	1.50	6.52	4.08	3.21	4.70	2.70	4.83	4.62	4.00	5.16	4.06	5.96	4.29	2.88	4.66	2.92	3.65
t_{FF}	3.30	3.00	7.52	5.36	4.81	5.95	4.01	5.31	5.18	4.22	5.35	4.79	6.10	4.92	3.94	4.43	2.62	3.38
t_{PS}	3.53	3.33	7.48	5.35	4.95	6.16	4.34	5.32	5.34	4.21	5.22	4.67	6.17	4.87	3.78	4.61	2.50	3.37
t_C	2.13	2.06	5.81	2.96	2.51	4.20	2.17	0.83	0.92	1.35	2.66	2.06	3.88	2.46	1.35	4.09	0.24	0.60
t_q	0.14	-0.07	4.54	1.82	1.17	0.58	-2.02	-0.49	-0.58	1.72	2.19	1.70	2.85	1.90	1.10	3.44	0.26	0.34
t_a	2.47	2.40	6.57	4.95	4.30	4.01	1.47	2.14	2.38	3.34	4.53	4.39	5.04	4.60	3.93	4.33	1.59	2.15
$ \alpha $	0.20	0.20	0.31	0.17	0.13	0.25	0.18	0.30	0.26	0.20	0.21	0.19	0.27	0.22	0.19	0.29	0.19	0.17
$ \alpha_{FF} $	0.14	0.11	0.36	0.19	0.14	0.24	0.15	0.26	0.24	0.13	0.15	0.12	0.23	0.16	0.11	0.24	0.07	0.06
$ \alpha_{PS} $	0.14	0.11	0.36	0.19	0.14	0.23	0.15	0.26	0.24	0.13	0.15	0.12	0.24	0.17	0.11	0.25	0.07	0.06
$ \alpha_C $	0.14	0.15	0.23	0.15	0.14	0.21	0.17	0.12	0.13	0.11	0.13	0.14	0.15	0.15	0.14	0.24	0.11	0.09
$ \alpha_q $	0.10	0.10	0.23	0.16	0.19	0.11	0.09	0.12	0.13	0.09	0.11	0.11	0.14	0.11	0.11	0.20	0.06	0.03
$ \alpha_a $	0.09	0.06	0.33	0.18	0.13	0.19	0.11	0.15	0.14	0.12	0.13	0.11	0.22	0.15	0.10	0.23	0.13	0.14
p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_{FF}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
p_{PS}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02
p_C	0.05	0.01	0.00	0.00	0.02	0.00	0.00	0.03	0.11	0.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.29
p_q	0.66	0.18	0.00	0.01	0.05	0.00	0.00	0.08	0.20	0.02	0.00	0.01	0.00	0.00	0.00	0.00	0.02	0.63
p_a	0.20	0.17	0.00	0.00	0.00	0.00	0.00	0.03	0.02	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.08

	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
	llr1	llr6	llr12	lle1	lle6	Cm1	Cm6	Cm12	Sim1	Sim6	Sim12	Cim1	Cim6	Cim12	Bm	Bmj	Bm ^q 12	Am
m	0.89	0.45	0.36	0.78	0.32	0.53	0.19	0.16	1.15	0.30	0.22	1.00	0.39	0.33	0.74	0.53	0.52	0.62
α	0.96	0.50	0.39	0.79	0.35	0.55	0.22	0.18	1.17	0.31	0.23	1.02	0.42	0.34	0.94	0.70	0.68	0.77
α_{FF}	0.97	0.52	0.42	0.96	0.51	0.55	0.20	0.17	1.17	0.32	0.24	1.02	0.42	0.37	0.24	0.02	-0.03	0.02
α_{PS}	1.04	0.51	0.41	0.98	0.55	0.60	0.22	0.18	1.23	0.34	0.24	1.06	0.44	0.37	0.21	-0.04	-0.09	-0.02
α_{C}	0.82	0.20	0.11	0.75	0.29	0.43	0.00	0.02	0.99	-0.03	-0.06	0.80	0.10	0.08	0.17	0.30	0.29	0.01
α_q	0.90	0.29	0.19	0.50	0.09	0.38	-0.02	0.01	0.95	0.08	0.04	0.87	0.19	0.16	0.08	0.19	0.16	-0.18
a	0.98	0.47	0.37	0.78	0.40	0.53	0.12	0.12	1.17	0.29	0.22	1.06	0.39	0.36	0.01	-0.12	-0.17	-0.25
t_m	4.41	4.35	4.52	5.01	2.37	2.78	2.31	2.69	5.24	2.41	2.52	4.12	3.63	4.13	3.24	2.14	2.18	2.49
t_α	4.75	4.97	4.97	4.86	2.55	2.88	2.78	2.96	5.19	2.55	2.53	4.30	3.96	4.36	4.19	2.95	2.90	3.01
t_{FF}	4.54	5.02	5.20	5.90	3.73	2.82	2.38	2.90	4.80	2.33	2.60	4.04	3.55	4.19	2.59	0.11	-0.21	0.16
t_{PS}	4.65	4.68	4.91	5.68	3.85	3.09	2.60	3.15	5.00	2.41	2.58	4.05	3.53	4.08	2.24	-0.32	-0.73	-0.16
t_{C}	4.15	2.18	1.77	4.56	2.13	1.94	0.05	0.26	4.30	-0.22	-0.78	3.50	1.04	1.14	1.46	2.30	2.46	0.05
t_q	3.50	1.79	1.57	2.76	0.58	1.48	-0.16	0.08	2.76	0.34	0.26	2.53	0.99	1.21	0.37	0.88	0.74	-0.76
t_a	4.15	3.44	3.50	4.52	2.56	2.38	0.99	1.62	3.98	1.53	1.85	3.45	2.46	3.10	0.08	-0.86	-1.33	-1.89
$ \alpha $	0.30	0.19	0.17	0.19	0.14	0.14	0.10	0.09	0.34	0.15	0.14	0.32	0.16	0.14	0.25	0.21	0.21	0.22
$ \alpha_{\text{FF}} $	0.27	0.10	0.09	0.21	0.13	0.14	0.08	0.08	0.29	0.08	0.07	0.29	0.10	0.08	0.05	0.04	0.03	0.05
$ \alpha_{\text{PS}} $	0.28	0.10	0.09	0.21	0.14	0.15	0.09	0.09	0.31	0.08	0.07	0.31	0.11	0.08	0.04	0.04	0.02	0.05
$ \alpha_{\text{C}} $	0.25	0.09	0.08	0.13	0.08	0.12	0.07	0.09	0.28	0.11	0.11	0.26	0.13	0.12	0.13	0.13	0.13	0.13
$ \alpha_q $	0.25	0.07	0.06	0.10	0.05	0.19	0.21	0.22	0.25	0.09	0.08	0.26	0.12	0.11	0.14	0.14	0.15	0.14
$ \alpha_a $	0.27	0.14	0.14	0.20	0.14	0.17	0.07	0.06	0.29	0.09	0.09	0.30	0.09	0.09	0.03	0.05	0.06	0.07
p	0.00	0.00	0.00	0.00	0.10	0.11	0.05	0.05	0.00	0.02	0.02	0.00	0.00	0.00	0.01	0.02	0.02	0.02
p_{FF}	0.00	0.00	0.00	0.00	0.01	0.16	0.16	0.10	0.00	0.07	0.06	0.00	0.00	0.00	0.59	0.75	0.71	0.36
p_{PS}	0.00	0.00	0.00	0.00	0.01	0.09	0.11	0.07	0.00	0.08	0.06	0.00	0.00	0.00	0.68	0.73	0.90	0.17
p_{C}	0.00	0.03	0.04	0.00	0.42	0.42	0.69	0.70	0.00	0.08	0.15	0.00	0.05	0.10	0.10	0.00	0.00	0.10
p_q	0.00	0.02	0.02	0.11	0.95	0.07	0.13	0.13	0.00	0.00	0.00	0.00	0.01	0.01	0.20	0.07	0.00	0.05
p_a	0.00	0.00	0.00	0.00	0.03	0.13	0.28	0.24	0.00	0.00	0.00	0.00	0.00	0.00	0.91	0.37	0.08	0.26
	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72
	Rev1	Rev6	Rev12	Ep	Ep ^q 1	Ep ^q 6	Ep ^q 12	Cp	Cp ^q 1	Cp ^q 6	Cp ^q 12	Op	Op ^q 1	Op ^q 6	Op ^q 12	Nop	Nop ^q 6	Nop ^q 12
m	-0.62	-0.53	-0.50	0.61	1.12	0.80	0.53	0.74	0.83	0.53	0.54	0.41	0.38	0.37	0.31	0.66	0.46	0.45
α	-0.69	-0.60	-0.57	0.80	1.22	0.93	0.67	0.92	0.94	0.67	0.69	0.64	0.52	0.55	0.49	0.89	0.76	0.76
α_{FF}	-0.27	-0.23	-0.23	0.31	0.80	0.53	0.28	0.27	0.38	0.10	0.12	0.24	0.36	0.39	0.33	0.59	0.48	0.48
α_{PS}	-0.25	-0.20	-0.21	0.26	0.74	0.48	0.22	0.24	0.29	0.01	0.03	0.23	0.38	0.42	0.36	0.59	0.49	0.49
α_{C}	-0.22	-0.19	-0.19	0.29	1.04	0.69	0.36	0.19	0.77	0.40	0.26	0.20	0.62	0.56	0.41	0.44	0.46	0.41
α_q	-0.30	-0.26	-0.20	0.23	0.62	0.33	0.10	0.09	0.43	0.14	0.07	0.23	0.38	0.33	0.19	0.21	0.01	-0.04
a	-0.21	-0.20	-0.20	0.18	0.64	0.33	0.09	0.02	0.14	-0.11	-0.10	0.17	0.18	0.19	0.14	0.25	0.10	0.09
t_m	-3.44	-3.10	-3.05	3.12	5.41	4.45	3.19	3.30	3.49	2.38	2.62	2.08	2.22	2.47	2.19	3.65	1.98	2.04
t_α	-3.79	-3.54	-3.56	4.21	5.82	5.32	4.19	4.19	3.95	3.06	3.36	3.84	3.09	3.85	3.74	5.52	3.36	3.57
t_{FF}	-1.76	-1.60	-1.71	2.86	5.07	4.13	2.60	2.45	2.13	0.69	0.92	1.83	2.68	3.51	3.32	4.93	3.89	4.01
t_{PS}	-1.61	-1.40	-1.53	2.30	4.58	3.59	2.00	2.13	1.62	0.06	0.23	1.82	2.87	3.83	3.67	5.03	3.99	4.12
t_{C}	-1.35	-1.24	-1.28	2.56	7.48	6.09	3.64	1.39	5.43	2.88	2.02	1.38	4.71	5.13	3.99	3.59	3.21	2.83
t_q	-1.91	-1.55	-1.18	1.29	2.45	1.67	0.58	0.38	1.54	0.56	0.31	1.26	1.87	2.51	1.76	1.53	0.07	-0.22
t_a	-1.38	-1.41	-1.43	1.48	3.90	2.65	0.87	0.16	0.78	-0.71	-0.82	1.25	1.20	1.70	1.40	2.11	0.85	0.76
$ \alpha $	0.26	0.26	0.25	0.26	0.34	0.28	0.23	0.28	0.29	0.25	0.25	0.29	0.23	0.25	0.24	0.27	0.23	0.23
$ \alpha_{\text{FF}} $	0.10	0.09	0.09	0.08	0.20	0.13	0.08	0.08	0.11	0.06	0.06	0.12	0.15	0.14	0.14	0.14	0.13	0.13
$ \alpha_{\text{PS}} $	0.09	0.09	0.09	0.08	0.19	0.13	0.08	0.07	0.09	0.05	0.05	0.11	0.15	0.14	0.13	0.13	0.13	0.13
$ \alpha_{\text{C}} $	0.14	0.15	0.15	0.15	0.28	0.20	0.16	0.14	0.21	0.15	0.14	0.18	0.23	0.22	0.21	0.19	0.16	0.16
$ \alpha_q $	0.08	0.09	0.09	0.07	0.16	0.09	0.07	0.07	0.14	0.09	0.09	0.10	0.15	0.11	0.09	0.11	0.14	0.15
$ \alpha_a $	0.04	0.04	0.04	0.04	0.15	0.08	0.05	0.05	0.09	0.06	0.04	0.08	0.12	0.10	0.08	0.06	0.06	0.05
p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.01	0.00
p_{FF}	0.02	0.02	0.00	0.10	0.00	0.00	0.12	0.18	0.16	0.40	0.80	0.00	0.01	0.01	0.01	0.00	0.02	0.00
p_{PS}	0.11	0.10	0.01	0.12	0.00	0.00	0.16	0.36	0.48	0.60	0.88	0.00	0.01	0.00	0.01	0.00	0.01	0.00
p_{C}	0.04	0.02	0.00	0.07	0.00	0.00	0.00	0.15	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_q	0.15	0.24	0.16	0.50	0.00	0.00	0.03	0.29	0.00	0.00	0.02	0.03	0.00	0.00	0.01	0.03	0.01	0.00
p_a	0.45	0.38	0.15	0.56	0.00	0.00	0.15	0.58	0.04	0.05	0.51	0.03	0.01	0.02	0.04	0.02	0.29	0.02

	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
	Sg	Em	Em ^{q1}	Em ^{q6}	Em ^{q12}	Sp	Sp ^{q1}	Sp ^{q6}	Sp ^{q12}	Ocp	Ocp ^{q1}	Ir	Vhp	Ebp	Ndp	Dur	Aci	I/A
m	-0.41	-0.80	-0.97	-0.54	-0.55	0.77	0.77	0.67	0.64	0.65	0.59	-0.64	0.50	0.61	0.32	-0.65	-0.31	-0.69
α	-0.56	-1.00	-1.15	-0.77	-0.79	0.91	0.85	0.78	0.75	0.84	0.69	-0.70	0.64	0.78	0.35	-0.88	-0.32	-0.83
α_{FF}	-0.27	-0.47	-0.80	-0.41	-0.41	0.18	0.12	0.06	0.05	0.20	0.27	-0.09	0.16	0.05	-0.10	-0.36	-0.30	-0.55
α_{PS}	-0.28	-0.45	-0.77	-0.38	-0.39	0.09	0.03	-0.03	-0.04	0.18	0.24	-0.04	0.09	0.02	-0.13	-0.32	-0.33	-0.57
α_C	-0.20	-0.34	-1.03	-0.54	-0.42	0.19	0.58	0.39	0.25	0.07	0.59	-0.15	0.18	0.01	0.00	-0.19	-0.18	-0.46
α_q	-0.02	-0.10	-0.57	-0.10	-0.05	-0.29	-0.02	-0.16	-0.27	-0.01	0.14	-0.21	0.13	-0.17	0.14	0.01	-0.12	-0.29
a	-0.05	-0.10	-0.37	0.04	0.04	-0.36	-0.39	-0.46	-0.48	-0.09	-0.07	-0.02	0.03	-0.25	0.00	-0.06	-0.24	-0.34
t_m	-2.96	-3.70	-3.61	-2.26	-2.48	2.79	2.53	2.37	2.35	2.94	1.98	-3.29	2.71	2.46	2.10	-2.96	-3.64	-4.74
t_α	-4.33	-4.72	-4.31	-3.23	-3.51	3.19	2.78	2.71	2.70	3.85	2.24	-3.54	3.41	3.09	2.21	-4.28	-3.75	-5.86
t_{FF}	-2.56	-3.65	-3.89	-2.31	-2.69	0.99	0.54	0.27	0.24	1.68	1.22	-0.79	1.30	0.41	-0.94	-2.73	-3.48	-4.65
t_{PS}	-2.56	-3.42	-3.79	-2.20	-2.62	0.50	0.11	-0.15	-0.19	1.59	1.13	-0.34	0.73	0.15	-1.28	-2.43	-3.81	-4.76
t_C	-1.82	-2.35	-5.62	-3.36	-2.78	1.03	2.86	2.08	1.36	0.46	3.08	-1.18	1.51	0.10	0.02	-1.29	-2.04	-3.79
t_q	-0.16	-0.45	-1.82	-0.38	-0.22	-1.11	-0.05	-0.53	-0.98	-0.07	0.40	-1.34	0.71	-0.73	1.02	0.07	-1.27	-2.46
t_a	-0.58	-0.71	-1.87	0.28	0.33	-2.43	-1.83	-2.69	-3.20	-0.61	-0.31	-0.19	0.23	-2.07	-0.03	-0.50	-2.72	-3.21
$ \alpha $	0.23	0.30	0.38	0.30	0.30	0.26	0.26	0.25	0.24	0.32	0.28	0.27	0.24	0.23	0.16	0.28	0.22	0.28
$ \alpha_{FF} $	0.12	0.11	0.23	0.14	0.13	0.06	0.05	0.04	0.04	0.13	0.18	0.08	0.07	0.04	0.06	0.10	0.09	0.15
$ \alpha_{PS} $	0.12	0.10	0.22	0.13	0.12	0.05	0.04	0.03	0.03	0.11	0.16	0.07	0.07	0.04	0.08	0.09	0.09	0.15
$ \alpha_C $	0.17	0.15	0.30	0.20	0.18	0.13	0.16	0.14	0.14	0.20	0.26	0.16	0.14	0.13	0.06	0.14	0.17	0.19
$ \alpha_q $	0.11	0.09	0.18	0.10	0.11	0.14	0.13	0.15	0.17	0.14	0.18	0.09	0.07	0.13	0.06	0.10	0.10	0.14
$ \alpha_a $	0.07	0.05	0.15	0.07	0.04	0.10	0.10	0.11	0.12	0.05	0.10	0.04	0.04	0.07	0.13	0.04	0.07	0.08
p	0.00	0.00	0.00	0.00	0.00	0.02	0.08	0.08	0.07	0.00	0.02	0.00	0.00	0.02	0.23	0.00	0.00	0.00
p_{FF}	0.00	0.00	0.00	0.07	0.14	0.44	0.94	0.92	0.79	0.00	0.09	0.19	0.36	0.80	0.74	0.05	0.00	0.00
p_{PS}	0.00	0.01	0.00	0.13	0.28	0.56	0.89	0.89	0.72	0.00	0.13	0.31	0.39	0.76	0.55	0.10	0.00	0.00
p_C	0.00	0.01	0.00	0.00	0.01	0.07	0.02	0.08	0.08	0.00	0.00	0.04	0.17	0.13	0.91	0.04	0.00	0.00
p_q	0.16	0.01	0.00	0.00	0.01	0.08	0.01	0.02	0.03	0.02	0.01	0.70	0.22	0.03	0.39	0.04	0.14	0.01
p_a	0.29	0.11	0.00	0.16	0.47	0.05	0.04	0.01	0.01	0.02	0.05	0.69	0.58	0.24	0.23	0.34	0.09	0.01
	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108
	Ia ^{q1}	Ia ^{q6}	Ia ^{q12}	dPia	Noa	dNoa	dLno	Ig	2Ig	3Ig	Nsi	dLi	Cei	Cdi	Ivg	Ivc	Oa	Ta
m	-0.81	-0.84	-0.78	-0.64	-0.58	-0.74	-0.57	-0.41	-0.46	-0.34	-0.82	-0.46	-0.67	-0.25	-0.48	-0.50	-0.28	-0.43
α	-0.94	-0.98	-0.91	-0.75	-0.63	-0.84	-0.63	-0.48	-0.54	-0.44	-0.99	-0.53	-0.92	-0.31	-0.58	-0.58	-0.29	-0.50
α_{FF}	-0.56	-0.65	-0.61	-0.62	-0.78	-0.70	-0.52	-0.32	-0.39	-0.28	-0.82	-0.40	-0.72	-0.28	-0.46	-0.49	-0.28	-0.34
α_{PS}	-0.54	-0.65	-0.61	-0.66	-0.81	-0.71	-0.52	-0.33	-0.40	-0.26	-0.82	-0.40	-0.70	-0.29	-0.46	-0.54	-0.27	-0.32
α_C	-0.56	-0.56	-0.50	-0.51	-0.66	-0.58	-0.43	-0.23	-0.25	-0.19	-0.64	-0.26	-0.57	-0.23	-0.34	-0.43	-0.29	-0.36
α_q	-0.54	-0.48	-0.39	-0.35	-0.74	-0.44	-0.31	-0.04	-0.12	-0.03	-0.28	-0.15	-0.31	-0.16	-0.28	-0.43	-0.50	-0.42
a	-0.37	-0.45	-0.41	-0.46	-0.82	-0.55	-0.40	-0.14	-0.25	-0.14	-0.37	-0.27	-0.47	-0.23	-0.37	-0.51	-0.47	-0.36
t_m	-4.52	-5.25	-5.58	-5.10	-3.56	-5.79	-4.99	-4.37	-4.11	-2.96	-5.59	-4.76	-4.09	-3.03	-4.26	-4.36	-2.27	-3.74
t_α	-5.13	-6.16	-6.68	-6.42	-4.00	-7.04	-5.60	-5.19	-5.07	-4.20	-7.32	-5.79	-6.94	-4.12	-5.44	-5.14	-2.34	-4.50
t_{FF}	-3.81	-5.06	-5.34	-5.30	-4.63	-5.85	-4.48	-3.78	-3.77	-2.84	-6.29	-4.77	-7.00	-3.78	-4.17	-4.47	-2.25	-3.22
t_{PS}	-3.73	-5.24	-5.51	-5.53	-4.79	-5.96	-4.39	-3.87	-3.88	-2.61	-6.22	-4.81	-7.03	-3.69	-4.20	-4.95	-2.15	-3.06
t_C	-3.77	-4.00	-3.96	-4.37	-4.13	-4.94	-3.95	-2.49	-2.38	-1.87	-4.70	-3.05	-4.98	-3.12	-3.17	-3.84	-2.03	-2.79
t_q	-3.57	-3.31	-3.06	-2.85	-3.45	-3.55	-2.54	-0.45	-1.17	-0.32	-2.14	-1.69	-2.40	-2.04	-2.34	-3.55	-3.82	-3.85
t_a	-2.59	-3.56	-3.61	-4.44	-5.17	-5.04	-3.72	-1.61	-2.60	-1.50	-3.23	-3.28	-4.35	-3.02	-3.51	-5.24	-4.36	-3.61
$ \alpha $	0.30	0.31	0.31	0.25	0.24	0.29	0.21	0.20	0.23	0.23	0.27	0.22	0.28	0.22	0.25	0.21	0.22	0.22
$ \alpha_{FF} $	0.16	0.17	0.17	0.16	0.17	0.19	0.12	0.09	0.12	0.11	0.16	0.11	0.16	0.08	0.12	0.11	0.13	0.09
$ \alpha_{PS} $	0.14	0.16	0.16	0.16	0.17	0.18	0.12	0.09	0.12	0.10	0.16	0.11	0.15	0.08	0.12	0.11	0.13	0.08
$ \alpha_C $	0.19	0.21	0.23	0.21	0.21	0.22	0.17	0.15	0.17	0.16	0.20	0.15	0.20	0.14	0.18	0.17	0.18	0.17
$ \alpha_q $	0.16	0.18	0.20	0.17	0.17	0.17	0.11	0.12	0.11	0.11	0.15	0.11	0.11	0.06	0.10	0.12	0.16	0.12
$ \alpha_a $	0.11	0.12	0.13	0.13	0.15	0.13	0.08	0.04	0.06	0.04	0.11	0.06	0.08	0.08	0.07	0.10	0.12	0.07
p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_{FF}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.00
p_{PS}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.03	0.00	0.00	0.00	0.00
p_C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_q	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.40	0.59	0.79	0.00	0.11	0.00	0.23	0.01	0.00	0.00	0.00
p_a	0.03	0.00	0.00	0.00	0.00	0.00	0.01	0.80	0.28	0.90	0.00	0.09	0.00	0.02	0.00	0.00	0.00	0.04

	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126
	dWc	dCoa	dCol	dNco	dNca	dFin	dLti	dFnl	dBe	Dac	Poa	Pta	Pda	Nxf	Nef	Ndf	Roe1	Roe6
m	-0.36	-0.55	-0.41	-0.68	-0.66	0.32	-0.27	-0.46	-0.64	-0.32	-0.41	-0.50	-0.31	-0.66	-0.54	-0.41	0.97	0.66
α	-0.43	-0.68	-0.53	-0.75	-0.73	0.31	-0.36	-0.51	-0.77	-0.32	-0.49	-0.60	-0.36	-0.87	-0.79	-0.46	1.09	0.79
α_{FF}	-0.40	-0.47	-0.28	-0.60	-0.56	0.44	-0.28	-0.45	-0.47	-0.37	-0.34	-0.47	-0.37	-0.70	-0.56	-0.40	1.16	0.86
α_{PS}	-0.40	-0.48	-0.31	-0.61	-0.57	0.46	-0.31	-0.47	-0.48	-0.38	-0.33	-0.44	-0.36	-0.70	-0.56	-0.42	1.14	0.86
α_C	-0.33	-0.38	-0.21	-0.50	-0.46	0.43	-0.12	-0.39	-0.42	-0.39	-0.23	-0.43	-0.26	-0.53	-0.42	-0.33	0.86	0.55
α_q	-0.39	-0.25	-0.01	-0.30	-0.25	0.47	0.02	-0.17	-0.34	-0.46	-0.15	-0.34	-0.18	-0.22	-0.08	-0.12	0.07	-0.22
a	-0.46	-0.33	-0.07	-0.41	-0.35	0.55	-0.07	-0.28	-0.34	-0.45	-0.24	-0.34	-0.26	-0.33	-0.14	-0.22	0.54	0.23
t_m	-3.30	-4.25	-3.31	-5.59	-5.42	2.80	-2.31	-5.23	-4.39	-3.32	-3.75	-4.85	-4.30	-3.82	-2.84	-4.15	4.53	3.39
t_α	-4.07	-5.68	-4.51	-6.25	-6.05	2.71	-3.25	-5.83	-5.57	-3.26	-4.60	-6.45	-5.03	-5.77	-4.58	-4.78	5.23	4.08
t_{FF}	-3.81	-4.51	-2.99	-5.11	-4.77	3.68	-2.59	-4.90	-4.17	-3.85	-3.83	-5.29	-4.94	-5.39	-4.10	-4.01	5.90	4.68
t_{PS}	-3.74	-4.66	-3.33	-5.14	-4.82	3.92	-2.78	-5.30	-4.27	-3.98	-3.58	-5.03	-4.88	-5.47	-4.15	-4.40	5.69	4.60
t_C	-3.19	-3.57	-2.06	-4.40	-4.02	3.60	-1.06	-4.16	-3.49	-3.46	-2.50	-4.62	-3.37	-4.24	-2.86	-3.43	4.30	2.86
t_q	-3.61	-2.39	-0.13	-2.53	-2.13	3.18	0.16	-1.91	-2.90	-4.23	-1.54	-3.71	-2.00	-1.67	-0.54	-1.29	0.52	-1.59
t_a	-5.17	-3.60	-0.86	-3.70	-3.17	4.72	-0.69	-3.21	-3.04	-5.03	-2.80	-3.70	-3.20	-2.60	-1.10	-2.28	4.06	1.74
$ \alpha $	0.22	0.24	0.22	0.24	0.24	0.21	0.18	0.21	0.25	0.19	0.20	0.20	0.16	0.27	0.27	0.20	0.26	0.23
$ \alpha_{FF} $	0.12	0.12	0.10	0.15	0.14	0.13	0.09	0.13	0.09	0.11	0.12	0.09	0.11	0.17	0.14	0.10	0.24	0.19
$ \alpha_{PS} $	0.12	0.12	0.10	0.15	0.14	0.13	0.10	0.13	0.10	0.11	0.12	0.08	0.11	0.16	0.13	0.10	0.23	0.19
$ \alpha_C $	0.19	0.18	0.16	0.20	0.19	0.19	0.15	0.19	0.17	0.17	0.18	0.18	0.16	0.21	0.19	0.19	0.19	0.17
$ \alpha_q $	0.16	0.14	0.12	0.15	0.15	0.15	0.16	0.14	0.12	0.14	0.16	0.14	0.14	0.13	0.10	0.14	0.12	0.14
$ \alpha_a $	0.10	0.08	0.06	0.10	0.10	0.12	0.09	0.13	0.07	0.10	0.13	0.09	0.11	0.09	0.05	0.12	0.11	0.07
p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_{FF}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_{PS}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_q	0.00	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.00
p_a	0.00	0.00	0.02	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.14	0.00	0.00	0.00
	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144
	dRoe1	dRoe6	dRoe12	Roa1	Roa6	dRoa1	dRoa6	dRoa12	Cto	Rna ^{q1}	Rna ^{q6}	Rna ^{q12}	Pm ^{q1}	Ato ^{q1}	Ato ^{q6}	Ato ^{q12}	Cto ^{q1}	Cto ^{q6}
m	0.87	0.44	0.24	0.89	0.62	0.85	0.42	0.24	0.36	0.87	0.59	0.47	0.63	0.74	0.57	0.46	0.79	0.68
α	0.91	0.46	0.27	1.02	0.76	0.85	0.42	0.25	0.27	1.02	0.74	0.61	0.84	0.67	0.49	0.38	0.79	0.68
α_{FF}	1.00	0.57	0.38	1.07	0.84	0.93	0.53	0.35	0.25	1.12	0.83	0.70	0.81	0.80	0.64	0.55	0.65	0.55
α_{PS}	0.99	0.57	0.39	1.09	0.87	0.94	0.56	0.38	0.21	1.16	0.86	0.73	0.76	0.85	0.68	0.58	0.67	0.58
α_C	0.73	0.35	0.19	0.80	0.54	0.63	0.30	0.16	0.24	0.93	0.64	0.48	0.63	0.62	0.48	0.41	0.55	0.41
α_q	0.49	0.10	-0.02	0.11	-0.14	0.38	0.05	-0.05	-0.14	0.24	-0.03	-0.11	0.00	0.31	0.18	0.13	-0.13	-0.24
a	0.92	0.50	0.32	0.50	0.26	0.83	0.46	0.29	-0.16	0.49	0.19	0.11	0.29	0.46	0.31	0.23	-0.01	-0.11
t_m	6.60	4.03	2.62	4.06	3.02	5.65	3.21	2.24	2.01	3.98	2.89	2.43	2.61	4.21	3.24	2.60	3.58	3.30
t_α	7.38	4.52	3.19	4.73	3.74	6.05	3.43	2.43	1.49	4.74	3.60	3.23	3.52	3.77	2.81	2.17	3.34	3.06
t_{FF}	8.49	5.93	4.64	5.28	4.43	6.91	4.56	3.54	1.47	5.64	4.40	4.06	3.89	5.03	4.06	3.44	3.06	2.74
t_{PS}	7.70	5.54	4.52	5.36	4.65	6.71	4.62	3.79	1.21	5.90	4.60	4.31	3.61	5.37	4.27	3.65	3.20	2.88
t_C	5.99	3.85	2.54	3.85	2.74	4.57	2.60	1.71	1.39	4.48	3.38	2.64	2.93	3.86	2.94	2.44	2.59	2.02
t_q	3.49	0.92	-0.23	0.78	-1.17	2.43	0.34	-0.41	-0.75	1.45	-0.24	-0.79	0.02	1.93	1.16	0.82	-0.69	-1.41
t_a	7.28	4.83	3.56	3.48	1.85	5.77	3.64	2.59	-1.18	3.67	1.63	0.93	1.82	2.94	2.09	1.64	-0.07	-0.69
$ \alpha $	0.29	0.21	0.18	0.25	0.24	0.30	0.22	0.20	0.17	0.31	0.26	0.23	0.21	0.14	0.14	0.14	0.16	0.16
$ \alpha_{FF} $	0.27	0.16	0.11	0.23	0.19	0.27	0.17	0.12	0.09	0.25	0.18	0.14	0.14	0.16	0.13	0.10	0.11	0.09
$ \alpha_{PS} $	0.28	0.16	0.12	0.24	0.20	0.28	0.17	0.12	0.10	0.26	0.19	0.15	0.15	0.17	0.13	0.10	0.11	0.09
$ \alpha_C $	0.20	0.15	0.16	0.17	0.18	0.20	0.15	0.17	0.15	0.22	0.17	0.16	0.14	0.14	0.15	0.16	0.12	0.15
$ \alpha_q $	0.13	0.11	0.13	0.14	0.17	0.15	0.12	0.14	0.12	0.14	0.15	0.17	0.13	0.12	0.13	0.15	0.10	0.15
$ \alpha_a $	0.24	0.12	0.07	0.13	0.08	0.25	0.14	0.10	0.09	0.10	0.09	0.09	0.11	0.14	0.11	0.10	0.06	0.06
p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.08	0.03	0.01	0.00
p_{FF}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.09	0.06
p_{PS}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.09	0.06
p_C	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.10	0.02
p_q	0.00	0.01	0.00	0.00	0.00	0.00	0.04	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.30	0.03
p_a	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.03	0.04	0.00	0.00	0.00	0.00	0.51	0.43

	145 Cto ^a 12	146 Gpa	147 Gla ^a 1	148 Gla ^a 6	149 Gla ^a 12	150 Ope	151 Ole ^a 1	152 Ole ^a 6	153 Ole ^a 12	154 Opa	155 Ola ^a 1	156 Ola ^a 6	157 Ola ^a 12	158 Cop	159 Cla	160 Cla ^a 1	161 Cla ^a 6	162 Cla ^a 12
m	0.55	0.62	0.81	0.56	0.50	0.47	0.96	0.53	0.42	0.64	1.02	0.67	0.57	0.76	0.63	0.88	0.73	0.68
α	0.55	0.61	0.83	0.57	0.50	0.60	1.08	0.66	0.54	0.75	1.11	0.78	0.68	0.91	0.72	1.00	0.83	0.79
α_{FF}	0.46	0.60	0.85	0.59	0.53	0.46	0.90	0.48	0.37	0.92	1.27	0.93	0.84	1.07	0.96	1.04	0.85	0.82
α_{PS}	0.48	0.56	0.89	0.62	0.55	0.45	0.91	0.50	0.38	0.96	1.30	0.96	0.86	1.11	0.99	1.03	0.86	0.83
α_C	0.30	0.50	0.76	0.48	0.41	0.29	0.78	0.32	0.18	0.64	1.05	0.66	0.54	0.84	0.78	0.91	0.69	0.63
α_q	-0.31	-0.08	0.12	-0.12	-0.10	-0.42	0.08	-0.35	-0.44	0.12	0.50	0.12	0.07	0.54	0.61	0.62	0.36	0.35
a	-0.17	-0.01	0.22	-0.03	-0.03	-0.29	0.30	-0.14	-0.23	0.34	0.75	0.39	0.33	0.72	0.74	0.71	0.51	0.52
t_m	2.78	3.52	4.39	3.29	3.02	2.12	4.12	2.48	2.00	3.27	4.78	3.45	3.04	4.92	4.27	6.01	5.65	5.63
t_α	2.59	3.21	4.27	3.14	2.85	2.62	4.56	2.98	2.55	3.80	5.09	3.88	3.61	6.54	5.15	6.87	6.31	6.65
t_{FF}	2.30	3.47	4.58	3.40	3.20	2.20	4.28	2.43	1.92	5.33	6.55	5.19	4.97	8.27	7.62	8.20	7.66	7.78
t_{PS}	2.45	3.21	4.81	3.59	3.38	2.11	4.30	2.50	2.02	5.58	6.61	5.30	5.13	8.43	7.74	7.94	7.53	7.74
t_C	1.48	2.98	4.17	2.81	2.47	1.32	3.91	1.57	0.83	3.31	5.36	3.61	3.02	6.57	5.95	7.08	5.70	5.44
t_q	-1.88	-0.54	0.72	-0.78	-0.72	-1.81	0.44	-1.80	-2.17	0.47	3.21	0.74	0.37	3.12	3.20	5.49	2.60	2.31
t_a	-1.12	-0.09	1.61	-0.20	-0.27	-2.25	2.08	-1.02	-1.68	1.74	4.35	2.33	1.88	4.93	4.91	6.35	4.35	4.12
$ \alpha $	0.15	0.15	0.19	0.17	0.16	0.20	0.28	0.23	0.20	0.21	0.28	0.23	0.20	0.27	0.24	0.31	0.25	0.24
$ \alpha_{FF} $	0.08	0.14	0.19	0.16	0.13	0.11	0.20	0.14	0.11	0.19	0.29	0.22	0.17	0.23	0.21	0.30	0.23	0.21
$ \alpha_{PS} $	0.08	0.14	0.20	0.17	0.15	0.11	0.20	0.14	0.10	0.19	0.29	0.22	0.17	0.24	0.21	0.29	0.23	0.21
$ \alpha_C $	0.17	0.16	0.18	0.16	0.15	0.15	0.21	0.18	0.17	0.19	0.25	0.19	0.17	0.24	0.22	0.28	0.21	0.20
$ \alpha_q $	0.17	0.15	0.14	0.17	0.17	0.14	0.13	0.16	0.18	0.14	0.14	0.13	0.14	0.15	0.15	0.20	0.15	0.15
$ \alpha_a $	0.08	0.09	0.13	0.12	0.11	0.08	0.10	0.08	0.08	0.11	0.20	0.13	0.12	0.17	0.16	0.21	0.15	0.15
p	0.00	0.00	0.00	0.01	0.01	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_{FF}	0.00	0.00	0.00	0.01	0.01	0.51	0.00	0.03	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_{PS}	0.00	0.00	0.00	0.00	0.01	0.59	0.00	0.05	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_C	0.00	0.00	0.00	0.00	0.00	0.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_q	0.00	0.00	0.01	0.00	0.00	0.10	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_a	0.04	0.02	0.01	0.02	0.03	0.16	0.07	0.14	0.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	163 F	164 F ^a 1	165 F ^a 6	166 F ^a 12	167 Fp6	168 O	169 O ^a 1	170 G	171 Sg ^a 12	172 Oca	173 Ioca	174 Adm	175 gAd	176 Rdm	177 Rdm ^a 1	178 Rdm ^a 6	179 Rdm ^a 12	180 Ol
m	0.56	0.93	0.70	0.54	-0.57	-0.30	-0.35	0.55	-0.29	0.46	0.37	0.67	-0.41	1.00	1.32	1.14	1.27	0.42
α	0.80	1.20	0.95	0.78	-0.94	-0.40	-0.47	0.73	-0.41	0.51	0.45	0.85	-0.50	0.87	1.15	1.04	1.20	0.48
α_{FF}	0.71	1.04	0.79	0.62	-1.08	-0.56	-0.64	0.82	-0.08	0.62	0.47	0.28	-0.32	0.75	0.99	0.86	1.00	0.41
α_{PS}	0.69	1.00	0.77	0.59	-1.11	-0.53	-0.66	0.82	-0.06	0.63	0.44	0.25	-0.31	0.74	0.93	0.80	0.97	0.39
α_C	0.48	0.89	0.62	0.42	-0.29	-0.45	-0.58	0.69	-0.18	0.69	0.40	0.28	-0.22	0.75	1.60	1.35	1.31	0.41
α_q	0.23	0.41	0.16	0.01	0.13	-0.26	-0.17	0.40	-0.15	0.48	0.40	-0.15	0.07	0.90	1.60	1.35	1.28	-0.02
a	0.47	0.70	0.47	0.32	-0.50	-0.35	-0.31	0.50	0.10	0.46	0.46	-0.20	0.00	0.80	0.93	0.79	0.86	0.02
t_m	2.57	3.82	3.29	2.75	-2.08	-2.25	-2.06	2.87	-2.04	2.30	3.91	2.33	-2.41	3.99	3.73	3.31	3.99	2.25
t_α	4.45	5.15	4.59	4.14	-4.10	-3.11	-2.84	4.17	-2.98	2.54	5.06	2.93	-2.93	3.48	3.38	3.03	3.63	2.52
t_{FF}	4.38	5.48	4.85	4.46	-5.09	-4.51	-4.31	5.09	-0.75	3.15	5.05	1.28	-1.87	3.26	3.02	2.73	3.52	2.17
t_{PS}	4.27	5.24	4.66	4.38	-5.44	-4.24	-4.41	5.12	-0.57	3.12	4.73	1.19	-1.85	3.29	2.94	2.66	3.54	2.02
t_C	2.76	4.83	3.74	2.80	-1.52	-3.46	-3.54	3.98	-1.49	3.40	3.92	1.19	-1.02	3.18	4.83	4.65	4.67	2.25
t_q	1.27	1.98	0.92	0.04	0.38	-1.72	-1.22	2.05	-1.27	1.89	3.16	-0.51	0.25	3.23	4.00	3.92	4.74	-0.13
t_a	2.75	3.69	2.99	2.33	-1.64	-2.54	-2.40	2.84	0.98	2.17	4.59	-0.93	-0.01	3.28	2.61	2.43	3.17	0.11
$ \alpha $	0.25	0.33	0.26	0.22	0.26	0.19	0.22	0.22	0.21	0.15	0.13	0.24	0.28	0.19	0.22	0.21	0.25	0.13
$ \alpha_{FF} $	0.19	0.29	0.22	0.18	0.21	0.12	0.12	0.23	0.10	0.17	0.14	0.08	0.14	0.17	0.23	0.21	0.26	0.06
$ \alpha_{PS} $	0.19	0.28	0.21	0.18	0.21	0.11	0.12	0.23	0.10	0.16	0.13	0.07	0.13	0.17	0.24	0.22	0.26	0.07
$ \alpha_C $	0.17	0.26	0.20	0.17	0.15	0.17	0.16	0.20	0.16	0.21	0.15	0.24	0.29	0.29	0.37	0.36	0.43	0.16
$ \alpha_q $	0.10	0.14	0.10	0.12	0.15	0.13	0.13	0.18	0.13	0.18	0.17	0.23	0.21	0.38	0.57	0.56	0.61	0.13
$ \alpha_a $	0.13	0.19	0.12	0.10	0.09	0.08	0.08	0.11	0.06	0.14	0.10	0.12	0.08	0.24	0.39	0.36	0.39	0.07
p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.06	0.00	0.01	0.10	0.06	0.00	0.13
p_{FF}	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.50	0.03	0.02	0.20	0.12	0.00	0.12
p_{PS}	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.02	0.02	0.00	0.56	0.02	0.02	0.32	0.12	0.00	0.06
p_C	0.00	0.00	0.00	0.02	0.01	0.00	0.01	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00	0.00	0.00	0.03
p_q	0.13	0.03	0.54	0.67	0.00	0.01	0.00	0.01	0.01	0.02	0.00	0.04	0.19	0.00	0.00	0.00	0.00	0.05
p_a	0.01	0.00	0.09	0.27	0.01	0.04	0.01	0.03	0.12	0.05	0.00	0.21	0.56	0.01	0.01	0.04	0.01	0.10

	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198
	Ol ^{q1}	Ol ^{q6}	Ol ^{q12}	Hn	Parc	dSi	Rer	Eprd	Ala	Alm ^{q1}	Alm ^{q6}	Alm ^{q12}	R _a ¹	R _n ¹	R _a ^[2,5]	R _n ^[2,5]	R _a ^[6,10]	R _n ^[6,10]
m	0.55	0.53	0.49	-0.49	0.29	0.17	0.37	-0.62	-0.46	0.62	0.64	0.58	0.60	0.70	0.54	-0.83	0.65	-0.42
α	0.57	0.56	0.53	-0.63	0.32	0.19	0.37	-0.78	-0.71	0.76	0.77	0.70	0.52	0.76	0.50	-1.02	0.63	-0.52
α_{FF}	0.53	0.50	0.48	-0.37	0.30	0.18	0.37	-1.06	-0.34	0.17	0.21	0.17	0.63	1.05	0.57	-0.55	0.65	-0.32
α_{PS}	0.56	0.51	0.49	-0.39	0.28	0.18	0.38	-1.07	-0.35	0.12	0.17	0.13	0.67	1.10	0.55	-0.52	0.65	-0.33
α_{C}	0.57	0.52	0.48	-0.28	0.29	0.16	0.35	-0.86	-0.26	0.19	0.12	0.05	0.45	-0.25	0.68	-0.53	0.81	-0.22
α_q	0.16	0.13	0.11	-0.05	0.25	0.19	0.32	-0.58	0.00	0.00	-0.08	-0.16	0.51	0.08	0.73	-0.46	0.82	0.02
a	0.19	0.16	0.15	-0.11	0.24	0.23	0.37	-0.85	-0.06	-0.14	-0.12	-0.15	0.64	1.01	0.64	-0.35	0.74	-0.14
t_m	2.62	2.60	2.50	-3.60	2.31	2.05	3.73	-3.53	-2.29	2.51	2.88	2.76	3.40	2.32	4.04	-3.98	5.77	-2.68
t_α	2.66	2.73	2.67	-4.73	2.50	2.29	3.70	-4.84	-4.01	2.87	3.17	3.11	2.90	2.71	3.75	-5.28	5.75	-3.40
t_{FF}	2.50	2.42	2.40	-3.64	2.22	2.10	3.70	-7.80	-3.04	1.10	1.56	1.29	3.83	3.64	4.32	-3.81	5.67	-2.41
t_{PS}	2.59	2.46	2.44	-3.86	2.03	2.06	3.70	-7.50	-3.09	0.77	1.29	1.04	4.03	3.75	4.16	-3.60	5.72	-2.42
t_{C}	2.79	2.57	2.45	-2.72	2.07	1.81	3.38	-5.92	-2.34	1.14	0.83	0.33	2.61	-1.56	4.74	-3.57	6.08	-1.57
t_q	0.75	0.63	0.54	-0.46	1.63	2.03	2.86	-3.81	0.02	0.01	-0.46	-0.86	2.59	0.18	4.76	-2.77	5.05	0.12
t_a	0.96	0.84	0.83	-1.23	1.56	2.65	3.50	-5.98	-0.58	-0.97	-0.92	-1.12	3.75	2.61	4.55	-2.27	5.53	-1.03
$ \alpha $	0.16	0.16	0.14	0.22	0.13	0.21	0.10	0.27	0.25	0.24	0.25	0.23	0.19	0.18	0.23	0.31	0.26	0.26
$ \alpha_{\text{FF}} $	0.10	0.09	0.08	0.10	0.14	0.09	0.07	0.23	0.09	0.05	0.06	0.06	0.15	0.21	0.14	0.14	0.17	0.11
$ \alpha_{\text{PS}} $	0.10	0.10	0.08	0.10	0.14	0.09	0.07	0.23	0.08	0.04	0.05	0.05	0.16	0.22	0.14	0.13	0.16	0.12
$ \alpha_{\text{C}} $	0.13	0.15	0.17	0.16	0.35	0.16	0.18	0.21	0.17	0.10	0.13	0.15	0.14	0.12	0.19	0.16	0.23	0.13
$ \alpha_q $	0.11	0.12	0.13	0.12	0.34	0.08	0.21	0.15	0.12	0.15	0.18	0.18	0.13	0.18	0.17	0.10	0.20	0.06
$ \alpha_a $	0.06	0.06	0.05	0.08	0.16	0.08	0.10	0.20	0.07	0.08	0.07	0.07	0.13	0.23	0.16	0.06	0.18	0.09
p	0.01	0.01	0.04	0.00	0.06	0.00	0.02	0.00	0.00	0.04	0.01	0.00	0.00	0.01	0.00	0.00	0.00	0.00
p_{FF}	0.04	0.01	0.05	0.00	0.08	0.00	0.04	0.00	0.00	0.87	0.50	0.04	0.00	0.00	0.00	0.00	0.00	0.17
p_{PS}	0.03	0.01	0.03	0.00	0.14	0.00	0.03	0.00	0.00	0.95	0.50	0.05	0.00	0.00	0.00	0.00	0.00	0.14
p_{C}	0.02	0.01	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.39	0.14	0.01	0.01	0.00	0.00	0.00	0.00	0.15
p_q	0.01	0.02	0.05	0.00	0.00	0.02	0.00	0.00	0.00	0.08	0.10	0.02	0.02	0.00	0.00	0.01	0.00	0.40
p_a	0.07	0.08	0.19	0.01	0.01	0.02	0.01	0.00	0.00	0.41	0.56	0.08	0.01	0.00	0.00	0.05	0.00	0.29
	R _a ^[11,15]	R _a ^[16,20]	Ivc1	Ivq1	Sv1	Srev	Dtv1	Dtv6	Dtv12	Ami6	Ami12	Lm ⁶ 1	Lm ⁶ 6	Lm ⁶ 12	Lm ¹² 1	Lm ¹² 6	Lm ¹² 12	Mdr1
m	0.44	0.49	-0.69	-0.63	-0.45	-0.52	-0.36	-0.45	-0.45	0.37	0.38	0.53	0.52	0.51	0.50	0.52	0.46	-0.67
α	0.46	0.51	-1.21	-1.12	-0.56	-0.36	-0.49	-0.58	-0.58	0.32	0.32	0.91	0.90	0.88	0.87	0.88	0.81	-1.12
α_{FF}	0.46	0.52	-0.97	-0.88	-0.52	-0.31	-0.11	-0.20	-0.21	0.05	0.06	0.60	0.60	0.59	0.60	0.62	0.57	-0.88
α_{PS}	0.47	0.55	-0.93	-0.86	-0.57	-0.26	-0.11	-0.20	-0.21	0.05	0.06	0.56	0.56	0.57	0.56	0.60	0.55	-0.86
α_{C}	0.46	0.57	-0.75	-0.67	-0.48	-0.60	-0.15	-0.16	-0.12	0.04	0.05	0.54	0.46	0.39	0.46	0.42	0.34	-0.75
α_q	0.37	0.59	-0.14	-0.08	-0.14	-0.57	0.04	-0.01	0.02	0.15	0.14	0.12	0.06	-0.02	0.11	0.07	-0.01	-0.18
a	0.43	0.53	-0.34	-0.27	-0.16	-0.39	0.07	-0.03	-0.04	0.11	0.11	0.14	0.15	0.16	0.20	0.22	0.20	-0.31
t_m	4.09	4.50	-2.10	-1.97	-2.20	-2.40	-2.52	-3.19	-3.31	2.57	2.72	2.36	2.33	2.35	2.33	2.46	2.26	-2.22
t_α	4.35	4.67	-4.70	-4.47	-2.69	-1.72	-3.40	-4.04	-4.11	2.28	2.42	4.79	4.92	4.93	4.94	5.06	4.85	-4.66
t_{FF}	3.92	4.74	-5.21	-4.98	-2.58	-1.29	-1.07	-2.14	-2.25	0.68	0.84	3.57	3.81	3.88	3.88	4.10	3.81	-4.91
t_{PS}	4.00	4.88	-4.96	-4.78	-2.87	-1.07	-1.01	-2.05	-2.24	0.63	0.79	3.27	3.51	3.66	3.58	3.90	3.66	-4.68
t_{C}	3.76	5.06	-3.70	-3.47	-2.25	-2.67	-1.32	-1.55	-1.17	0.51	0.61	3.03	2.70	2.22	2.79	2.54	2.02	-3.78
t_q	2.74	4.60	-0.69	-0.42	-0.65	-1.62	0.31	-0.08	0.16	2.11	1.96	0.61	0.32	-0.09	0.57	0.36	-0.05	-0.86
t_a	3.39	4.66	-2.28	-1.97	-0.81	-1.33	0.66	-0.29	-0.46	1.50	1.70	0.93	1.03	1.02	1.36	1.46	1.29	-2.32
$ \alpha $	0.25	0.25	0.32	0.30	0.20	0.18	0.14	0.15	0.15	0.10	0.10	0.22	0.22	0.22	0.22	0.22	0.22	0.30
$ \alpha_{\text{FF}} $	0.13	0.14	0.22	0.21	0.16	0.13	0.06	0.07	0.07	0.04	0.04	0.12	0.13	0.13	0.12	0.12	0.12	0.20
$ \alpha_{\text{PS}} $	0.13	0.15	0.22	0.20	0.16	0.12	0.06	0.07	0.08	0.04	0.04	0.11	0.12	0.12	0.12	0.12	0.12	0.19
$ \alpha_{\text{C}} $	0.16	0.17	0.22	0.20	0.18	0.19	0.08	0.10	0.13	0.10	0.13	0.15	0.15	0.16	0.14	0.14	0.15	0.19
$ \alpha_q $	0.10	0.15	0.16	0.15	0.16	0.18	0.13	0.15	0.17	0.15	0.17	0.13	0.15	0.17	0.12	0.15	0.16	0.13
$ \alpha_a $	0.12	0.15	0.14	0.13	0.09	0.14	0.06	0.06	0.05	0.06	0.05	0.08	0.07	0.06	0.07	0.07	0.06	0.10
p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00
p_{FF}	0.00	0.00	0.00	0.00	0.01	0.02	0.14	0.02	0.05	0.39	0.39	0.02	0.00	0.00	0.04	0.00	0.01	0.00
p_{PS}	0.00	0.00	0.00	0.00	0.01	0.06	0.19	0.04	0.12	0.57	0.48	0.05	0.01	0.00	0.05	0.00	0.01	0.00
p_{C}	0.00	0.00	0.00	0.00	0.01	0.01	0.01	0.00	0.00	0.08	0.03	0.01	0.00	0.00	0.07	0.00	0.00	0.00
p_q	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.01	0.01	0.06	0.05	0.01	0.06	0.00	0.00	0.02
p_a	0.00	0.00	0.00	0.00	0.06	0.03	0.17	0.05	0.18	0.27	0.34	0.50	0.24	0.05	0.34	0.03	0.07	0.03

Table 5 : Betas for the q -factor Model and the Five-factor Model, Significant Anomalies, NYSE-VW, January 1967 to December 2014, 576 Months

For each high-minus-low decile, β_{MKT} , β_{ME} , $\beta_{\text{I/A}}$, and β_{ROE} are the loadings on the market, size, investment, and ROE factors in the q -factor model, respectively, and $t_{\beta_{\text{MKT}}}$, $t_{\beta_{\text{ME}}}$, $t_{\beta_{\text{I/A}}}$, and $t_{\beta_{\text{ROE}}}$ are their t -statistics. b , s , h , r , and c are the loadings on MKT, SMB, HML, RMW, and CMA in the five-factor model, and t_b , t_s , t_h , t_r , and t_c are their t -statistics. All t -statistics are adjusted for heteroscedasticity and autocorrelations. Table A1 describes the symbols.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Sue1	Abr1	Abr6	Abr12	Re1	Re6	R^6_1	R^6_6	R^6_{12}	R^{11}_1	R^{11}_6	Im1	Im6	Im12	Rs1	dEf1	dEf6	dEf12
β_{MKT}	-0.04	-0.06	-0.03	-0.02	-0.06	-0.06	-0.21	-0.08	-0.02	-0.13	-0.05	-0.20	-0.07	-0.04	-0.05	0.02	0.06	0.03
β_{ME}	-0.04	0.07	0.09	0.07	-0.19	-0.17	0.21	0.22	0.07	0.32	0.16	0.15	0.24	0.15	-0.12	-0.10	-0.03	-0.08
$\beta_{\text{I/A}}$	-0.09	-0.13	-0.17	-0.26	0.07	-0.09	0.06	-0.01	-0.20	0.10	-0.11	0.05	0.10	-0.16	-0.41	-0.18	-0.31	-0.34
β_{ROE}	0.86	0.26	0.17	0.16	1.28	1.07	1.17	0.99	0.83	1.43	1.27	0.79	0.83	0.66	0.60	0.80	0.79	0.68
$t_{\beta_{\text{MKT}}}$	-0.93	-1.39	-1.32	-0.75	-0.93	-1.13	-2.39	-1.13	-0.34	-1.38	-0.63	-2.50	-1.17	-0.81	-0.98	0.45	1.26	0.76
$t_{\beta_{\text{ME}}}$	-0.64	0.75	1.89	1.86	-2.20	-1.86	1.01	1.27	0.51	1.50	0.89	0.75	1.51	1.12	-2.39	-1.03	-0.36	-1.28
$t_{\beta_{\text{I/A}}}$	-0.95	-1.28	-2.40	-4.27	0.45	-0.61	0.18	-0.04	-1.11	0.33	-0.47	0.19	0.45	-0.86	-4.77	-1.25	-2.47	-3.57
$t_{\beta_{\text{ROE}}}$	11.24	3.12	2.87	3.71	9.71	8.96	4.09	5.33	5.88	5.67	6.52	3.91	5.01	4.44	7.96	7.13	7.86	8.95
b	-0.08	-0.08	-0.06	-0.06	-0.16	-0.15	-0.28	-0.17	-0.11	-0.23	-0.16	-0.24	-0.13	-0.11	-0.09	-0.07	-0.04	-0.06
s	-0.19	-0.04	0.01	0.02	-0.42	-0.38	-0.11	-0.08	-0.17	-0.05	-0.19	-0.05	-0.01	-0.07	-0.25	-0.30	-0.24	-0.24
h	-0.39	-0.19	-0.12	-0.14	-0.18	-0.26	-0.54	-0.53	-0.49	-0.71	-0.69	-0.50	-0.37	-0.39	-0.48	-0.27	-0.22	-0.28
r	0.47	-0.11	-0.13	-0.11	0.53	0.38	0.21	0.09	0.05	0.32	0.19	0.19	0.11	-0.03	0.26	0.09	0.05	0.11
c	0.18	0.12	-0.07	-0.18	0.00	-0.01	0.53	0.37	0.11	0.66	0.35	0.57	0.39	0.13	-0.01	-0.06	-0.24	-0.19
t_b	-1.82	-1.97	-2.56	-2.50	-1.98	-2.29	-2.66	-1.89	-1.54	-2.02	-1.68	-2.89	-1.72	-1.67	-1.88	-1.15	-0.73	-1.38
t_s	-2.70	-0.57	0.22	0.42	-3.86	-4.10	-0.63	-0.54	-1.45	-0.28	-1.25	-0.28	-0.04	-0.57	-4.08	-3.12	-2.90	-3.76
t_h	-3.86	-1.82	-1.86	-2.85	-1.08	-1.76	-1.79	-2.46	-3.06	-2.55	-3.11	-2.31	-1.83	-2.43	-5.73	-1.83	-1.75	-3.00
t_r	3.90	-1.22	-1.80	-2.23	3.41	2.65	0.64	0.37	0.27	1.08	0.83	0.67	0.48	-0.16	3.13	0.63	0.43	1.33
t_c	1.04	0.83	-0.62	-2.01	0.01	-0.03	1.22	1.20	0.42	1.65	1.08	1.68	1.27	0.51	-0.09	-0.27	-1.13	-1.30
	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
	Nei1	Nei6	52w6	ϵ^6_6	ϵ^6_{12}	ϵ^{11}_1	ϵ^{11}_6	ϵ^{11}_{12}	Sm1	Ilr1	Ilr6	Ilr12	Ile1	Cm1	Cm12	Sim1	Cim1	Cim6
β_{MKT}	0.01	-0.01	-0.44	-0.03	-0.02	0.01	0.01	0.01	-0.03	-0.19	-0.11	-0.05	-0.05	0.07	0.02	0.04	0.01	-0.04
β_{ME}	-0.08	-0.09	-0.36	0.11	0.05	0.12	0.10	0.02	-0.19	-0.10	0.08	0.08	0.00	-0.17	0.09	0.02	-0.18	0.11
$\beta_{\text{I/A}}$	-0.32	-0.42	0.52	0.08	0.00	0.19	0.10	0.01	0.14	0.08	0.01	-0.03	-0.18	0.21	0.00	0.16	0.19	0.19
β_{ROE}	0.65	0.60	1.24	0.25	0.29	0.40	0.39	0.34	-0.01	0.08	0.35	0.33	0.62	-0.04	0.13	0.23	0.19	0.28
$t_{\beta_{\text{MKT}}}$	0.46	-0.33	-6.35	-0.73	-0.40	0.12	0.17	0.23	-0.44	-2.67	-3.27	-2.12	-0.89	0.90	0.54	0.50	0.12	-1.22
$t_{\beta_{\text{ME}}}$	-2.03	-2.61	-2.20	1.48	0.73	1.75	1.19	0.30	-1.85	-0.99	0.97	1.29	0.04	-1.95	1.47	0.18	-1.81	1.47
$t_{\beta_{\text{I/A}}}$	-4.46	-6.32	2.46	0.75	-0.05	1.48	0.84	0.14	0.74	0.47	0.10	-0.36	-1.40	1.18	0.05	0.70	0.89	1.22
$t_{\beta_{\text{ROE}}}$	11.51	11.64	6.53	2.63	4.22	3.30	3.87	4.00	-0.07	0.59	4.17	5.11	6.11	-0.27	2.28	1.46	1.13	2.89
b	-0.04	-0.05	-0.50	-0.05	-0.05	-0.02	-0.04	-0.04	-0.08	-0.21	-0.13	-0.08	-0.08	0.03	0.01	0.00	-0.01	-0.05
s	-0.16	-0.16	-0.65	-0.01	-0.06	-0.02	-0.06	-0.11	-0.20	-0.10	0.00	0.00	-0.15	-0.19	0.04	-0.08	-0.18	0.02
h	-0.34	-0.35	-0.38	-0.14	-0.21	-0.24	-0.22	-0.28	0.17	-0.11	-0.16	-0.16	-0.57	-0.05	-0.14	0.09	0.00	-0.13
r	0.44	0.42	0.57	-0.10	-0.06	0.00	-0.10	-0.09	-0.18	-0.03	0.11	0.07	0.25	-0.14	0.01	-0.12	0.07	0.05
c	-0.09	-0.15	0.70	0.21	0.15	0.40	0.21	0.19	-0.22	0.12	0.17	0.11	0.35	0.12	0.18	-0.06	0.15	0.29
t_b	-1.42	-2.07	-5.52	-1.08	-1.14	-0.29	-0.66	-0.80	-1.07	-3.04	-3.56	-2.52	-1.57	0.45	0.26	-0.02	-0.12	-1.61
t_s	-3.43	-3.55	-4.67	-0.09	-1.03	-0.26	-0.82	-1.80	-1.93	-1.03	0.01	-0.04	-1.99	-1.86	0.82	-0.70	-1.77	0.34
t_h	-5.34	-5.72	-1.65	-1.47	-2.51	-1.87	-2.00	-3.06	1.23	-0.90	-1.76	-2.00	-4.80	-0.37	-2.89	0.55	-0.04	-1.52
t_r	6.39	6.92	2.24	-1.05	-0.80	0.03	-0.89	-1.01	-1.19	-0.21	0.88	0.83	2.11	-0.86	0.17	-0.48	0.32	0.36
t_c	-0.86	-1.46	2.22	1.38	1.08	2.12	1.22	1.36	-0.89	0.57	1.24	0.96	2.10	0.51	2.28	-0.22	0.60	1.78
	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
	Cim12	Bm	Bmj	Bm ^q 12	Rev1	Rev6	Rev12	Ep	Ep ^q 1	Ep ^q 6	Ep ^q 12	Efp1	Cp	Cp ^q 1	Cp ^q 6	Cp ^q 12	Nop	Em
β_{MKT}	-0.01	0.00	-0.05	0.02	0.05	0.08	0.09	-0.09	0.00	-0.03	-0.06	-0.19	0.00	0.08	0.00	-0.03	-0.17	0.12
β_{ME}	0.10	0.41	0.31	0.32	-0.63	-0.60	-0.60	0.28	0.29	0.25	0.27	-0.09	0.23	0.18	0.17	0.22	-0.34	-0.17
$\beta_{\text{I/A}}$	0.07	1.33	1.32	1.22	-1.18	-1.04	-0.95	1.01	0.82	0.84	0.82	0.79	1.26	0.99	0.97	1.01	1.05	-0.95
β_{ROE}	0.27	-0.55	-0.82	-0.94	0.72	0.66	0.50	-0.07	0.13	0.17	0.13	-0.07	-0.39	-0.61	-0.56	-0.45	0.04	0.14
$t_{\beta_{\text{MKT}}}$	-0.67	0.10	-1.21	0.47	1.05	1.52	1.68	-1.60	-0.01	-0.55	-1.13	-2.78	-0.02	1.24	-0.01	-0.65	-3.46	2.37
$t_{\beta_{\text{ME}}}$	1.70	5.04	3.29	3.06	-7.76	-7.72	-8.37	2.41	2.16	2.01	2.34	-0.65	1.89	1.31	1.37	1.99	-4.34	-2.08
$t_{\beta_{\text{I/A}}}$	0.63	13.09	11.07	9.42	-10.63	-9.77	-8.48	6.55	4.77	6.10	6.37	4.76	9.36	6.12	6.74	7.57	10.23	-7.24
$t_{\beta_{\text{ROE}}}$	4.10	-6.64	-9.67	-8.85	7.44	6.89	4.75	-0.55	0.90	1.36	1.23	-0.44	-3.33	-4.30	-4.70	-4.16	0.37	1.20

<i>b</i>	-0.04	0.08	0.03	0.11	-0.02	0.02	0.03	-0.03	0.05	0.02	-0.01	-0.10	0.08	0.16	0.08	0.05	-0.09	0.05
<i>s</i>	0.01	0.45	0.40	0.46	-0.65	-0.59	-0.55	0.32	0.28	0.26	0.29	0.02	0.30	0.32	0.30	0.32	-0.27	-0.24
<i>h</i>	-0.12	1.15	1.16	1.23	-0.48	-0.36	-0.29	1.30	1.05	0.96	0.94	1.16	1.28	1.16	1.16	1.15	0.47	-0.90
<i>r</i>	0.00	-0.31	-0.41	-0.42	0.39	0.46	0.42	0.22	0.31	0.39	0.38	0.22	0.00	-0.03	-0.03	0.02	0.54	-0.25
<i>c</i>	0.14	0.25	0.23	0.04	-0.77	-0.78	-0.76	-0.29	-0.22	-0.12	-0.10	-0.35	0.02	-0.14	-0.15	-0.09	0.55	-0.13
<i>t_b</i>	-1.62	2.47	0.66	2.35	-0.34	0.36	0.63	-0.79	1.09	0.43	-0.19	-1.67	2.29	2.63	1.65	1.20	-2.52	1.25
<i>t_s</i>	0.16	9.84	5.99	7.16	-6.55	-6.40	-6.80	5.77	4.15	4.15	5.00	0.28	5.97	3.71	4.92	6.78	-4.42	-4.09
<i>t_h</i>	-1.87	15.74	11.28	11.85	-3.81	-2.97	-2.59	13.28	10.20	10.25	11.69	10.09	15.78	9.58	12.02	15.09	5.76	-9.02
<i>t_r</i>	0.00	-4.47	-3.92	-4.21	4.51	4.54	3.89	2.80	3.38	4.82	5.51	2.22	0.02	-0.22	-0.25	0.29	7.50	-2.52
<i>t_c</i>	1.15	2.36	1.31	0.24	-4.61	-4.82	-4.83	-2.28	-1.32	-0.94	-0.97	-2.08	0.15	-0.81	-1.07	-0.82	4.52	-1.00
	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72
	Em ^{q1}	Em ^{q6}	Em ^{q12}	Sp	Sp ^{q1}	Sp ^{q6}	Sp ^{q12}	Ocp	Ocp ^{q1}	Ir	Vhp	Vfp	Ebp	Dur	Acı	I/A	Ia ^{q6}	Ia ^{q12}
β_{MKT}	0.07	0.09	0.11	0.09	0.13	0.09	0.07	-0.02	0.11	-0.03	-0.04	-0.05	0.05	0.07	0.01	0.03	0.07	0.04
β_{ME}	0.02	-0.02	-0.07	0.62	0.59	0.61	0.64	0.16	0.13	-0.57	0.24	0.15	0.51	-0.27	-0.29	-0.13	-0.18	-0.21
$\beta_{\text{I/A}}$	-0.67	-0.67	-0.71	1.14	1.07	1.11	1.08	1.37	1.19	-1.16	0.91	0.50	1.19	-0.97	0.13	-1.37	-1.35	-1.36
β_{ROE}	0.03	0.03	-0.01	-0.30	-0.56	-0.51	-0.39	-0.50	-0.58	0.65	-0.10	0.18	-0.59	0.18	-0.20	0.16	0.34	0.21
$t_{\beta_{\text{MKT}}}$	1.26	1.77	2.33	1.77	1.88	1.47	1.31	-0.33	1.23	-0.64	-0.62	-0.84	1.01	1.14	0.18	1.06	2.23	1.41
$t_{\beta_{\text{ME}}}$	0.19	-0.18	-0.76	4.50	3.43	3.98	4.54	1.40	0.64	-7.94	2.01	1.46	6.51	-1.97	-4.98	-2.31	-3.30	-4.28
$t_{\beta_{\text{I/A}}}$	-4.56	-5.53	-5.95	9.49	5.92	7.02	7.99	9.73	5.54	-10.70	5.82	3.04	12.49	-6.69	1.04	-16.72	-12.31	-13.62
$t_{\beta_{\text{ROE}}}$	0.25	0.23	-0.09	-2.87	-3.13	-3.37	-3.20	-4.31	-2.96	7.39	-0.78	1.51	-7.49	1.42	-2.26	2.54	4.37	3.11
<i>b</i>	0.01	0.02	0.05	0.17	0.24	0.20	0.17	0.09	0.22	-0.10	0.02	0.02	0.11	0.01	0.03	-0.01	0.00	-0.01
<i>s</i>	-0.11	-0.17	-0.21	0.71	0.76	0.78	0.77	0.27	0.42	-0.59	0.31	0.23	0.55	-0.32	-0.26	-0.09	-0.15	-0.15
<i>h</i>	-0.72	-0.68	-0.71	1.05	1.16	1.10	1.04	1.19	1.18	-0.95	1.22	0.84	1.04	-1.22	0.16	-0.17	-0.29	-0.22
<i>r</i>	-0.37	-0.42	-0.43	0.20	0.22	0.24	0.23	-0.02	0.13	0.42	0.25	0.27	-0.33	-0.09	-0.03	0.02	0.10	0.06
<i>c</i>	0.05	0.02	0.01	0.17	0.05	0.13	0.17	0.21	-0.06	-0.33	-0.31	-0.22	0.20	0.21	-0.02	-1.14	-1.11	-1.14
<i>t_b</i>	0.12	0.49	1.13	4.97	4.01	3.81	3.93	2.30	2.79	-3.01	0.44	0.36	3.30	0.20	0.66	-0.31	-0.01	-0.53
<i>t_s</i>	-1.41	-2.32	-2.88	13.86	9.76	11.54	13.51	5.39	3.72	-9.26	4.93	2.22	11.65	-5.39	-4.00	-1.44	-2.77	-3.46
<i>t_h</i>	-6.53	-6.22	-6.77	11.13	7.84	8.77	9.84	15.80	6.82	-11.83	13.43	6.43	17.22	-12.13	1.54	-2.53	-4.54	-4.48
<i>t_r</i>	-3.61	-4.18	-4.40	2.49	1.79	2.40	2.83	-0.26	0.90	4.58	3.06	2.14	-6.02	-1.04	-0.35	0.28	1.26	0.90
<i>t_c</i>	0.28	0.15	0.07	1.45	0.26	0.85	1.37	1.62	-0.23	-2.83	-2.60	-1.13	2.17	1.79	-0.10	-11.07	-8.57	-11.00
	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
	dPia	Noa	dNoa	dLno	Ig	2Ig	Nsi	dli	Cei	Ivg	Ivc	Oa	dWc	dCoa	dNco	dNca	dFin	dFnl
β_{MKT}	0.04	-0.01	0.00	-0.08	-0.02	0.07	0.04	0.03	0.22	-0.02	0.05	0.06	0.03	0.05	-0.02	-0.05	-0.03	0.03
β_{ME}	-0.09	0.11	0.03	-0.16	-0.15	-0.30	0.15	-0.17	0.28	0.07	0.00	0.31	0.35	-0.04	-0.08	-0.10	-0.11	-0.06
$\beta_{\text{I/A}}$	-0.82	-0.07	-1.05	-0.81	-0.75	-0.73	-0.67	-0.64	-1.04	-0.94	-0.67	-0.02	-0.33	-1.15	-0.78	-0.87	-0.30	-0.42
β_{ROE}	0.14	0.00	0.02	0.02	-0.06	-0.07	-0.28	-0.21	-0.12	0.04	0.20	0.26	0.14	0.13	0.00	0.03	0.03	-0.14
$t_{\beta_{\text{MKT}}}$	1.14	-0.14	-0.11	-1.60	-0.70	1.88	1.07	1.01	6.28	-0.66	1.44	1.83	0.62	1.97	-0.59	-1.42	-1.08	1.00
$t_{\beta_{\text{ME}}}$	-1.86	1.04	0.55	-2.34	-2.64	-4.76	2.15	-3.68	4.25	1.70	-0.08	5.06	4.32	-0.85	-1.61	-1.94	-2.19	-1.50
$t_{\beta_{\text{I/A}}}$	-8.63	-0.44	-9.49	-6.86	-10.47	-9.36	-7.67	-7.58	-13.74	-12.85	-6.21	-0.23	-3.20	-16.21	-10.85	-11.77	-2.54	-5.58
$t_{\beta_{\text{ROE}}}$	1.83	0.04	0.25	0.15	-0.90	-1.01	-4.39	-2.99	-1.57	0.59	2.26	4.13	2.18	2.10	0.00	0.41	0.45	-2.05
<i>b</i>	0.03	0.00	-0.01	-0.09	-0.03	0.05	0.00	0.02	0.17	-0.02	0.05	0.08	0.04	0.02	-0.02	-0.05	-0.06	0.02
<i>s</i>	-0.03	0.16	0.07	-0.10	-0.13	-0.23	0.09	-0.13	0.25	0.11	0.07	0.33	0.36	0.00	-0.03	-0.06	-0.13	-0.03
<i>h</i>	0.02	0.46	-0.17	0.04	-0.10	0.03	-0.07	-0.27	-0.39	-0.09	0.01	0.00	-0.11	-0.23	0.03	0.04	-0.25	0.16
<i>r</i>	0.22	0.08	-0.03	0.02	-0.11	-0.04	-0.68	-0.24	-0.41	0.07	0.36	0.41	0.17	0.04	0.02	0.00	-0.09	-0.19
<i>c</i>	-0.79	-0.53	-0.73	-0.82	-0.57	-0.75	-0.62	-0.29	-0.62	-0.73	-0.62	0.04	-0.09	-0.88	-0.73	-0.78	-0.09	-0.54
<i>t_b</i>	0.95	-0.05	-0.29	-1.99	-0.91	1.53	-0.17	0.74	5.90	-0.66	1.43	2.32	0.89	0.56	-0.64	-1.51	-1.85	0.74
<i>t_s</i>	-0.65	2.31	1.24	-1.44	-2.52	-5.05	1.74	-2.82	5.12	2.12	1.31	6.39	4.95	0.05	-0.58	-1.05	-2.87	-0.70
<i>t_h</i>	0.21	5.21	-2.22	0.50	-1.53	0.41	-1.15	-4.05	-6.00	-1.10	0.08	-0.01	-1.23	-3.68	0.53	0.59	-4.09	2.54
<i>t_r</i>	3.51	0.62	-0.30	0.18	-1.36	-0.42	-10.64	-2.82	-5.89	0.74	4.00	6.49	1.63	0.44	0.26	0.05	-1.03	-2.22
<i>t_c</i>	-7.37	-3.68	-5.54	-5.53	-5.28	-6.05	-7.09	-2.72	-6.40	-7.32	-4.62	0.38	-0.62	-8.33	-7.40	-7.43	-0.78	-4.86

	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108
	Dac	Poa	Pta	Pda	Ndf	Roe1	dRoe1	dRoe6	dRoe12	Roa1	dRoa1	dRoa6	Rna ^{q1}	Ato ^{q1}	Ato ^{q6}	Ato ^{q12}	Cto ^{q1}	Cto ^{q6}
β_{MKT}	0.01	-0.01	0.06	0.05	0.06	-0.08	0.03	0.04	0.01	-0.13	0.11	0.09	-0.14	0.11	0.09	0.08	0.12	0.12
β_{ME}	0.19	0.14	0.17	0.05	-0.12	-0.37	-0.06	-0.02	-0.01	-0.37	0.09	0.11	-0.44	0.43	0.38	0.33	0.33	0.32
$\beta_{\text{I/A}}$	0.23	-0.94	-0.87	-0.18	-0.44	0.12	0.23	0.21	0.14	-0.08	0.25	0.19	-0.14	-0.49	-0.61	-0.69	-0.14	-0.21
β_{ROE}	0.19	0.07	0.05	-0.09	-0.26	1.49	0.58	0.56	0.52	1.34	0.59	0.59	1.29	0.55	0.53	0.47	0.83	0.77
$t_{\beta_{\text{MKT}}}$	0.32	-0.35	1.69	1.32	1.76	-2.22	0.64	0.97	0.26	-4.17	2.44	1.96	-3.48	1.87	1.69	1.52	2.08	2.29
$t_{\beta_{\text{ME}}}$	3.27	3.36	2.66	0.63	-2.34	-6.34	-0.88	-0.42	-0.16	-6.34	1.30	1.59	-8.60	5.44	5.41	5.64	3.03	3.34
$t_{\beta_{\text{I/A}}}$	2.38	-11.07	-8.94	-1.34	-5.80	1.24	2.75	2.60	2.53	-0.95	2.15	2.13	-1.40	-4.70	-5.95	-6.82	-1.31	-2.04
$t_{\beta_{\text{ROE}}}$	3.05	1.39	0.65	-0.97	-3.79	19.40	6.76	6.02	8.01	17.49	5.18	5.51	19.43	5.73	7.03	6.73	10.37	10.61
b	0.02	-0.04	0.03	0.04	0.04	-0.11	-0.02	-0.02	-0.04	-0.15	0.06	0.04	-0.15	0.11	0.08	0.07	0.15	0.15
s	0.20	0.18	0.14	0.08	-0.10	-0.46	-0.25	-0.18	-0.13	-0.47	-0.13	-0.08	-0.42	0.47	0.42	0.39	0.43	0.43
h	0.11	-0.18	-0.24	0.17	0.06	-0.26	-0.27	-0.26	-0.18	-0.25	-0.37	-0.32	-0.29	-0.68	-0.69	-0.67	-0.35	-0.36
r	0.28	-0.01	-0.21	-0.13	-0.38	1.42	0.02	0.06	0.15	1.25	-0.01	0.05	1.33	0.72	0.67	0.63	1.24	1.19
c	0.15	-0.72	-0.56	-0.33	-0.50	0.22	0.38	0.33	0.19	0.04	0.56	0.42	0.02	0.31	0.19	0.09	0.29	0.23
t_b	0.73	-1.41	0.84	1.20	1.08	-2.43	-0.46	-0.39	-1.00	-3.69	1.14	0.82	-3.33	2.18	1.82	1.53	3.12	3.35
t_s	3.64	4.53	2.15	1.21	-1.59	-5.86	-2.92	-2.52	-2.14	-5.95	-1.56	-1.05	-6.43	5.66	6.39	6.62	6.66	7.24
t_h	1.61	-3.24	-2.49	1.89	0.67	-2.51	-2.24	-2.32	-1.98	-3.05	-2.89	-2.58	-2.82	-5.47	-7.76	-8.68	-4.17	-4.84
t_r	3.83	-0.26	-2.54	-1.12	-4.34	12.27	0.19	0.61	1.92	10.73	-0.12	0.49	12.33	7.81	7.73	8.04	15.29	15.81
t_c	1.33	-8.28	-4.62	-2.32	-3.88	1.37	2.39	2.39	1.66	0.25	3.30	2.54	0.10	2.06	1.54	0.76	2.75	2.25
	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126
	Cto ^{q12}	Gpa	Gla ^{q1}	Gla ^{q6}	Gla ^{q12}	Ole ^{q1}	Ole ^{q6}	Ola ^{q1}	Ola ^{q6}	Ola ^{q12}	Cop	Cla	Cla ^{q1}	Cla ^{q6}	Cla ^{q12}	F ^{q1}	F ^{q6}	F ^{q12}
β_{MKT}	0.11	0.04	0.00	0.02	0.01	-0.05	-0.06	-0.11	-0.10	-0.13	-0.23	-0.21	-0.08	-0.04	-0.07	-0.07	-0.03	-0.05
β_{ME}	0.30	0.03	0.11	0.06	0.05	-0.24	-0.29	-0.33	-0.37	-0.37	-0.60	-0.62	-0.32	-0.32	-0.31	-0.33	-0.40	-0.41
$\beta_{\text{I/A}}$	-0.27	-0.31	-0.28	-0.37	-0.45	0.38	0.33	-0.24	-0.31	-0.42	-0.06	-0.31	-0.13	-0.13	-0.19	0.44	0.33	0.32
β_{ROE}	0.72	0.55	0.66	0.60	0.53	1.15	1.05	1.08	0.98	0.89	0.49	0.40	0.48	0.45	0.40	0.73	0.67	0.65
$t_{\beta_{\text{MKT}}}$	2.06	0.95	-0.10	0.82	0.23	-1.08	-1.43	-2.44	-3.08	-4.28	-5.84	-5.46	-1.87	-1.46	-2.79	-1.03	-0.70	-0.99
$t_{\beta_{\text{ME}}}$	3.48	0.69	2.20	1.23	1.11	-2.25	-3.31	-3.82	-5.62	-5.68	-7.78	-8.65	-4.77	-5.77	-6.01	-3.16	-4.55	-4.82
$t_{\beta_{\text{I/A}}}$	-2.68	-3.21	-3.03	-4.56	-5.22	2.63	2.64	-2.09	-3.25	-4.60	-0.66	-3.35	-1.07	-1.24	-2.14	3.07	2.80	3.18
$t_{\beta_{\text{ROE}}}$	10.25	7.66	12.26	10.83	8.96	10.91	9.99	13.43	14.45	12.18	7.88	6.00	5.25	7.12	7.34	6.97	6.90	7.11
b	0.13	0.04	-0.01	0.02	0.00	-0.02	-0.02	-0.12	-0.11	-0.14	-0.22	-0.21	-0.09	-0.06	-0.09	-0.12	-0.09	-0.10
s	0.41	0.11	0.15	0.09	0.11	-0.24	-0.26	-0.36	-0.38	-0.35	-0.63	-0.64	-0.35	-0.36	-0.35	-0.37	-0.43	-0.41
h	-0.38	-0.47	-0.51	-0.46	-0.40	0.06	0.06	-0.51	-0.45	-0.40	-0.46	-0.45	-0.35	-0.29	-0.30	0.06	0.04	0.16
r	1.11	0.89	0.84	0.76	0.72	1.45	1.40	1.04	0.98	0.93	0.57	0.42	0.41	0.36	0.31	0.62	0.53	0.56
c	0.18	0.20	0.18	0.06	-0.08	0.24	0.19	0.24	0.13	-0.03	0.44	0.14	0.21	0.14	0.08	0.19	0.10	-0.05
t_b	2.94	1.06	-0.21	0.55	0.05	-0.48	-0.76	-2.41	-2.83	-3.94	-6.34	-6.36	-2.31	-1.86	-3.36	-1.77	-1.66	-2.03
t_s	7.01	2.23	2.69	1.80	2.05	-3.55	-4.84	-4.88	-5.56	-4.99	-11.27	-12.49	-5.77	-6.70	-7.98	-3.27	-4.40	-4.84
t_h	-5.07	-4.63	-5.80	-5.61	-5.15	0.63	0.89	-4.74	-5.18	-4.56	-5.42	-5.71	-4.62	-4.97	-5.37	0.49	0.42	1.86
t_r	15.30	9.85	11.54	10.10	9.88	13.95	14.81	6.79	7.66	7.31	5.39	4.65	2.88	3.32	3.38	3.97	3.83	5.88
t_c	1.72	1.59	1.42	0.50	-0.68	1.62	1.63	1.41	0.96	-0.25	4.23	1.33	1.36	1.20	0.80	1.07	0.57	-0.31

	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144
	Fp6	Tbi ^q 12	Oca	Ioca	Adm	Rdm	Rdm ^q 1	Rdm ^q 6	Rdm ^q 12	Ol	Ol ^q 1	Ol ^q 6	Ol ^q 12	Hs	Etr	Rer	Eprd	Etl
β_{MKT}	0.41	-0.07	-0.16	-0.06	0.07	0.16	0.01	-0.08	-0.08	-0.04	-0.10	-0.13	-0.13	-0.17	0.01	0.05	0.10	0.01
β_{ME}	0.40	-0.17	0.22	0.25	0.48	0.62	0.14	0.52	0.62	0.30	0.27	0.32	0.32	-0.08	0.12	-0.13	0.35	0.29
$\beta_{\text{I/A}}$	0.10	-0.14	0.27	0.36	1.36	0.17	0.61	0.69	0.82	0.11	0.04	0.05	0.04	0.28	0.04	-0.15	0.41	-0.13
β_{ROE}	-1.54	0.05	0.55	0.51	-0.30	-0.62	-1.02	-0.90	-0.70	0.55	0.67	0.62	0.59	-0.03	0.18	0.01	-0.62	0.05
$t_{\beta_{\text{MKT}}}$	5.87	-2.10	-2.41	-1.88	0.76	2.45	0.05	-0.96	-1.04	-0.80	-1.84	-2.66	-2.85	-3.35	0.36	0.93	1.63	0.25
$t_{\beta_{\text{ME}}}$	2.19	-3.37	2.89	5.60	2.75	6.37	0.71	3.56	4.72	3.18	3.34	3.64	4.03	-0.96	2.19	-1.28	4.21	3.18
$t_{\beta_{\text{I/A}}}$	0.39	-2.07	2.05	3.73	5.94	0.95	1.99	3.17	4.35	0.95	0.33	0.38	0.34	1.69	0.40	-1.17	3.59	-0.87
$t_{\beta_{\text{ROE}}}$	-8.64	0.65	4.38	7.32	-1.49	-4.26	-3.50	-4.82	-4.62	5.07	6.80	5.82	5.75	-0.21	2.29	0.10	-6.46	0.54
b	0.45	-0.07	-0.15	-0.07	0.14	0.22	0.24	0.11	0.09	-0.02	-0.07	-0.10	-0.11	-0.12	0.01	0.05	0.16	-0.01
s	0.61	-0.08	0.21	0.21	0.62	0.57	0.35	0.54	0.58	0.37	0.29	0.35	0.33	-0.02	0.04	-0.06	0.44	0.25
h	0.62	0.05	-0.35	-0.09	1.02	0.00	0.27	0.37	0.24	0.03	-0.11	-0.07	-0.06	0.38	-0.04	-0.09	0.61	-0.26
r	-1.04	0.25	0.82	0.52	0.44	-0.56	-0.26	-0.54	-0.44	0.89	0.94	0.91	0.84	0.25	0.05	0.13	-0.37	-0.12
c	-0.50	-0.21	0.59	0.36	0.12	0.45	0.58	0.69	0.93	0.08	0.16	0.15	0.12	-0.04	0.06	-0.04	-0.02	0.13
t_b	4.87	-2.32	-2.48	-1.84	2.36	3.49	1.82	1.22	1.13	-0.59	-1.49	-2.22	-2.48	-2.75	0.20	0.95	2.90	-0.28
t_s	3.85	-1.67	2.92	4.13	6.41	6.32	1.61	3.42	4.38	5.67	4.29	4.88	5.22	-0.34	0.80	-0.65	6.54	3.79
t_h	2.67	0.67	-2.82	-1.00	6.94	0.02	1.05	1.96	1.72	0.25	-1.22	-0.77	-0.71	3.71	-0.50	-0.87	6.05	-2.54
t_r	-3.77	3.74	5.83	4.81	3.76	-3.04	-1.02	-2.19	-2.14	10.68	9.71	10.05	9.64	1.47	0.54	1.28	-3.64	-1.00
t_c	-1.42	-1.93	3.45	2.57	0.55	2.12	1.21	2.09	3.72	0.61	1.20	1.13	0.90	-0.19	0.52	-0.24	-0.15	0.94
	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	
	Alm ^q 1	Alm ^q 6	Alm ^q 12	R_a^1	$R_a^{[2,5]}$	$R_n^{[2,5]}$	$R_a^{[6,10]}$	$R_n^{[6,10]}$	$R_a^{[11,15]}$	$R_a^{[16,20]}$	Sv1	Dtv6	Dtv12	Ami12	Ts1	Isff1	Isq1	
β_{MKT}	0.07	0.06	0.07	0.23	0.06	0.19	-0.03	0.16	-0.01	-0.07	0.04	0.14	0.13	-0.03	0.03	-0.01	-0.02	
β_{ME}	0.67	0.71	0.72	-0.14	-0.18	-0.27	0.03	-0.31	-0.07	-0.07	0.35	-1.08	-1.14	1.30	0.06	0.17	0.19	
$\beta_{\text{I/A}}$	0.83	0.77	0.70	-0.15	-0.28	-1.32	-0.37	-0.81	-0.03	-0.04	-0.14	-0.38	-0.36	0.15	-0.08	0.01	-0.06	
β_{ROE}	-0.44	-0.33	-0.24	0.18	0.05	0.38	-0.23	-0.28	0.10	0.00	-0.44	0.33	0.29	-0.36	-0.15	-0.04	-0.13	
$t_{\beta_{\text{MKT}}}$	1.83	2.04	2.23	4.14	1.06	3.03	-0.64	3.06	-0.25	-1.37	0.65	4.54	5.12	-1.14	1.13	-0.27	-0.66	
$t_{\beta_{\text{ME}}}$	7.56	10.54	11.47	-1.28	-1.75	-2.08	0.31	-3.40	-0.83	-1.21	2.58	-17.69	-31.71	42.18	1.41	4.25	2.54	
$t_{\beta_{\text{I/A}}}$	8.07	9.20	8.45	-0.97	-2.46	-9.56	-2.22	-5.88	-0.23	-0.34	-0.80	-5.64	-7.11	2.95	-0.83	0.13	-0.74	
$t_{\beta_{\text{ROE}}}$	-5.96	-5.55	-3.73	1.25	0.47	2.77	-1.97	-2.30	1.09	-0.01	-3.45	6.94	7.17	-8.56	-3.04	-0.77	-2.40	
b	0.14	0.12	0.12	0.19	0.07	0.11	-0.05	0.13	-0.02	-0.07	0.04	0.11	0.11	-0.02	0.02	-0.02	-0.03	
s	0.74	0.74	0.73	-0.13	-0.12	-0.34	0.03	-0.29	-0.10	-0.04	0.29	-1.13	-1.16	1.33	0.07	0.17	0.18	
h	0.73	0.64	0.57	-0.08	0.12	-0.78	-0.04	-0.54	0.04	-0.05	0.03	-0.34	-0.25	0.16	-0.09	-0.06	-0.10	
r	-0.19	-0.21	-0.20	0.10	0.18	-0.13	-0.31	-0.38	-0.01	0.07	-0.56	0.10	0.16	-0.29	-0.17	-0.09	-0.18	
c	0.18	0.21	0.21	-0.18	-0.39	-0.57	-0.32	-0.26	-0.11	0.01	-0.15	-0.12	-0.19	0.08	-0.02	0.04	0.06	
t_b	3.62	3.97	3.72	3.43	1.20	1.89	-0.89	2.68	-0.49	-1.39	0.61	4.23	4.85	-0.78	0.66	-0.72	-0.86	
t_s	12.88	14.70	14.33	-1.23	-1.49	-3.22	0.49	-3.61	-1.35	-0.64	2.70	-26.70	-35.27	39.34	1.55	4.02	3.33	
t_h	7.69	7.99	6.53	-0.54	1.05	-5.37	-0.34	-5.11	0.36	-0.61	0.20	-5.39	-5.30	3.44	-1.59	-1.22	-1.43	
t_r	-2.87	-4.13	-3.68	0.51	1.53	-1.34	-2.26	-3.43	-0.07	0.72	-3.95	2.00	3.36	-6.63	-2.17	-1.13	-2.14	
t_c	1.54	2.20	2.15	-0.83	-2.70	-3.08	-1.73	-1.49	-0.80	0.08	-0.73	-1.69	-2.92	1.28	-0.21	0.35	0.61	

Table 6 : Betas for the q -factor Model and the Five-factor Model, Significant Anomalies, ABM-EW, January 1967 to December 2014, 576 Months

For each high-minus-low decile, $\beta_{\text{MKT}}, \beta_{\text{ME}}, \beta_{\text{I/A}},$ and β_{ROE} are the loadings on the market, size, investment, and ROE factors in the q -factor model, respectively, and $t_{\beta_{\text{MKT}}}, t_{\beta_{\text{ME}}}, t_{\beta_{\text{I/A}}},$ and $t_{\beta_{\text{ROE}}}$ are their t -statistics. $b, s, h, r,$ and c are the loadings on MKT, SMB, HML, RMW, and CMA in the five-factor model, and $t_b, t_s, t_h, t_r,$ and t_c are their t -statistics. All t -statistics are adjusted for heteroscedasticity and autocorrelations. Table A1 describes the symbols.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Sue1	Sue6	Abr1	Abr6	Abr12	Re1	Re6	Re12	R^6_1	R^6_6	R^6_{12}	R^{11}_1	R^{11}_6	Im1	Im6	Im12	Rs1	Rs6
β_{MKT}	0.01	0.02	-0.07	-0.01	0.00	0.01	0.02	-0.02	-0.23	-0.05	-0.02	-0.10	-0.03	-0.15	-0.06	-0.03	-0.03	-0.03
β_{ME}	-0.05	-0.09	0.11	0.13	0.09	0.05	0.00	-0.03	0.52	0.46	0.31	0.52	0.41	0.21	0.33	0.24	-0.10	-0.07
$\beta_{\text{I/A}}$	-0.05	-0.11	0.00	-0.05	-0.12	-0.14	-0.10	-0.15	-0.01	0.10	-0.19	-0.07	-0.29	0.18	0.17	-0.09	-0.18	-0.27
β_{ROE}	0.89	0.82	0.21	0.25	0.21	0.93	0.86	0.69	1.13	1.19	0.97	1.33	1.25	0.64	0.70	0.56	0.76	0.71
$t_{\beta_{\text{MKT}}}$	0.21	1.18	-2.05	-0.47	-0.20	0.20	0.56	-0.53	-2.26	-0.58	-0.27	-1.05	-0.36	-1.84	-0.95	-0.65	-1.09	-1.11
$t_{\beta_{\text{ME}}}$	-1.31	-2.80	1.46	2.22	2.36	0.85	-0.07	-0.77	1.84	2.32	2.01	2.17	2.05	1.04	1.89	1.79	-2.49	-2.41
$t_{\beta_{\text{I/A}}}$	-0.80	-2.18	0.01	-0.52	-1.88	-1.32	-1.06	-1.82	-0.02	0.33	-1.00	-0.22	-1.15	0.61	0.77	-0.52	-2.78	-3.68
$t_{\beta_{\text{ROE}}}$	14.01	18.81	2.79	3.18	4.41	8.70	10.76	12.57	3.99	4.98	6.18	5.07	5.75	3.19	4.04	4.06	14.03	12.74
b	-0.04	-0.03	-0.09	-0.04	-0.03	-0.07	-0.05	-0.07	-0.29	-0.13	-0.10	-0.21	-0.14	-0.17	-0.10	-0.08	-0.06	-0.06
s	-0.20	-0.21	0.02	0.04	0.03	-0.11	-0.15	-0.13	0.19	0.12	0.03	0.14	0.05	0.03	0.10	0.05	-0.20	-0.16
h	-0.39	-0.36	-0.20	-0.14	-0.11	-0.09	-0.12	-0.15	-0.66	-0.58	-0.60	-0.85	-0.83	-0.44	-0.34	-0.36	-0.36	-0.41
r	0.48	0.48	-0.07	-0.02	0.01	0.30	0.31	0.32	0.19	0.23	0.14	0.17	0.11	0.14	0.10	-0.02	0.54	0.51
c	0.23	0.12	0.24	0.03	-0.07	-0.23	-0.14	-0.13	0.62	0.53	0.29	0.63	0.38	0.67	0.48	0.23	0.10	0.10
t_b	-1.12	-0.84	-2.61	-1.56	-1.63	-1.12	-0.95	-1.94	-2.58	-1.34	-1.33	-1.88	-1.45	-2.09	-1.33	-1.34	-1.87	-1.82
t_s	-3.55	-4.14	0.29	0.76	0.85	-1.30	-2.24	-2.43	0.91	0.77	0.28	0.80	0.31	0.17	0.71	0.44	-4.10	-3.91
t_h	-4.06	-4.55	-2.36	-1.97	-2.24	-0.58	-1.08	-2.06	-1.94	-2.19	-3.04	-2.84	-3.22	-2.06	-1.58	-2.20	-4.91	-7.16
t_r	6.96	7.90	-0.80	-0.22	0.19	2.94	3.57	4.74	0.43	0.69	0.59	0.46	0.37	0.42	0.35	-0.12	6.73	6.66
t_c	1.85	1.12	1.97	0.22	-0.83	-1.07	-0.77	-0.97	1.26	1.31	0.95	1.43	0.96	2.03	1.57	0.97	1.05	1.00
	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
	Tes1	Tes6	dEf1	dEf6	dEf12	Nei1	Nei6	52w6	52w12	ϵ^6_1	ϵ^6_6	ϵ^6_{12}	ϵ^{11}_1	ϵ^{11}_6	ϵ^{11}_{12}	Sm1	Sm6	Sm12
β_{MKT}	0.13	0.11	-0.01	0.04	0.03	0.03	0.02	-0.44	-0.37	-0.08	-0.04	-0.02	0.02	0.01	0.01	-0.03	-0.04	0.01
β_{ME}	0.09	0.11	0.07	0.03	0.00	-0.02	-0.03	-0.07	-0.14	0.10	0.11	0.05	0.12	0.07	0.01	-0.13	0.19	0.14
$\beta_{\text{I/A}}$	-0.28	-0.34	-0.29	-0.24	-0.25	-0.08	-0.09	0.74	0.54	0.07	0.05	-0.04	0.08	-0.03	-0.11	0.15	0.12	0.07
β_{ROE}	0.50	0.49	0.58	0.62	0.51	0.80	0.78	1.47	1.36	0.35	0.39	0.38	0.50	0.49	0.41	0.01	0.26	0.26
$t_{\beta_{\text{MKT}}}$	4.45	4.02	-0.20	1.15	1.08	1.67	1.01	-5.49	-5.68	-1.35	-0.91	-0.57	0.32	0.19	0.20	-0.53	-1.23	0.37
$t_{\beta_{\text{ME}}}$	1.89	2.51	0.61	0.47	0.01	-0.92	-1.23	-0.37	-1.02	1.21	1.53	0.80	1.71	1.02	0.23	-1.52	3.08	3.59
$t_{\beta_{\text{I/A}}}$	-2.52	-3.80	-2.05	-2.49	-3.59	-2.03	-1.62	2.50	2.45	0.60	0.56	-0.43	0.62	-0.22	-0.99	0.89	1.02	0.84
$t_{\beta_{\text{ROE}}}$	6.08	8.35	5.75	7.69	9.15	27.16	18.19	6.21	7.08	3.17	3.90	4.84	4.51	4.96	4.56	0.09	3.14	4.89
b	0.09	0.07	-0.08	-0.03	-0.03	0.00	-0.01	-0.48	-0.43	-0.09	-0.07	-0.06	-0.02	-0.04	-0.04	-0.06	-0.05	-0.01
s	0.01	0.04	-0.09	-0.13	-0.13	-0.11	-0.11	-0.36	-0.40	-0.03	-0.03	-0.07	-0.03	-0.08	-0.11	-0.13	0.12	0.08
h	-0.32	-0.27	-0.37	-0.21	-0.21	-0.25	-0.26	-0.24	-0.22	-0.25	-0.17	-0.17	-0.24	-0.20	-0.22	0.18	-0.02	-0.04
r	0.21	0.24	0.00	0.05	0.08	0.65	0.66	0.84	0.77	-0.01	-0.03	0.00	0.03	-0.01	-0.03	-0.13	0.07	0.07
c	-0.01	-0.15	0.05	-0.13	-0.13	0.10	0.11	0.82	0.57	0.35	0.18	0.07	0.26	0.07	0.01	-0.13	0.15	0.08
t_b	2.66	2.13	-1.61	-0.67	-0.89	-0.11	-0.28	-4.70	-4.78	-1.46	-1.37	-1.30	-0.34	-0.75	-0.93	-0.99	-1.43	-0.38
t_s	0.14	0.73	-1.10	-1.97	-2.44	-2.56	-2.59	-2.42	-3.11	-0.39	-0.44	-1.28	-0.35	-1.10	-1.86	-1.48	2.32	2.14
t_h	-4.71	-4.10	-2.78	-1.98	-3.24	-3.97	-4.16	-0.94	-0.97	-1.86	-1.47	-1.85	-1.82	-1.78	-2.24	1.71	-0.32	-0.69
t_r	2.27	2.99	0.03	0.57	1.22	11.04	10.34	2.57	2.94	-0.11	-0.26	0.00	0.22	-0.09	-0.31	-1.00	0.73	1.09
t_c	-0.10	-1.21	0.24	-0.82	-1.20	1.02	0.99	2.05	1.58	1.71	1.06	0.44	1.27	0.39	0.06	-0.67	1.08	0.87
	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
	Ilr1	Ilr6	Ilr12	Ile1	Ile6	Cm1	Cm6	Cm12	Sim1	Sim6	Sim12	Cim1	Cim6	Cim12	Bm	Bmj	Bmq12	Am
β_{MKT}	-0.13	-0.10	-0.04	0.01	-0.03	0.01	-0.02	0.00	0.00	-0.03	-0.01	-0.02	-0.04	-0.01	-0.14	-0.12	-0.09	0.00
β_{ME}	-0.01	0.14	0.11	0.08	0.06	0.03	0.14	0.12	0.07	0.22	0.17	0.05	0.15	0.10	0.09	-0.07	-0.09	0.05
$\beta_{\text{I/A}}$	0.09	0.03	0.00	-0.10	-0.16	0.14	0.12	0.02	0.17	0.04	-0.01	0.09	0.10	0.03	1.78	1.69	1.66	1.94
β_{ROE}	0.03	0.27	0.29	0.52	0.55	0.13	0.24	0.21	0.19	0.27	0.26	0.15	0.24	0.23	-0.13	-0.55	-0.52	-0.09
$t_{\beta_{\text{MKT}}}$	-1.92	-2.57	-1.53	0.18	-0.78	0.19	-0.81	-0.23	0.00	-0.61	-0.35	-0.33	-1.04	-0.55	-2.64	-2.12	-1.32	-0.06
$t_{\beta_{\text{ME}}}$	-0.05	1.44	1.63	0.94	0.91	0.26	2.78	3.13	0.52	1.89	2.59	0.30	1.51	1.38	0.83	-0.51	-0.53	0.37
$t_{\beta_{\text{I/A}}}$	0.53	0.21	0.02	-0.95	-1.99	0.75	1.08	0.40	0.72	0.19	-0.09	0.38	0.63	0.33	8.62	9.46	9.33	8.58
$t_{\beta_{\text{ROE}}}$	0.23	2.88	4.15	6.60	7.92	0.85	3.36	4.49	1.12	2.11	3.03	0.94	2.41	3.43	-0.76	-3.91	-3.78	-0.49

b	-0.14	-0.11	-0.06	-0.02	-0.06	-0.02	-0.03	-0.02	-0.02	-0.04	-0.03	-0.04	-0.05	-0.04	-0.05	-0.02	0.03	0.09
s	-0.05	0.05	0.03	-0.05	-0.07	-0.02	0.10	0.08	-0.04	0.12	0.10	-0.04	0.04	0.01	0.12	0.05	0.07	0.11
h	-0.12	-0.16	-0.14	-0.51	-0.37	-0.05	-0.01	-0.04	-0.07	-0.14	-0.10	-0.07	-0.14	-0.12	1.28	1.32	1.42	1.45
r	-0.16	0.01	0.05	0.26	0.22	-0.01	0.14	0.10	-0.12	0.00	0.03	-0.20	-0.05	-0.05	0.35	0.14	0.26	0.49
c	0.20	0.20	0.14	0.38	0.14	0.11	0.09	0.05	0.18	0.18	0.08	0.15	0.23	0.12	0.48	0.38	0.24	0.42
t_b	-2.25	-2.90	-1.89	-0.36	-1.62	-0.29	-1.24	-0.83	-0.33	-0.90	-0.84	-0.72	-1.45	-1.38	-1.93	-0.28	0.54	2.87
t_s	-0.52	0.74	0.52	-0.71	-1.16	-0.17	1.98	2.31	-0.32	1.39	1.77	-0.38	0.64	0.10	3.10	0.67	0.96	2.33
t_h	-0.91	-1.59	-1.76	-5.70	-4.71	-0.41	-0.09	-0.73	-0.41	-0.97	-1.04	-0.41	-1.26	-1.49	23.74	11.22	11.88	21.75
t_r	-0.92	0.10	0.43	2.36	2.46	-0.08	1.59	1.57	-0.47	-0.02	0.23	-0.79	-0.31	-0.47	3.68	1.10	2.39	5.16
t_c	1.07	1.44	1.27	2.71	1.16	0.55	0.69	0.52	0.68	0.88	0.53	0.67	1.46	1.03	4.37	1.90	1.40	3.58
	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72
	Rev1	Rev6	Rev12	Ep	Ep ^{q1}	Ep ^{q6}	Ep ^{q12}	Cp	Cp ^{q1}	Cp ^{q6}	Cp ^{q12}	Op	Op ^{q1}	Op ^{q6}	Op ^{q12}	Nop	Nop ^{q6}	Nop ^{q12}
β_{MKT}	0.06	0.07	0.07	-0.19	-0.02	-0.08	-0.12	-0.12	-0.02	-0.08	-0.09	-0.28	-0.10	-0.14	-0.14	-0.20	-0.14	-0.13
β_{ME}	-0.36	-0.29	-0.26	-0.02	0.00	-0.03	-0.01	0.04	-0.08	-0.07	-0.01	-0.21	-0.25	-0.19	-0.14	-0.25	-0.32	-0.32
$\beta_{\text{I/A}}$	-1.12	-0.98	-0.91	1.14	0.92	0.94	0.92	1.60	1.22	1.22	1.25	1.07	0.53	0.60	0.64	1.09	1.11	1.12
β_{ROE}	0.44	0.36	0.25	-0.01	0.20	0.20	0.17	0.01	-0.17	-0.13	-0.04	-0.15	-0.12	-0.07	0.02	0.27	0.39	0.45
$t_{\beta_{\text{MKT}}}$	1.46	1.68	1.62	-3.38	-0.27	-1.30	-2.23	-1.74	-0.21	-1.03	-1.28	-6.09	-1.82	-3.51	-3.88	-6.48	-2.71	-2.71
$t_{\beta_{\text{ME}}}$	-5.61	-4.07	-3.26	-0.17	0.01	-0.25	-0.05	0.32	-0.42	-0.37	-0.04	-2.96	-2.65	-3.19	-2.66	-5.38	-4.78	-5.37
$t_{\beta_{\text{I/A}}}$	-9.11	-6.98	-6.11	8.10	5.15	6.41	6.89	8.38	5.73	5.96	6.71	7.28	3.96	6.36	7.01	9.69	6.91	7.49
$t_{\beta_{\text{ROE}}}$	4.05	2.90	1.91	-0.10	1.43	1.51	1.46	0.08	-0.87	-0.72	-0.23	-1.16	-0.95	-0.81	0.26	2.74	2.82	3.46
b	0.02	0.03	0.04	-0.12	0.03	-0.03	-0.07	-0.03	0.08	0.01	0.00	-0.23	-0.03	-0.09	-0.10	-0.15	-0.12	-0.12
s	-0.36	-0.28	-0.23	0.02	0.05	0.03	0.05	0.10	0.11	0.12	0.14	-0.20	-0.12	-0.11	-0.08	-0.25	-0.26	-0.28
h	-0.45	-0.45	-0.44	1.09	1.04	0.96	0.92	1.34	1.29	1.32	1.28	0.79	0.33	0.28	0.29	0.41	0.56	0.51
r	0.21	0.23	0.16	0.37	0.56	0.60	0.56	0.57	0.71	0.68	0.66	0.07	0.30	0.26	0.25	0.57	0.60	0.61
c	-0.65	-0.51	-0.43	0.00	-0.18	-0.09	-0.05	0.21	-0.10	-0.15	-0.06	0.16	0.17	0.31	0.28	0.59	0.39	0.43
t_b	0.34	0.81	0.99	-3.99	0.68	-0.83	-2.29	-1.06	1.46	0.22	-0.13	-7.12	-0.70	-2.47	-3.46	-5.23	-3.36	-3.62
t_s	-4.37	-3.67	-3.15	0.33	0.70	0.47	1.01	2.04	1.41	1.89	2.59	-3.68	-1.94	-2.14	-1.79	-4.87	-4.29	-4.97
t_h	-3.95	-4.30	-4.30	16.32	11.01	12.73	15.34	20.18	11.17	12.94	17.06	10.88	3.49	3.60	4.61	6.23	7.56	7.31
t_r	1.69	1.58	1.09	4.97	4.67	7.47	9.03	6.46	5.37	6.85	8.56	0.94	3.70	4.85	4.99	6.57	7.17	7.95
t_c	-3.74	-2.69	-2.22	0.03	-1.46	-0.93	-0.55	1.80	-0.66	-1.11	-0.59	1.47	1.21	3.53	3.25	5.44	3.17	3.39
	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
	Sg	Em	Em ^{q1}	Em ^{q6}	Em ^{q12}	Sp	Sp ^{q1}	Sp ^{q6}	Sp ^{q12}	Ocp	Ocp ^{q1}	Ir	Vhp	Ebp	Ndp	Dur	Aci	I/A
β_{MKT}	0.07	0.16	0.04	0.10	0.12	0.02	0.09	0.05	0.04	-0.12	0.10	0.00	-0.12	-0.05	0.03	0.22	0.01	0.07
β_{ME}	0.14	-0.05	0.32	0.25	0.16	0.21	0.06	0.11	0.16	0.01	-0.25	-0.24	0.06	0.08	0.08	-0.09	-0.15	0.03
$\beta_{\text{I/A}}$	-1.22	-1.40	-0.99	-1.06	-1.08	1.84	1.63	1.68	1.68	1.57	1.26	-1.36	0.97	1.87	0.94	-1.23	-0.15	-1.25
β_{ROE}	0.09	-0.29	-0.29	-0.35	-0.41	0.34	0.03	0.10	0.20	0.08	-0.01	0.42	-0.02	-0.05	-0.48	-0.39	-0.14	0.15
$t_{\beta_{\text{MKT}}}$	3.05	2.43	0.55	1.36	1.88	0.24	0.94	0.57	0.48	-1.95	1.07	0.05	-1.83	-0.72	0.92	3.16	0.33	3.36
$t_{\beta_{\text{ME}}}$	3.24	-0.44	1.68	1.45	1.11	1.00	0.24	0.45	0.75	0.05	-1.08	-2.39	0.46	0.53	1.02	-0.72	-3.87	0.64
$t_{\beta_{\text{I/A}}}$	-16.46	-7.19	-4.50	-5.38	-5.85	8.12	5.94	6.99	7.58	7.55	5.74	-10.74	7.03	8.96	8.18	-7.50	-2.21	-15.21
$t_{\beta_{\text{ROE}}}$	1.28	-1.77	-1.74	-2.26	-2.90	1.70	0.13	0.44	0.99	0.46	-0.03	3.80	-0.13	-0.32	-5.84	-2.14	-2.19	2.16
b	0.03	0.08	-0.04	0.03	0.05	0.13	0.24	0.19	0.17	-0.02	0.19	-0.07	-0.05	0.05	0.08	0.15	0.02	0.04
s	0.17	-0.11	0.12	0.06	0.00	0.35	0.31	0.33	0.35	0.07	0.07	-0.31	0.15	0.15	0.12	-0.10	-0.10	0.09
h	-0.30	-1.04	-0.91	-0.91	-0.89	1.36	1.47	1.42	1.36	1.11	1.07	-1.07	1.09	1.38	0.92	-1.11	0.10	-0.20
r	-0.07	-0.86	-1.01	-1.04	-1.01	1.27	1.28	1.26	1.26	0.58	0.74	-0.03	0.47	0.55	-0.23	-0.78	-0.02	0.03
c	-0.86	-0.31	0.08	0.07	0.00	0.44	0.24	0.31	0.35	0.36	0.02	-0.29	-0.16	0.44	-0.05	-0.10	-0.24	-0.97
t_b	1.52	2.41	-0.78	0.69	1.61	3.76	3.92	3.81	4.05	-0.66	3.10	-2.10	-1.71	1.62	2.59	3.93	0.68	1.49
t_s	4.00	-2.24	1.58	0.91	-0.04	6.74	3.25	4.46	6.03	1.19	0.75	-5.68	2.81	3.32	2.81	-1.68	-2.82	1.99
t_h	-5.65	-15.17	-8.59	-9.41	-11.56	18.12	9.05	10.90	13.66	14.91	7.63	-13.18	15.56	22.78	14.48	-12.05	1.70	-3.06
t_r	-1.05	-9.02	-8.69	-11.01	-13.20	16.17	7.35	9.81	14.00	5.56	5.10	-0.43	6.36	7.27	-4.37	-7.09	-0.34	0.29
t_c	-8.96	-2.41	0.53	0.58	0.00	3.96	1.11	1.95	3.02	2.51	0.09	-2.49	-1.54	4.55	-0.49	-0.66	-2.64	-9.07

	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108
	Ia ^{q1}	Ia ^{q6}	Ia ^{q12}	dPia	Noa	dNoa	dLno	Ig	2Ig	3Ig	Nsi	dIi	Cei	Cdi	Ivg	Ivc	Oa	Ta
β_{MKT}	0.10	0.10	0.07	0.10	0.13	0.07	0.04	0.00	0.04	0.08	0.12	0.04	0.24	0.08	0.07	0.06	0.01	0.06
β_{ME}	-0.02	0.02	0.03	-0.01	-0.03	0.02	-0.09	0.01	-0.05	0.00	0.05	-0.01	0.29	0.02	0.08	0.10	0.21	0.07
$\beta_{\text{I/A}}$	-1.27	-1.30	-1.24	-0.80	0.01	-0.83	-0.63	-0.76	-0.68	-0.68	-0.86	-0.51	-0.93	-0.23	-0.69	-0.55	-0.19	-0.67
β_{ROE}	0.41	0.26	0.18	0.02	0.18	0.04	0.04	-0.08	-0.10	-0.09	-0.46	-0.19	-0.31	-0.05	0.06	0.18	0.42	0.40
$t_{\beta_{\text{MKT}}}$	2.56	3.29	3.15	3.30	2.15	2.89	1.52	0.23	1.51	2.51	4.43	1.46	7.24	3.58	3.16	2.80	0.24	1.76
$t_{\beta_{\text{ME}}}$	-0.24	0.23	0.46	-0.12	-0.23	0.42	-1.80	0.30	-1.13	-0.06	0.87	-0.20	6.48	0.37	2.01	2.38	3.97	1.48
$t_{\beta_{\text{I/A}}}$	-7.17	-8.99	-13.04	-8.52	0.07	-10.06	-7.95	-11.69	-8.02	-7.55	-8.45	-7.50	-10.49	-3.93	-7.84	-6.25	-1.75	-8.11
$t_{\beta_{\text{ROE}}}$	3.63	2.79	2.67	0.26	1.53	0.64	0.62	-1.30	-1.19	-1.04	-5.72	-2.64	-3.62	-1.60	0.95	3.12	4.88	4.78
b	0.03	0.05	0.03	0.09	0.14	0.06	0.03	-0.01	0.04	0.07	0.08	0.04	0.22	0.08	0.07	0.07	0.02	0.05
s	0.00	0.09	0.11	0.07	0.10	0.12	-0.02	0.05	0.00	0.02	0.03	0.02	0.33	0.06	0.12	0.17	0.25	0.08
h	-0.33	-0.23	-0.20	0.07	0.62	0.03	0.08	-0.09	-0.12	-0.18	-0.13	-0.20	-0.39	0.01	-0.04	-0.01	-0.02	-0.10
r	0.13	0.14	0.11	0.07	0.52	0.10	0.10	-0.13	-0.08	-0.13	-0.81	-0.21	-0.44	-0.01	0.11	0.33	0.62	0.39
c	-0.98	-1.06	-0.99	-0.82	-0.65	-0.82	-0.68	-0.63	-0.49	-0.41	-0.70	-0.23	-0.42	-0.22	-0.58	-0.46	-0.11	-0.52
t_b	0.59	1.57	1.43	3.00	3.84	2.45	1.28	-0.66	1.16	2.18	3.00	1.25	7.86	3.49	2.82	2.64	0.79	1.24
t_s	-0.02	1.60	2.50	1.65	1.66	2.90	-0.44	1.41	0.00	0.52	0.53	0.54	5.74	1.59	3.10	5.36	5.90	1.52
t_h	-3.06	-3.42	-3.73	1.04	6.16	0.54	1.29	-1.90	-1.53	-2.54	-2.20	-3.10	-5.38	0.33	-0.62	-0.08	-0.25	-1.22
t_r	0.73	1.17	1.41	1.05	4.60	1.40	1.60	-1.94	-0.93	-1.32	-9.90	-2.89	-5.71	-0.16	1.65	5.67	8.59	4.94
t_c	-5.43	-7.54	-9.30	-8.82	-5.37	-8.90	-8.61	-6.77	-3.57	-3.30	-7.65	-2.10	-4.49	-3.17	-6.54	-5.70	-0.93	-3.66
	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126
	dWc	dCoa	dCol	dNco	dNca	dFin	dLti	dFnl	dBe	Dac	Poa	Pta	Pda	Nxf	Nef	Ndf	Roe1	Roe6
β_{MKT}	0.08	0.08	0.06	0.02	0.02	-0.03	0.10	0.01	0.10	0.01	0.00	0.09	0.05	0.20	0.23	0.03	-0.02	-0.03
β_{ME}	0.15	0.15	0.07	-0.05	-0.11	0.02	-0.01	-0.01	0.03	0.06	0.19	0.09	0.00	0.23	0.33	-0.04	-0.12	-0.11
$\beta_{\text{I/A}}$	-0.33	-1.01	-1.05	-0.80	-0.87	-0.21	-0.44	-0.46	-1.15	-0.02	-0.75	-0.57	-0.17	-0.89	-0.99	-0.38	0.24	0.25
β_{ROE}	0.16	0.09	0.01	-0.04	-0.03	-0.09	-0.25	-0.16	0.26	0.22	0.00	0.03	-0.16	-0.39	-0.45	-0.23	1.50	1.48
$t_{\beta_{\text{MKT}}}$	3.46	3.94	2.27	0.68	0.68	-0.92	3.08	0.32	4.83	0.36	0.00	3.67	2.22	7.41	7.11	1.15	-0.45	-0.74
$t_{\beta_{\text{ME}}}$	2.79	5.42	2.02	-1.21	-2.93	0.23	-0.23	-0.34	0.81	1.26	5.53	2.30	0.06	5.09	6.22	-1.04	-1.19	-1.26
$t_{\beta_{\text{I/A}}}$	-4.10	-15.43	-15.05	-10.70	-12.13	-1.78	-4.93	-8.17	-14.35	-0.19	-11.76	-8.94	-2.70	-8.03	-7.77	-6.01	2.40	1.79
$t_{\beta_{\text{ROE}}}$	3.37	1.80	0.24	-0.69	-0.50	-1.25	-3.18	-4.24	3.72	2.50	-0.07	0.49	-3.50	-4.37	-3.89	-4.65	21.32	14.04
b	0.09	0.06	0.03	0.00	0.00	-0.06	0.08	0.00	0.06	0.02	-0.01	0.08	0.05	0.16	0.17	0.02	-0.02	-0.03
s	0.22	0.22	0.10	0.03	-0.04	-0.08	0.02	0.04	0.08	0.09	0.23	0.07	0.04	0.27	0.32	0.02	-0.16	-0.14
h	0.04	-0.21	-0.26	0.04	0.03	-0.36	0.01	0.04	-0.30	0.18	-0.23	-0.16	0.11	-0.13	-0.31	0.04	-0.03	-0.05
r	0.34	0.08	-0.12	-0.04	-0.07	-0.39	-0.29	-0.18	0.13	0.35	0.00	-0.13	-0.11	-0.60	-0.83	-0.24	1.58	1.58
c	-0.31	-0.71	-0.71	-0.80	-0.84	0.15	-0.46	-0.45	-0.80	-0.18	-0.43	-0.37	-0.30	-0.68	-0.58	-0.40	0.21	0.23
t_b	4.21	2.66	1.10	-0.05	-0.08	-1.99	2.32	0.00	2.75	0.79	-0.36	2.79	2.07	5.82	6.21	0.59	-0.50	-0.76
t_s	5.82	6.13	3.12	0.65	-1.06	-1.83	0.48	1.23	1.74	2.34	6.75	1.89	1.35	5.29	5.95	0.34	-2.03	-2.08
t_h	0.87	-3.97	-5.05	0.69	0.46	-5.12	0.06	0.62	-5.52	2.54	-4.99	-2.70	2.69	-2.02	-4.76	0.66	-0.26	-0.49
t_r	7.55	1.41	-2.14	-0.58	-1.00	-5.23	-3.15	-3.59	1.48	4.92	0.06	-2.35	-2.25	-6.25	-8.99	-3.60	16.48	15.15
t_c	-4.39	-8.84	-9.07	-8.54	-9.40	1.68	-3.47	-5.94	-6.61	-1.48	-5.81	-3.81	-5.15	-6.21	-5.09	-4.07	1.38	1.28
	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144
	dRoe1	dRoe6	dRoe12	Roa1	Roa6	dRoa1	dRoa6	dRoa12	Cto	Rna ^{q1}	Rna ^{q6}	Rna ^{q12}	Pm ^{q1}	Ato ^{q1}	Ato ^{q6}	Ato ^{q12}	Cto ^{q1}	Cto ^{q6}
β_{MKT}	0.04	0.04	0.00	-0.04	-0.07	0.08	0.07	0.03	0.15	-0.05	-0.05	-0.07	-0.15	0.10	0.09	0.07	0.13	0.12
β_{ME}	-0.05	-0.06	-0.03	-0.12	-0.12	0.01	-0.02	0.01	0.44	-0.19	-0.17	-0.11	-0.36	0.45	0.45	0.44	0.28	0.34
$\beta_{\text{I/A}}$	0.18	0.13	0.04	0.15	0.12	0.23	0.11	0.02	-0.18	-0.02	0.01	-0.07	0.38	-0.25	-0.32	-0.41	0.47	0.43
β_{ROE}	0.54	0.49	0.46	1.40	1.41	0.57	0.52	0.46	0.65	1.28	1.23	1.18	1.17	0.64	0.63	0.61	1.02	1.01
$t_{\beta_{\text{MKT}}}$	1.17	1.55	-0.01	-1.12	-2.06	2.32	2.07	1.19	3.04	-0.99	-1.26	-2.08	-2.72	2.11	2.02	1.78	2.18	2.42
$t_{\beta_{\text{ME}}}$	-1.03	-1.38	-0.86	-1.29	-1.64	0.24	-0.31	0.23	4.30	-1.61	-1.69	-1.45	-3.23	8.70	9.68	9.65	1.81	2.70
$t_{\beta_{\text{I/A}}}$	1.49	1.37	0.58	1.30	0.94	1.85	1.12	0.34	-1.49	-0.16	0.05	-0.47	2.29	-2.46	-3.29	-4.40	2.99	2.82
$t_{\beta_{\text{ROE}}}$	4.79	6.00	9.13	18.72	15.12	4.87	5.53	8.45	6.91	13.12	12.11	10.37	10.15	8.97	9.38	9.75	8.95	9.15

b	-0.01	-0.01	-0.05	-0.03	-0.06	0.03	0.01	-0.02	0.18	-0.04	-0.04	-0.06	-0.13	0.10	0.09	0.07	0.20	0.18
s	-0.22	-0.21	-0.15	-0.16	-0.17	-0.19	-0.19	-0.12	0.58	-0.09	-0.07	-0.02	-0.36	0.44	0.46	0.47	0.37	0.42
h	-0.26	-0.26	-0.21	-0.07	-0.15	-0.25	-0.27	-0.19	-0.10	-0.04	-0.06	-0.06	0.31	-0.57	-0.56	-0.56	0.04	-0.03
r	0.01	0.02	0.07	1.49	1.49	0.00	0.00	0.10	1.17	1.44	1.45	1.38	1.39	0.72	0.75	0.75	1.58	1.56
c	0.31	0.24	0.11	0.16	0.21	0.37	0.27	0.09	0.03	-0.11	-0.05	-0.10	0.03	0.41	0.33	0.24	0.46	0.48
t_b	-0.31	-0.44	-1.63	-0.67	-1.42	0.71	0.31	-0.47	5.67	-1.09	-1.21	-1.92	-2.45	2.27	2.21	1.93	4.60	5.41
t_s	-3.27	-3.84	-3.26	-2.24	-2.42	-2.59	-3.14	-2.25	13.31	-1.46	-1.34	-0.47	-4.55	8.38	9.28	9.56	5.12	7.60
t_h	-2.28	-2.97	-2.79	-0.70	-1.55	-2.19	-2.83	-2.36	-1.61	-0.49	-1.03	-0.94	2.84	-7.56	-7.98	-7.81	0.39	-0.42
t_r	0.08	0.24	1.09	14.66	13.59	0.00	-0.01	1.54	15.96	14.81	17.28	15.26	12.77	8.32	9.70	9.27	13.61	17.26
t_c	2.49	2.28	1.09	1.04	1.14	2.63	2.12	0.79	0.24	-0.67	-0.32	-0.61	0.17	3.43	2.80	2.01	3.66	3.74
	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162
	Cto ^a 12	Gpa	Gla ^q 1	Gla ^q 6	Gla ^q 12	Ope	Ole ^q 1	Ole ^q 6	Ole ^q 12	Opa	Ola ^q 1	Ola ^q 6	Ola ^q 12	Cop	Cla	Cla ^q 1	Cla ^q 6	Cla ^q 12
β_{MKT}	0.11	0.11	0.11	0.10	0.08	0.00	0.00	0.01	-0.01	-0.11	-0.04	-0.06	-0.09	-0.18	-0.14	-0.08	-0.06	-0.08
β_{ME}	0.36	0.25	0.03	0.09	0.15	-0.11	-0.15	-0.15	-0.10	0.04	0.00	0.05	0.06	-0.19	-0.17	-0.09	-0.01	-0.01
$\beta_{\text{I/A}}$	0.35	0.23	0.13	0.12	0.01	0.65	0.66	0.68	0.64	-0.01	-0.20	-0.16	-0.22	0.09	-0.32	0.18	0.24	0.18
β_{ROE}	0.99	0.84	0.98	0.93	0.86	1.15	1.13	1.13	1.11	1.01	1.08	1.10	1.07	0.61	0.52	0.49	0.55	0.54
$t_{\beta_{\text{MKT}}}$	2.30	2.81	1.82	2.21	1.93	-0.04	0.06	0.21	-0.18	-2.54	-1.17	-1.99	-2.62	-4.64	-3.19	-2.80	-2.38	-2.92
$t_{\beta_{\text{ME}}}$	3.36	3.06	0.20	0.75	1.61	-0.83	-0.82	-0.96	-0.69	0.38	0.05	0.67	0.82	-2.07	-1.66	-1.83	-0.28	-0.24
$t_{\beta_{\text{I/A}}}$	2.47	1.72	1.00	0.99	0.06	2.77	3.67	3.44	3.00	-0.02	-1.64	-0.98	-1.12	0.63	-2.04	1.65	1.60	1.25
$t_{\beta_{\text{ROE}}}$	9.73	7.81	8.95	9.10	9.07	6.44	8.10	7.15	6.52	5.61	14.38	10.03	8.19	5.98	4.46	6.90	5.31	5.46
b	0.16	0.14	0.12	0.12	0.10	0.05	0.05	0.06	0.04	-0.09	-0.04	-0.06	-0.09	-0.17	-0.14	-0.07	-0.05	-0.07
s	0.43	0.36	0.20	0.24	0.29	0.00	-0.08	-0.09	-0.04	-0.01	-0.02	0.02	0.03	-0.26	-0.23	-0.11	-0.05	-0.07
h	-0.09	-0.20	-0.08	-0.13	-0.15	0.32	0.40	0.36	0.36	-0.56	-0.39	-0.38	-0.38	-0.46	-0.54	-0.19	-0.17	-0.19
r	1.49	1.43	1.41	1.35	1.26	1.88	1.61	1.62	1.59	1.20	1.05	1.08	1.05	0.61	0.44	0.54	0.55	0.49
c	0.45	0.46	0.04	0.11	0.06	0.32	0.13	0.19	0.15	0.66	0.24	0.24	0.19	0.58	0.28	0.41	0.42	0.39
t_b	5.00	5.12	3.28	4.27	3.74	1.65	1.29	1.78	1.18	-1.84	-0.78	-1.39	-1.91	-4.62	-3.52	-1.98	-1.72	-2.34
t_s	8.35	7.79	3.00	4.42	6.10	0.09	-1.32	-1.78	-0.91	-0.09	-0.32	0.23	0.39	-3.88	-3.26	-2.15	-1.00	-1.46
t_h	-1.25	-2.93	-0.76	-1.64	-2.31	5.51	4.44	5.16	5.66	-4.59	-3.90	-4.00	-3.45	-4.66	-5.40	-2.89	-3.16	-2.98
t_r	17.06	18.38	13.13	16.41	17.07	21.06	14.63	16.59	16.47	5.66	7.43	7.15	6.19	4.22	2.87	5.10	4.70	4.20
t_c	3.18	3.88	0.28	0.86	0.53	2.33	1.06	1.35	0.94	2.91	1.42	1.17	0.82	4.13	1.85	3.45	2.80	2.64
	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
	F	F ^q 1	F ^q 6	F ^q 12	Fp6	O	O ^q 1	G	Sg ^q 12	Oca	Ioca	Adm	gAd	Rdm	Rdm ^q 1	Rdm ^q 6	Rdm ^q 12	Ol
β_{MKT}	-0.29	-0.08	-0.07	-0.07	0.40	0.15	0.06	-0.17	0.10	-0.14	-0.15	-0.08	-0.04	0.11	0.10	0.00	0.02	-0.07
β_{ME}	-0.24	-0.41	-0.30	-0.24	-0.10	0.14	0.37	-0.39	0.12	0.12	0.01	0.13	0.12	0.61	-0.07	0.09	0.21	0.27
$\beta_{\text{I/A}}$	0.38	0.60	0.56	0.55	-0.12	0.30	0.26	0.12	-1.01	-0.10	0.05	1.61	-1.06	0.25	0.64	0.73	0.87	0.29
β_{ROE}	0.69	0.81	0.81	0.80	-1.50	-0.53	-0.78	0.54	0.38	0.09	0.04	0.22	-0.09	-0.43	-1.05	-0.93	-0.70	0.49
$t_{\beta_{\text{MKT}}}$	-5.67	-1.38	-1.42	-1.44	5.27	4.30	1.65	-3.89	3.23	-2.54	-6.07	-1.15	-0.91	1.64	0.94	0.05	0.29	-1.49
$t_{\beta_{\text{ME}}}$	-3.18	-3.20	-3.45	-3.40	-0.54	2.67	4.23	-5.40	2.01	1.33	0.14	0.97	1.37	4.42	-0.34	0.68	1.74	4.02
$t_{\beta_{\text{I/A}}}$	2.49	4.87	4.99	4.27	-0.39	2.54	2.51	0.67	-9.65	-0.46	0.47	8.81	-5.08	0.97	2.47	3.13	3.79	2.28
$t_{\beta_{\text{ROE}}}$	5.72	7.69	7.58	5.87	-6.61	-5.74	-9.17	3.66	4.30	0.59	0.50	1.16	-0.67	-2.41	-3.90	-4.00	-3.32	4.76
b	-0.29	-0.13	-0.13	-0.13	0.43	0.14	0.07	-0.16	0.02	-0.14	-0.16	0.02	-0.06	0.15	0.36	0.24	0.22	-0.03
s	-0.28	-0.39	-0.30	-0.24	0.13	0.16	0.33	-0.39	0.03	0.13	-0.02	0.22	0.12	0.47	0.11	0.21	0.25	0.31
h	0.27	0.37	0.33	0.38	0.53	0.28	0.13	-0.06	-0.45	-0.45	-0.08	0.94	-0.09	-0.30	0.17	0.19	0.13	-0.09
r	0.68	0.75	0.72	0.69	-0.97	-0.55	-0.80	0.63	-0.12	0.25	-0.02	0.92	-0.29	-0.46	-0.22	-0.26	-0.20	0.86
c	0.00	0.00	0.00	-0.06	-0.70	-0.06	0.15	0.12	-0.61	0.37	0.08	0.54	-0.83	0.77	0.69	0.79	0.96	0.42
t_b	-5.08	-2.48	-2.57	-2.56	4.36	4.06	1.84	-3.71	0.63	-2.71	-6.15	0.39	-1.42	2.64	2.82	2.17	2.11	-0.95
t_s	-3.02	-4.61	-4.26	-3.31	0.84	3.33	5.06	-5.22	0.64	1.60	-0.55	2.74	1.58	4.84	0.52	1.19	1.66	5.58
t_h	2.24	3.40	4.07	4.28	2.16	3.47	1.53	-0.57	-4.92	-3.94	-1.27	9.63	-1.04	-2.32	0.64	0.82	0.65	-1.13
t_r	5.30	5.25	6.91	6.72	-3.29	-4.88	-7.91	3.95	-1.54	1.86	-0.19	8.68	-2.57	-2.03	-0.84	-1.01	-0.80	8.92
t_c	0.01	-0.02	0.02	-0.29	-1.71	-0.53	1.28	0.69	-3.99	1.96	0.73	3.80	-4.40	2.83	1.36	1.83	2.46	3.06

	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198
	Ol ^{q1}	Ol ^{q6}	Ol ^{q12}	Hn	Parc	dSi	Rer	Eprd	Ala	Alm ^{q1}	Alm ^{q6}	Alm ^{q12}	R _a ¹	R _n ¹	R _a ^[2,5]	R _n ^[2,5]	R _a ^[6,10]	R _n ^[6,10]
β_{MKT}	-0.02	-0.04	-0.05	0.07	-0.02	-0.05	0.03	0.25	0.21	-0.04	0.00	0.00	0.17	-0.16	0.05	0.20	0.01	0.07
β_{ME}	0.26	0.30	0.29	0.09	0.08	0.02	-0.08	0.25	0.29	0.22	0.25	0.28	-0.09	0.61	-0.19	-0.01	-0.05	-0.03
$\beta_{\text{I/A}}$	0.11	0.15	0.13	-1.14	0.10	-0.10	0.10	0.37	-1.40	1.45	1.48	1.42	-0.12	-0.26	-0.27	-1.31	-0.14	-0.67
β_{ROE}	0.48	0.47	0.47	-0.02	0.01	0.08	0.02	-0.74	-0.10	0.03	0.13	0.18	0.14	1.10	-0.07	0.18	-0.18	-0.31
$t_{\beta_{\text{MKT}}}$	-0.35	-0.86	-1.09	2.45	-0.78	-1.66	0.98	5.06	4.97	-0.68	-0.09	-0.08	3.92	-1.44	1.25	3.39	0.16	1.62
$t_{\beta_{\text{ME}}}$	3.90	4.20	4.08	2.35	1.52	0.43	-2.01	2.98	3.56	1.45	2.16	2.77	-0.98	1.97	-2.45	-0.06	-1.12	-0.38
$t_{\beta_{\text{I/A}}}$	0.91	1.25	1.07	-15.63	1.27	-1.63	1.45	3.48	-13.31	8.64	9.08	10.48	-0.71	-0.69	-2.93	-9.04	-1.24	-5.66
$t_{\beta_{\text{ROE}}}$	4.59	4.74	4.69	-0.27	0.21	1.43	0.39	-8.22	-0.95	0.21	1.19	1.81	0.99	3.75	-0.95	1.31	-1.98	-3.11
b	0.02	0.00	-0.02	0.03	-0.02	-0.07	0.02	0.28	0.16	0.07	0.09	0.08	0.14	-0.29	0.05	0.13	0.00	0.07
s	0.30	0.34	0.32	0.12	0.06	0.04	-0.10	0.33	0.29	0.34	0.32	0.32	-0.10	0.20	-0.13	-0.08	-0.01	-0.03
h	-0.13	-0.09	-0.12	-0.25	-0.07	0.16	0.03	0.47	-0.68	0.96	0.84	0.78	-0.17	-0.99	0.07	-0.82	0.04	-0.46
r	0.82	0.81	0.78	-0.20	0.06	0.07	0.00	-0.54	-0.40	0.49	0.46	0.43	-0.01	-0.24	0.04	-0.30	-0.14	-0.43
c	0.28	0.28	0.28	-0.82	0.18	-0.33	0.00	-0.06	-0.60	0.49	0.61	0.61	-0.04	0.58	-0.36	-0.44	-0.19	-0.06
t_b	0.39	-0.09	-0.47	1.10	-0.60	-2.66	0.70	5.67	5.44	1.46	2.12	2.30	2.85	-2.48	1.23	2.96	0.05	1.90
t_s	4.67	5.76	5.62	2.81	1.27	0.99	-2.54	5.05	5.97	4.57	5.08	5.80	-1.07	1.01	-1.80	-0.98	-0.31	-0.41
t_h	-1.47	-1.13	-1.47	-4.79	-0.92	3.04	0.66	4.62	-10.57	9.03	9.40	10.45	-1.38	-3.00	0.81	-7.42	0.58	-5.86
t_r	9.43	9.72	8.85	-2.74	0.76	1.29	-0.08	-4.50	-4.76	4.86	5.00	5.89	-0.06	-0.59	0.40	-2.33	-1.97	-4.84
t_c	2.18	2.13	2.07	-7.92	1.42	-3.70	-0.03	-0.39	-4.85	3.11	3.63	4.27	-0.22	1.27	-2.89	-2.47	-1.65	-0.46
	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216
	R _a ^[11,15]	R _a ^[16,20]	Ivc1	Ivq1	Sv1	Srev	Dtv1	Dtv6	Dtv12	Ami6	Ami12	Lm ⁶ 1	Lm ⁶ 6	Lm ⁶ 12	Lm ¹² 1	Lm ¹² 6	Lm ¹² 12	Mdr1
β_{MKT}	-0.02	-0.05	0.56	0.54	0.00	-0.29	0.26	0.27	0.26	-0.07	-0.06	-0.47	-0.49	-0.47	-0.48	-0.46	-0.45	0.49
β_{ME}	0.00	0.01	0.80	0.79	0.20	0.04	-0.65	-0.70	-0.73	0.95	0.92	-0.22	-0.15	-0.11	-0.15	-0.12	-0.10	0.66
$\beta_{\text{I/A}}$	0.07	-0.08	-1.36	-1.34	-0.21	0.17	-0.74	-0.74	-0.73	0.20	0.21	1.17	1.14	1.15	1.07	1.07	1.04	-1.20
β_{ROE}	0.09	-0.06	-0.89	-0.85	-0.54	0.19	0.00	-0.06	-0.09	-0.25	-0.22	0.39	0.46	0.54	0.40	0.47	0.51	-0.78
$t_{\beta_{\text{MKT}}}$	-0.68	-1.49	9.48	9.66	0.02	-2.95	6.82	7.40	7.12	-3.06	-2.27	-8.00	-9.14	-8.82	-9.34	-8.95	-8.65	7.84
$t_{\beta_{\text{ME}}}$	-0.07	0.11	6.48	6.79	1.21	0.18	-7.00	-9.31	-10.94	24.75	19.83	-1.87	-1.67	-1.39	-1.82	-1.46	-1.32	4.73
$t_{\beta_{\text{I/A}}}$	0.89	-1.10	-7.46	-7.62	-1.16	0.62	-8.60	-9.05	-9.79	4.22	3.42	8.72	8.80	7.90	9.11	8.70	8.13	-6.40
$t_{\beta_{\text{ROE}}}$	1.55	-0.88	-5.34	-5.38	-4.23	0.97	0.07	-0.71	-1.10	-6.11	-4.53	2.97	3.55	3.84	3.31	3.74	3.92	-4.60
b	-0.03	-0.05	0.49	0.48	0.00	-0.30	0.22	0.23	0.23	-0.06	-0.05	-0.41	-0.43	-0.42	-0.44	-0.42	-0.42	0.43
s	-0.02	0.03	0.79	0.78	0.15	-0.06	-0.70	-0.72	-0.72	0.97	0.94	-0.16	-0.14	-0.14	-0.14	-0.13	-0.14	0.64
h	0.00	0.03	-0.64	-0.60	0.10	-0.32	-0.43	-0.42	-0.40	0.15	0.16	0.55	0.49	0.46	0.43	0.40	0.39	-0.63
r	0.03	0.03	-1.30	-1.26	-0.66	-0.12	-0.33	-0.31	-0.28	-0.17	-0.15	0.89	0.83	0.81	0.73	0.72	0.68	-1.22
c	0.05	-0.10	-0.66	-0.67	-0.27	0.52	-0.33	-0.34	-0.34	0.09	0.08	0.60	0.63	0.63	0.59	0.62	0.56	-0.54
t_b	-0.82	-1.34	9.72	9.66	0.02	-3.55	8.08	8.82	8.32	-2.90	-2.21	-8.74	-9.39	-8.79	-9.71	-8.99	-8.72	7.88
t_s	-0.42	0.61	14.17	14.09	1.25	-0.37	-12.79	-15.39	-16.79	21.49	20.90	-2.26	-2.18	-2.28	-2.21	-2.07	-2.27	10.64
t_h	-0.03	0.36	-5.43	-5.40	0.70	-1.39	-6.18	-7.02	-7.01	2.58	2.83	5.46	5.26	4.98	4.77	4.37	4.21	-4.83
t_r	0.38	0.47	-13.48	-13.34	-3.98	-0.41	-5.25	-6.17	-5.64	-3.92	-3.15	10.27	11.11	9.29	9.98	9.61	8.02	-12.39
t_c	0.59	-1.05	-4.30	-4.71	-1.43	1.93	-3.70	-3.87	-3.68	1.22	1.00	4.43	4.60	4.04	4.30	4.21	3.41	-3.59

A Constructing the q -factors

Following Hou, Xue, and Zhang (2015), we construct the size, investment, and ROE factors from a triple $2 \times 3 \times 3$ sort on size, investment-to-assets (I/A), and ROE from January 1972 onward. Size is the market equity, which is stock price per share times shares outstanding from CRSP, I/A is the annual change in total assets (Compustat annual item AT) divided by one-year-lagged total assets, and ROE is income before extraordinary items (Compustat quarterly item IBQ) divided by one-quarter-lagged book equity.³ At the end of June of each year t , we use the median NYSE size to split NYSE, Amex, and NASDAQ stocks into two groups, small and big. Independently, at the end of June of year t , we break stocks into three I/A groups using the NYSE breakpoints for the low 30%, middle 40%, and high 30% of the ranked values of I/A for the fiscal year ending in calendar year $t - 1$. Also, independently, at the beginning of each month, we sort all stocks into three groups based on the NYSE breakpoints for the low 30%, middle 40%, and high 30% of the ranked values of ROE. Earnings data in Compustat quarterly files are used in the months immediately after the most recent public quarterly earnings announcement dates (item RDQ). For a firm to enter the factor construction, we require the end of the fiscal quarter that corresponds to its announced earnings to be within six months prior to the portfolio formation month.

Taking the intersection of the two size, three I/A, and three ROE groups, we form 18 benchmark portfolios. Monthly value-weighted portfolio returns are calculated for the current month, and the portfolios are rebalanced monthly. The size factor is the difference (small-minus-big), each month, between the simple average of the returns on the nine small size portfolios and the simple average of the returns on the nine big size portfolios. The investment factor is the difference (low-minus-high), each month, between the simple average of the returns on the six low I/A portfolios and the simple average of the returns on the six high I/A portfolios. Finally, the ROE factor is the difference (high-minus-low), each month, between the simple average of the returns on the six high ROE portfolios and the simple average of the returns on the six low ROE portfolios.

We follow Hou, Xue, and Zhang (2017) to extend the sample backward to January 1967. Prior to January 1972, we use the most recent quarterly earnings from the fiscal quarter ending at least four months prior to the portfolio formation month. To expand the coverage for quarterly book equity, we use book equity from Compustat annual files and impute quarterly book equity with clean surplus accounting. We first use quarterly book equity from Compustat quarterly files whenever available, and then supplement the coverage for the fourth fiscal quarter with book equity from Compustat annual files.⁴ If neither estimate is available, we apply the clean surplus relation

³Book equity is shareholders' equity, plus balance sheet deferred taxes and investment tax credit (Compustat quarterly item TXDITCQ) if available, minus the book value of preferred stock (item PSTKQ). Depending on availability, we use stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the book value of preferred stock, or total assets (item ATQ) minus total liabilities (item LTQ) in that order as shareholders' equity.

⁴Following Davis, Fama, and French (2000), we measure annual book equity as stockholders' book equity, plus balance sheet deferred taxes and investment tax credit (Compustat annual item TXDITC) if available, minus the book value of preferred stock. Stockholders' equity is the value reported by Compustat (item SEQ), if available.

to impute the book equity. We first backward impute the beginning-of-quarter book equity as the end-of-quarter book equity minus quarterly earnings plus quarterly dividends.⁵ Because we impose a four-month lag between earnings and the holding period month (and the book equity in the denominator of ROE is one-quarter-lagged relative to earnings), all the Compustat data in the backward imputation are at least four-month lagged relative to the portfolio formation month.

If data are unavailable for the backward imputation, we impute the book equity for quarter t forward based on book equity from prior quarters. Let BEQ_{t-j} , with $1 \leq j \leq 4$, denote the latest available quarterly book equity as of quarter t , and $IBQ_{t-j+1,t}$ and $DVQ_{t-j+1,t}$ be the sum of quarterly earnings and quarterly dividends from quarter $t-j+1$ to t , respectively. BEQ_t can then be imputed as $BEQ_{t-j} + IBQ_{t-j+1,t} - DVQ_{t-j+1,t}$. We do not use prior book equity from more than four quarters ago to reduce imputation errors ($1 \leq j \leq 4$). We start the sample in January 1967 to ensure that all the 18 benchmark portfolios from the triple sort on size, I/A, and ROE have at least ten firms.

Following Beaver, McNichols, and Price (2007), we adjust monthly stock returns for delisting returns by compounding returns in the month before delisting with delisting returns from CRSP. When a delisting return is missing, we replace it with the mean of available delisting returns of the same delisting type and stock exchange in the prior 60 months (Hou, Xue, and Zhang 2017).

B Insignificant Anomalies with All-but-micro Breakpoints and Equal-weighted Returns

There are in total 221 insignificant anomalies. Eight out of 57 anomalies in the momentum category are insignificant. Standardized unexpected earnings (Sue), prior 11-month returns, revenue surprises (Rs), tax expense surprises (Tes), and the number of consecutive quarters with earnings increases (Nei) are all insignificant at the 12-month horizon. In the value-versus-growth category, 30 out of 68 anomalies are insignificant. Debt-to-market (Dm), assets-to-market (Am), and dividend yield (Dp) are insignificant in annual sorts and monthly sorts at all horizons, and analysts' earnings forecasts-to-price (Efp), enterprise book-to-price (Ebp), and net debt-to-price (Ndp) are insignificant in all monthly sorts. In the investment category, only two out of 38 anomalies are insignificant, including change in non-current operating liabilities (dNcl) and change in short-term investments (dSti). In the profitability category, 31 out of 78 anomalies are insignificant. In particular, Z-score and book leverage (Bl) are insignificant in both annual and monthly sorts.

Otherwise, we use the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption value (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock.

⁵Quarterly dividends are zero if dividends per share (item DVPSXQ) are zero. Otherwise, total dividends are dividends per share times beginning-of-quarter shares outstanding adjusted for stock splits during the quarter. Shares outstanding are from Compustat (quarterly item CSHOQ supplemented with annual item CSHO for fiscal quarter four) or CRSP (item SHROUT), and the share adjustment factor is from Compustat (quarterly item AJEXQ supplemented with annual item AJEX for fiscal quarter four) or CRSP (item CFACSHR).

In the intangibles category, 71 out of 100 anomalies are insignificant. R&D-to-sales (Rds), firm age (Age), analysts coverage (Ana), asset tangibility (Tan), cash flow volatility (Vcf), asset liquidity scaled by book assets (Ala), cash-to-assets (Cta), dispersion of analysts' earnings forecasts (Dis), dispersion in analysts' long-term growth forecasts (Dlg), and disparity between short- and long-term earnings growth forecasts (Dls) are insignificant in monthly sorts at all horizons. Corporate governance (Gind) and accrual quality (Acq) are insignificant in annual sorts.

Hou, Xue, and Zhang (2017) show that 89 out of 96 trading frictions variables (or 93%) are insignificant with NYSE breakpoints and value-weights. Similarly, 80 (or 83%) are insignificant with all-but-micro breakpoints and equal-weights, including total volatility (Tv), all versions of idiosyncratic volatilities (Iv, Ivc, Ivff, and Ivq), share turnover (Tur) and its coefficient of variation (Cvt), the coefficient of variation for dollar trading volume (Cvd), share price (Pps), prior 1-month turnover-adjusted number of zero daily trading volume (Lm^1), coskewness (Cs), downside beta (β^-), tail risk (Tail), all versions of liquidity betas (illiquidity-illiquidity, β^{lcc} , return-illiquidity, β^{lrc} , illiquidity-return, β^{lcr}), two versions of the bid-ask spread (Shl and Sba), and leverage beta (β^{Lev}). Total skewness (Ts), different versions of idiosyncratic skewness (Isc, Isff, and Isq), and maximum daily return (Mdr) are also mostly insignificant. In particular, all nine Acharya-Pedersen (2005) liquidity betas are insignificant. The average returns of their high-minus-low deciles vary from 0.01% to 0.17%, all of which are within 1.5 standard errors from zero. Similarly, the Adrian-Etula-Muir (2014) leverage beta is also insignificant. Across the 1-, 6-, and 12-month horizons, the high-minus-low decile earns on average 0.32%, 0.27%, and 0.24% ($t = 1.62, 1.37, \text{ and } 1.24$), respectively.

Table A1 : List of Anomaly Variables

The anomalies are grouped into six categories: (i) momentum; (ii) value-versus-growth; (iii) investment; (iv) profitability; (v) intangibles; and (vi) trading frictions. The number in parenthesis in a panel's title is the number of anomalies in the category. For each anomaly variable, we list its symbol, brief description, and its academic source. Hou, Xue, and Zhang (2017) detail variable definition and portfolio construction.

Panel A: Momentum (57)			
Sue1	Earnings surprise (1-month holding period), Foster, Olsen, and Shevlin (1984)	Sue6	Earnings surprise (6-month holding period), Foster, Olsen, and Shevlin (1984)
Sue12	Earnings surprise (12-month holding period), Foster, Olsen, and Shevlin (1984)	Abr1	Cumulative abnormal stock returns around earnings announcements (1-month holding period), Chan, Jegadeesh, and Lakonishok (1996)
Abr6	Cumulative abnormal stock returns around earnings announcements (6-month holding period), Chan, Jegadeesh, and Lakonishok (1996)	Abr12	Cumulative abnormal stock returns around earnings announcements (12-month holding period), Chan, Jegadeesh, and Lakonishok (1996)
Re1	Revisions in analysts' earnings forecasts (1-month holding period), Chan, Jegadeesh, and Lakonishok (1996)	Re6	Revisions in analysts' earnings forecasts (6-month holding period), Chan, Jegadeesh, and Lakonishok (1996)
Re12	Revisions in analysts' earnings forecasts (12-month holding period), Chan, Jegadeesh, and Lakonishok (1996)	R ⁶ 1	Price momentum (6-month prior returns, 1-month holding period), Jegadeesh and Titman (1993)
R ⁶ 6	Price momentum (6-month prior returns, 6-month holding period), Jegadeesh and Titman (1993)	R ⁶ 12	Price momentum (6-month prior returns, 12-month holding period), Jegadeesh and Titman (1993)
R ¹¹ 1	Price momentum (11-month prior returns, 1-month holding period), Fama and French (1996)	R ¹¹ 6	Price momentum (11-month prior returns, 6-month holding period), Fama and French (1996)
R ¹¹ 12	Price momentum, (11-month prior returns, 12-month holding period), Fama and French (1996)	Im1	Industry momentum, (1-month holding period), Moskowitz and Grinblatt (1999)
Im6	Industry momentum (6-month holding period), Moskowitz and Grinblatt (1999)	Im12	Industry momentum (12-month holding period), Moskowitz and Grinblatt (1999)
Rs1	Revenue surprise (1-month holding period), Jegadeesh and Livnat (2006)	Rs6	Revenue surprise (6-month holding period), Jegadeesh and Livnat (2006)
Rs12	Revenue surprise (12-month holding period), Jegadeesh and Livnat (2006)	Tes1	Tax expense surprise (1-month holding period), Thomas and Zhang (2011)
Tes6	Tax expense surprise (6-month holding period), Thomas and Zhang (2011)	Tes12	Tax expense surprise (12-month holding period), Thomas and Zhang (2011)
dEf1	Analysts' forecast change (1-month hold period), Hawkins, Chamberlin, and Daniel (1984)	dEf6	Analysts' forecast change (6-month hold period), Hawkins, Chamberlin, and Daniel (1984)
dEf12	Analysts' forecast change (12-month hold period), Hawkins, Chamberlin, and Daniel (1984)	Nei1	# of consecutive quarters with earnings increases (1-month holding period), Barth, Elliott, and Finn (1999)
Nei6	# consecutive quarters with earnings increases (6-month holding period), Barth, Elliott, and Finn (1999)	Nei12	# consecutive quarters with earnings increases (12-month holding period), Barth, Elliott, and Finn (1999)
52w1	52-week high (1-month holding period), George and Hwang (2004)	52w6	52-week high (6-month holding period), George and Hwang (2004)
52w12	52-week high (12-month holding period), George and Hwang (2004)	ε ⁶ 1	Six-month residual momentum (1-month holding period), Blitz, Huij, and Martens (2011)
ε ⁶ 6	Six-month residual momentum (6-month holding period), Blitz, Huij, and Martens (2011)	ε ⁶ 12	Six-month residual momentum (12-month holding period), Blitz, Huij, and Martens (2011)

$\epsilon^{11}1$	11-month residual momentum (1-month holding period), Blitz, Huij, and Martens (2011)	$\epsilon^{11}6$	11-month residual momentum (6-month holding period), Blitz, Huij, and Martens (2011)
$\epsilon^{11}12$	11-month residual momentum (12-month holding period), Blitz, Huij, and Martens (2011)	Sm1	Segment momentum (1-month holding period), Cohen and Lou (2012)
Sm6	Segment momentum (6-month holding period), Cohen and Lou (2012)	Sm12	Segment momentum (12-month holding period), Cohen and Lou (2012)
Ilr1	Industry lead-lag effect in prior returns (1-month holding period), Hou (2007)	Ilr6	Industry lead-lag effect in prior returns (6-month holding period), Hou (2007)
Ilr12	Industry lead-lag effect in prior returns (12-month holding period), Hou (2007)	Ile1	Industry lead-lag effect in earnings surprises (1-month holding period), Hou (2007)
Ile6	Industry lead-lag effect in earnings surprises (6-month holding period), Hou (2007)	Ile12	Industry lead-lag effect in earnings surprises (12-month holding period), Hou (2007)
Cm1	Customer momentum (1-month holding period), Cohen and Frazzini (2008)	Cm6	Customer momentum (6-month holding period), Cohen and Frazzini (2008)
Cm12	Customer momentum (12-month holding period), Cohen and Frazzini (2008)	Sim1	Supplier industries momentum (1-month holding period), Menzly and Ozbas (2010)
Sim6	Supplier industries momentum (6-month holding period), Menzly and Ozbas (2010)	Sim12	Supplier industries momentum (12-month holding period), Menzly and Ozbas (2010)
Cim1	Customer industries momentum (1-month holding period), Menzly and Ozbas (2010)	Cim6	Customer industries momentum (6-month holding period), Menzly and Ozbas (2010)
Cim12	Customer industries momentum (12-month holding period), Menzly and Ozbas (2010)		

Panel B: Value-versus-growth (68)

Bm	Book-to-market equity, Rosenberg, Reid, and Lanstein (1985)	Bmj	Book-to-June-end market equity, Asness and Frazzini (2013)
Bm ^q 1	Quarterly Book-to-market equity (1-month holding period)	Bm ^q 6	Quarterly Book-to-market equity (6-month holding period)
Bm ^q 12	Quarterly Book-to-market equity (12-month holding period)	Dm	Debt-to-market, Bhandari (1988)
Dm ^q 1	Quarterly Debt-to-market (1-month holding period)	Dm ^q 6	Quarterly Debt-to-market (6-month holding period)
Dm ^q 12	Quarterly Debt-to-market (12-month holding period)	Am	Assets-to-market, Fama and French (1992)
Am ^q 1	Quarterly Assets-to-market (1-month holding period)	Am ^q 6	Quarterly Assets-to-market (6-month holding period)
Am ^q 12	Quarterly Assets-to-market (12-month holding period)	Rev1	Reversal (1-month holding period) De Bondt and Thaler (1985)
Rev6	Reversal (6-month holding period), De Bondt and Thaler (1985)	Rev12	Reversal (12-month holding period) De Bondt and Thaler (1985)
Ep	Earnings-to-price, Basu (1983)	Ep ^q 1	Quarterly Earnings-to-price (1-month holding period)
Ep ^q 6	Quarterly Earnings-to-price (6-month holding period)	Ep ^q 12	Quarterly Earnings-to-price (12-month holding period)
Efp1	Analysts' earnings forecasts-to-price (1-month holding period), Elgers, Lo, and Pfeiffer (2001)	Efp6	Analysts' earnings forecasts-to-price (6-month holding period) Elgers, Lo, and Pfeiffer (2001)
Efp12	Analysts' earnings forecasts-to-price (12-month holding period), Elgers, Lo, and Pfeiffer (2001)	Cp	Cash flow-to-price, Lakonishok, Shleifer, and Vishny (1994)
Cp ^q 1	Quarterly Cash flow-to-price (1-month holding period)	Cp ^q 6	Quarterly Cash flow-to-price (6-month holding period)
Cp ^q 12	Quarterly Cash flow-to-price (12-month holding period)	Dp	Dividend yield, Litzenberger and Ramaswamy (1979)

Dp ^{q1}	Quarterly Dividend yield (1-month holding period)	Dp ^{q6}	Quarterly Dividend yield (6-month holding period)
Dp ^{q12}	Quarterly Dividend yield (12-month holding period)	Op	Payout yield, Boudoukh, Michaely, Richardson, and Roberts (2007)
Op ^{q1}	Quarterly Payout yield (1-month holding period)	Op ^{q6}	Quarterly Payout yield (6-month holding period)
Op ^{q12}	Quarterly Payout yield (12-month holding period)	Nop	Net payout yield, Boudoukh, Michaely, Richardson, and Roberts (2007)
Nop ^{q1}	Quarterly Net payout yield (1-month holding period)	Nop ^{q6}	Quarterly Net payout yield (6-month holding period)
Nop ^{q12}	Quarterly Net payout yield (12-month holding period)	Sr	Five-year sales growth rank, Lakonishok, Shleifer, and Vishny (1994)
Sg	Annual sales growth, Lakonishok, Shleifer, and Vishny (1994)	Em	Enterprise multiple, Loughran and Wellman (2011)
Em ^{q1}	Quarterly Enterprise multiple (1-month holding period)	Em ^{q6}	Quarterly Enterprise multiple (6-month holding period)
Em ^{q12}	Quarterly Enterprise multiple (12-month holding period)	Sp	Sales-to-price, Barbee, Mukherji, and Raines (1996)
Sp ^{q1}	Quarterly Sales-to-price (1-month holding period)	Sp ^{q6}	Quarterly Sales-to-price (6-month holding period)
Sp ^{q12}	Quarterly Sales-to-price (12-month holding period)	Ocp	Operating cash flow-to-price, Desai, Rajgopal, and Venkatachalam (2004)
Ocp ^{q1}	Quarterly Operating cash flow-to-price (1-month holding period)	Ocp ^{q6}	Quarterly Operating cash flow-to-price (6-month holding period)
Ocp ^{q12}	Quarterly Operating cash flow-to-price (12-month holding period)	Ir	Intangible return, Daniel and Titman (2006)
Vhp	Intrinsic value-to-market, Frankel and Lee (1998)	Vfp	Analysts-based intrinsic value-to-market, Frankel and Lee (1998)
Ebp	Enterprise book-to-price Penman, Richardson, and Tuna (2007)	Ebp ^{q1}	Quarterly enterprise book-to-price (1-month holding period)
Ebp ^{q6}	Quarterly enterprise book-to-price (6-month holding period)	Ebp ^{q12}	Quarterly enterprise book-to-price (12-month holding period)
Ndp	Net debt-to-price Penman, Richardson, and Tuna (2007)	Ndp ^{q1}	Quarterly net debt-to-price (1-month holding period)
Ndp ^{q6}	Quarterly net debt-to-price (6-month holding period)	Ndp ^{q12}	Quarterly net debt-to-price (12-month holding period)
Dur	Equity duration, Dechow, Sloan, and Soliman (2004)	Ltg1	Long-term growth forecasts of analysts (1-month holding period), La Porta (1996)
Ltg6	Long-term growth forecasts of analysts (6-month holding period), La Porta (1996)	Ltg12	Long-term growth forecasts of analysts (12-month holding period), La Porta (1996)

Panel C: Investment (38)

Aci	Abnormal corporate investment, Titman, Wei, and Xie (2004)	I/A	Investment-to-assets, Cooper, Gulen, and Schill (2008)
Ia ^{q1}	Quarterly Investment-to-assets (1-month holding period)	Ia ^{q6}	Quarterly Investment-to-assets (6-month holding period)
Ia ^{q12}	Quarterly Investment-to-assets (12-month holding period)	dPia	Changes in PPE and inventory/assets, Lyandres, Sun, and Zhang (2008)
Noa	Net operating assets, Hirshleifer, Hou, Teoh, and Zhang (2004)	dNoa	Changes in net operating assets, Hou, Xue, and Zhang (2015)
dLno	Change in long-term net operating assets, Fairfield, Whisenant, and Yohn (2003)	Ig	Investment growth, Xing (2008)
2Ig	Two-year investment growth, Anderson and Garcia-Feijoo (2006)	3Ig	Three-year investment growth, Anderson and Garcia-Feijoo (2006)
Nsi	Net stock issues, Pontiff and Woodgate (2008)	dIi	% change in investment – % change in industry investment, Abarbanell and Bushee (1998)
Cei	Composite equity issuance, Daniel and Titman (2006)	Cdi	Composite debt issuance, Lyandres, Sun, and Zhang (2008)

Ivg	Inventory growth, Belo and Lin (2011)	Ivc	Inventory changes, Thomas and Zhang (2002)
Oa	Operating accruals, Sloan (1996)	Ta	Total accruals, Richardson, Sloan, Soliman, and Tuna (2005)
dWc	Change in net non-cash working capital, Richardson, Sloan, Soliman, and Tuna (2005)	dCoa	Change in current operating assets, Richardson, Sloan, Soliman, and Tuna (2005)
dCol	Change in current operating liabilities, Richardson, Sloan, Soliman, and Tuna (2005)	dNco	Change in net non-current operating assets, Richardson, Sloan, Soliman, and Tuna (2005)
dNca	Change in non-current operating assets, Richardson, Sloan, Soliman, and Tuna (2005)	dNcl	Change in non-current operating liabilities, Richardson, Sloan, Soliman, and Tuna (2005)
dFin	Change in net financial assets, Richardson, Sloan, Soliman, and Tuna (2005)	dSti	Change in short-term investments, Richardson, Sloan, Soliman, and Tuna (2005)
dLti	Change in long-term investments, Richardson, Sloan, Soliman, and Tuna (2005)	dFnl	Change in financial liabilities, Richardson, Sloan, Soliman, and Tuna (2005)
dBe	Change in common equity, Richardson, Sloan, Soliman, and Tuna (2005)	Dac	Discretionary accruals, Xie (2001)
Poa	Percent operating accruals, Hafzalla, Lundholm, and Van Winkle (2011)	Pta	Percent total accruals, Hafzalla, Lundholm, and Van Winkle (2011)
Pda	Percent discretionary accruals	Nxf	Net external financing, Bradshaw, Richardson, and Sloan (2006)
Nef	Net equity financing, Bradshaw, Richardson, and Sloan (2006)	Ndf	Net debt financing, Bradshaw, Richardson, and Sloan (2006)

Panel D: Profitability (78)

Roe1	Return on equity (1-month holding period), Hou, Xue, and Zhang (2015)	Roe6	Return on equity (6-month holding period), Hou, Xue, and Zhang (2015)
Roe12	Return on equity (12-month holding period), Hou, Xue, and Zhang (2015)	dRoe1	Change in Roe (1-month holding period),
dRoe6	Change in Roe (6-month holding period)	dRoe12	Change in Roe (12-month holding period)
Roa1	Return on assets (1-month holding period), Balakrishnan, Bartov, and Faurel (2010)	Roa6	Return on assets (6-month holding period), Balakrishnan, Bartov, and Faurel (2010)
Roa12	Return on assets (12-month holding period), Balakrishnan, Bartov, and Faurel (2010)	dRoa1	Change in Roa (1-month holding period)
dRoa6	Change in Roa (6-month holding period)	dRoa12	Change in Roa (12-month holding period)
Rna	Return on net operating assets, Soliman (2008)	Pm	Profit margin, Soliman (2008)
Ato	Asset turnover, Soliman (2008)	Cto	Capital turnover, Haugen and Baker (1996)
Rna ^{q1}	Quarterly return on net operating assets (1-month holding period)	Rna ^{q6}	Quarterly return on net operating assets (6-month holding period)
Rna ^{q12}	Quarterly return on net operating assets (12-month holding period)	Pm ^{q1}	Quarterly profit margin (1-month holding period)
Pm ^{q6}	Quarterly profit margin (6-month holding period)	Pm ^{q12}	Quarterly profit margin (12-month holding period)
Ato ^{q1}	Quarterly asset turnover (1-month holding period)	Ato ^{q6}	Quarterly asset turnover (6-month holding period)
Ato ^{q12}	Quarterly asset turnover (12-month holding period)	Cto ^{q1}	Quarterly capital turnover (1-month holding period)
Cto ^{q6}	Quarterly capital turnover (6-month holding period)	Cto ^{q12}	Quarterly capital turnover (12-month holding period)
Gpa	Gross profits-to-assets, Novy-Marx (2013)	Gla	Gross profits-to-lagged assets
Gla ^{q1}	Gross profits-to-lagged assets (1-month holding period)	Gla ^{q6}	Gross profits-to-lagged assets (6-month holding period)
Gla ^{q12}	Gross profits-to-lagged assets (12-month holding period)	Ope	Operating profits-to-equity, Fama and French (2015)
Ole	Operating profits-to-lagged equity	Ole ^{q1}	Operating profits-to-lagged equity (1-month holding period)
Ole ^{q6}	Operating profits-to-lagged equity (6-month holding period)	Ole ^{q12}	Operating profits-to-lagged equity (12-month holding period)

Opa	Operating profits-to-assets, Ball, Gerakos, Linnainmaa, and Nikolaev (2015a)	Ola	Operating profits-to-lagged assets
Ola ^{q1}	Operating profits-to-lagged assets (1-month holding period)	Ola ^{q6}	Operating profits-to-lagged assets (6-month holding period)
Ola ^{q12}	Operating profits-to-lagged assets (12-month holding period)	Cop	Cash-based operating profitability, Ball, Gerakos, Linnainmaa, and Nikolaev (2015b)
Cla	Cash-based operating profits-to-lagged assets	Cla ^{q1}	Cash-based operating profits-to-lagged assets (1-month holding period)
Cla ^{q6}	Cash-based operating profits-to-lagged assets (6-month holding period)	Cla ^{q12}	Cash-based operating profits-to-lagged assets (12-month holding period)
F	Fundamental (F) score, Piotroski (2000)	F ^{q1}	Quarterly F-score (1-month holding period)
F ^{q6}	Quarterly F-score (6-month holding period)	F ^{q12}	Quarterly F-score (12-month holding period)
Fp1	Failure probability (1-month holding period), Campbell, Hilscher, and Szilagyi (2008)	Fp6	Failure probability (6-month holding period) Campbell, Hilscher, and Szilagyi (2008)
Fp12	Failure probability (12-month holding period), Campbell, Hilscher, and Szilagyi (2008)	O	O-score, Dichev (1998)
O ^{q1}	Quarterly O-score (1-month holding period)	O ^{q6}	Quarterly O-score (6-month holding period)
O ^{q12}	Quarterly O-score (12-month holding period)	Z	Z-score, Dichev (1998)
Z ^{q1}	Quarterly Z-score (1-month holding period)	Z ^{q6}	Quarterly Z-score (6-month holding period)
Z ^{q12}	Quarterly Z-score (12-month holding period)	G	Growth (G) score, Mohanram (2005)
Cr1	Credit ratings (1-month holding period) Avramov, Chordia, Jostova, and Philipov (2009)	Cr6	Credit ratings (6-month holding period) Avramov, Chordia, Jostova, and Philipov (2009)
Cr12	Credit ratings (12-month holding period) Avramov, Chordia, Jostova, and Philipov (2009)	Tbi	Taxable income-to-book income, Green, Hand, and Zhang (2013)
Tbi ^{q1}	Quarterly taxable income-to-book income (1-month holding period)	Tbi ^{q6}	Quarterly taxable income-to-book income (6-month holding period)
Tbi ^{q12}	Quarterly taxable income-to-book income (12-month holding period)	Bl	Book leverage, Fama and French (1992)
Bl ^{q1}	Quarterly book leverage (1-month holding period)	Bl ^{q6}	Quarterly book leverage (6-month holding period)
Bl ^{q12}	Quarterly book leverage (12-month holding period)	Sg ^{q1}	Quarterly sales growth (1-month holding period)
Sg ^{q6}	Quarterly sales growth (6-month holding period)	Sg ^{q12}	Quarterly sales growth (12-month holding period)

Panel E: Intangibles (100)

Oca	Organizational capital/assets, Eisfeldt and Papanikolaou (2013)	Ioca	Industry-adjusted organizational capital /assets, Eisfeldt and Papanikolaou (2013)
Adm	Advertising expense-to-market, Chan, Lakonishok, and Sougiannis (2001)	gAd	Growth in advertising expense, Lou (2014)
Rdm	R&D-to-market, Chan, Lakonishok, and Sougiannis (2001)	Rdm ^{q1}	Quarterly R&D-to-market (1-month holding period)
Rdm ^{q6}	Quarterly R&D-to-market (6-month holding period)	Rdm ^{q12}	Quarterly R&D-to-market (12-month holding period)
Rds	R&D-to-sales, Chan, Lakonishok, and Sougiannis (2001)	Rds ^{q1}	Quarterly R&D-to-sales (1-month holding period)
Rds ^{q6}	Quarterly R&D-to-sales (6-month holding period)	Rds ^{q12}	Quarterly R&D-to-sales (12-month holding period)
OI	Operating leverage, Novy-Marx (2011)	OI ^{q1}	Quarterly operating leverage (1-month holding period)
OI ^{q6}	Quarterly operating leverage (6-month holding period)	OI ^{q12}	Quarterly operating leverage (12-month holding period)
Hn	Hiring rate, Belo, Lin, and Bazdresch (2014)	Rca	R&D capital-to-assets, Li (2011)
Bca	Brand capital-to-assets, Belo, Lin, and Vitorino (2014)	Aop	Analysts optimism, Frankel and Lee (1998)
Pafe	Predicted analysts forecast error, Frankel and Lee (1998)	Parc	Patent-to-R&D capital, Hirshleifer, Hsu, and Li (2013)

Crd	Citations-to-R&D expense, Hirshleifer, Hsu, and Li (2013)	Hs	Industry concentration (sales), Hou and Robinson (2006)
Ha	Industry concentration (total assets), Hou and Robinson (2006)	He	Industry concentration (book equity), Hou and Robinson (2006)
Age1	Firm age (1-month holding period), Jiang, Lee, and Zhang (2005)	Age6	Firm age (6-month holding period), Jiang, Lee, and Zhang (2005)
Age12	Firm age (12-month holding period), Jiang, Lee, and Zhang (2005)	D1	Price delay based on R^2 , Hou and Moskowitz (2005)
D2	Price delay based on slopes, Hou and Moskowitz (2005)	D3	Price delay based on slopes adjusted for standard errors, Hou and Moskowitz (2005)
dSi	% change in sales – % change in inventory, Abarbanell and Bushee (1998)	dSa	% change in sales – % change in accounts receivable, Abarbanell and Bushee (1998)
dGs	% change in gross margin – % change in sales, Abarbanell and Bushee (1998)	dSs	% change in sales – % change in SG&A, Abarbanell and Bushee (1998)
Etr	Effective tax rate, Abarbanell and Bushee (1998)	Lfe	Labor force efficiency, Abarbanell and Bushee (1998)
Ana1	Analysts coverage (1-month holding period), Elgers, Lo, and Pfeiffer (2001)	Ana6	Analysts coverage (6-month holding period), Elgers, Lo, and Pfeiffer (2001)
Ana12	Analysts coverage (12-month holding period), Elgers, Lo, and Pfeiffer (2001)	Tan	Tangibility of assets, Hahn and Lee (2009)
Tan ^{q1}	Quarterly tangibility (1-month holding period)	Tan ^{q6}	Quarterly tangibility (6-month holding period)
Tan ^{q12}	Quarterly tangibility (12-month holding period)	Rer	Real estate ratio, Tuzel (2010)
Kz	Financial constraints (the Kaplan-Zingales index), Lamont, Polk, and Saa-Requejo (2001)	Kz ^{q1}	Quarterly Kaplan-Zingales index (1-month holding period)
Kz ^{q6}	Quarterly Kaplan-Zingales index (6-month holding period)	Kz ^{q12}	Quarterly Kaplan-Zingales index (12-month holding period)
Ww	Financial constraints (the Whited-Wu index), Whited and Wu (2006)	Ww ^{q1}	Quarterly Whited-Wu index (1-month holding period)
Ww ^{q6}	Quarterly Whited-Wu index (6-month holding period)	Ww ^{q12}	Quarterly Whited-Wu index (12-month holding period)
Sdd	Secured debt-to-total debt, Valta (2016)	Cdd	Convertible debt-to-total debt, Valta (2016)
Vcf1	Cash flow volatility (1-month holding period), Huang (2009)	Vcf6	Cash flow volatility (6-month holding period), Huang (2009)
Vcf12	Cash flow volatility (12-month holding period), Huang (2009)	Cta1	Cash-to-assets (1-month holding period), Palazzo (2012)
Cta6	Cash-to-assets (6-month holding period), Palazzo (2012)	Cta12	Cash-to-assets (12-month holding period), Palazzo (2012)
Gind	Corporate governance, Gompers, Ishii, and Metrick (2003)	Acq	Accrual quality, Francis, Lafond, Olsson, and Schipper (2005)
Eper	Earnings persistence, Francis, Lafond, Olsson, and Schipper (2004)	Eprd	Earnings predictability, Francis, Lafond, Olsson, and Schipper (2004)
Esm	Earnings smoothness, Francis, Lafond, Olsson, and Schipper (2004)	Evr	Value relevance of earnings, Francis, Lafond, Olsson, and Schipper (2004)
Etl	Earnings timeliness, Francis, Lafond, Olsson, and Schipper (2004)	Ecs	Earnings conservatism, Francis, Lafond, Olsson, and Schipper (2004)
Frm	Pension funding rate (scaled by market equity), Franzoni and Martin (2006)	Fra	Pension funding rate (scaled by assets), Franzoni and Martin (2006)
Ala	Asset liquidity (scaled by book assets) Ortiz-Molina and Phillips (2014)	Alm	Asset liquidity (scaled by market assets), Ortiz-Molina and Phillips (2014)
Ala ^{q1}	Quarterly asset liquidity (book assets) (1-month holding period)	Ala ^{q6}	Quarterly asset liquidity (book assets) (1-month holding period)
Ala ^{q12}	Quarterly asset liquidity (book assets) (12-month holding period)	Alm ^{q1}	Quarterly asset liquidity (market assets) (1-month holding period)
Alm ^{q6}	Quarterly asset liquidity (market assets) (6-month holding period)	Alm ^{q12}	Quarterly asset liquidity (market assets) (12-month holding period)

Dls1	Disparity between long- and short-term earnings growth forecasts (1-month holding period), Da and Warachka (2011)	Dls6	Disparity between long- and short-term earnings growth forecasts (6-month holding period), Da and Warachka (2011)
Dls12	Disparity between long- and short-term earnings growth forecasts (12-month holding period), Da and Warachka (2011)	Dis1	Dispersion of analysts' earnings forecasts (1-month holding period), Diether, Malloy, and Scherbina (2002)
Dis6	Dispersion of analysts' earnings forecasts (6-month holding period), Diether, Malloy, and Scherbina (2002)	Dis12	Dispersion of analysts' earnings forecasts (12-month holding period), Diether, Malloy, and Scherbina (2002)
Dlg1	Dispersion in analyst long-term growth forecasts (1-month holding period), Anderson, Ghysels, and Juergens (2005)	Dlg6	Dispersion in analyst long-term growth forecasts (6-month holding period), Anderson, Ghysels, and Juergens (2005)
Dlg12	Dispersion in analyst long-term growth forecasts (12-month holding period), Anderson, Ghysels, and Juergens (2005)	R_a^1	12-month-lagged return, Heston and Sadka (2008)
R_n^1	Year 1-lagged return, nonannual Heston and Sadka (2008)	$R_a^{[2,5]}$	Years 2–5 lagged returns, annual Heston and Sadka (2008)
$R_n^{[2,5]}$	Years 2–5 lagged returns, nonannual Heston and Sadka (2008)	$R_a^{[6,10]}$	Years 6–10 lagged returns, annual Heston and Sadka (2008)
$R_n^{[6,10]}$	Years 6–10 lagged returns, nonannual Heston and Sadka (2008)	$R_a^{[11,15]}$	Years 11–15 lagged returns, annual Heston and Sadka (2008)
$R_n^{[11,15]}$	Years 11–15 lagged returns, nonannual Heston and Sadka (2008)	$R_a^{[16,20]}$	Years 16–20 lagged returns, annual Heston and Sadka (2008)
$R_n^{[16,20]}$	Years 16–20 lagged returns, nonannual Heston and Sadka (2008)	Ob	Order backlog, Rajgopal, Shevlin, and Venkatachalam (2003)

Panel F: Trading frictions (96)

Me	Market equity, Banz (1981)	Iv	Idiosyncratic volatility, Ali, Hwang, and Trombley (2003)
Ivff1	Idiosyncratic volatility per the FF 3-factor model (1-month holding period), Ang, Hodrick, Xing, and Zhang (2006)	Ivff6	Idiosyncratic volatility per the FF 3-factor model (6-month holding period), Ang, Hodrick, Xing, and Zhang (2006)
Ivff12	Idiosyncratic volatility per the FF 3-factor model (12-month holding period), Ang, Hodrick, Xing, and Zhang (2006)	Ivc1	Idiosyncratic volatility per the CAPM (1-month holding period)
Ivc6	Idiosyncratic volatility per the CAPM (6-month holding period)	Ivc12	Idiosyncratic volatility per the CAPM (12-month holding period)
Ivq1	Idiosyncratic volatility per the q -factor model (1-month holding period)	Ivq6	Idiosyncratic volatility per the q -factor model (6-month holding period)
Ivq12	Idiosyncratic volatility per the q -factor model (12-month holding period), Ang, Hodrick, Xing, and Zhang (2006)	Tv1	Total volatility (1-month holding period), Ang, Hodrick, Xing, and Zhang (2006)
Tv6	Total volatility (6-month holding period), Ang, Hodrick, Xing, and Zhang (2006)	Tv12	Total volatility (12-month holding period), Ang, Hodrick, Xing, and Zhang (2006)
Sv1	Systematic volatility risk (1-month holding period), Ang, Hodrick, Xing, and Zhang (2006)	Sv6	Systematic volatility risk (6-month holding period), Ang, Hodrick, Xing, and Zhang (2006)
Sv12	Systematic volatility risk (12-month holding period), Ang, Hodrick, Xing, and Zhang (2006)	$\beta 1$	Market beta (1-month holding period) Fama and MacBeth (1973)
$\beta 6$	Market beta (6-month holding period) Fama and MacBeth (1973)	$\beta 12$	Market beta (12-month holding period) Fama and MacBeth (1973)
$\beta^{FP} 1$	The Frazzini-Pedersen (2014) beta (1-month holding period)	$\beta^{FP} 6$	The Frazzini-Pedersen (2014) beta (6-month holding period)
$\beta^{FP} 12$	The Frazzini-Pedersen (2014) beta (12-month holding period)	$\beta^D 1$	The Dimson (1979) beta (1-month holding period)

β^D6	The Dimson (1979) beta (6-month holding period)	β^D12	The Dimson (1979) beta (12-month holding period)
Tur1	Share turnover (1-month holding period), Datar, Naik, and Radcliffe (1998)	Tur6	Share turnover (6-month holding period), Datar, Naik, and Radcliffe (1998)
Tur12	Share turnover (12-month holding period), Datar, Naik, and Radcliffe (1998)	Cvt1	Coefficient of variation for share turnover (1-month holding period), Chordia, Subrahmanyam, and Anshuman (2001)
Cvt6	Coefficient of variation for share turnover (1-month holding period), Chordia, Subrahmanyam, and Anshuman (2001)	Cvt12	Coefficient of variation for share turnover (12-month holding period), Chordia, Subrahmanyam, and Anshuman (2001)
Dtv1	Dollar trading volume (1-month holding period), Brennan, Chordia, and Subrahmanyam (1998)	Dtv6	Dollar trading volume (6-month holding period), Brennan, Chordia, and Subrahmanyam (1998)
Dtv12	Dollar trading volume (12-month holding period), Brennan, Chordia, and Subrahmanyam (1998)	Cvd1	Coefficient of variation for dollar trading volume (1-month holding period), Chordia, Subrahmanyam, and Anshuman (2001)
Cvd6	Coefficient of variation for dollar trading volume (6-month holding period), Chordia, Subrahmanyam, and Anshuman (2001)	Cvd12	Coefficient of variation for dollar trading volume (12-month holding period), Chordia, Subrahmanyam, and Anshuman (2001)
Pps1	Share price (1-month holding period), Miller and Scholes (1982)	Pps6	Share price (6-month holding period), Miller and Scholes (1982)
Pps12	Share price (12-month holding period), Miller and Scholes (1982)	Ami1	Absolute return-to-volume (1-month holding period), Amihud (2002)
Ami6	Absolute return-to-volume (6-month holding period), Amihud (2002)	Ami12	Absolute return-to-volume (12-month holding period), Amihud (2002)
Lm ¹ 1	Prior 1-month turnover-adjusted number of zero daily trading volume (1-month holding period), Liu (2006)	Lm ¹ 6	Prior 1-month turnover-adjusted number of zero daily trading volume (6-month holding period), Liu (2006)
Lm ¹ 12	Prior 1-month turnover-adjusted number of zero daily trading volume (12-month holding period), Liu (2006)	Lm ⁶ 1	Prior 6-month turnover-adjusted number of zero daily trading volume (1-month holding period), Liu (2006)
Lm ⁶ 6	Prior 6-month turnover-adjusted number of zero daily trading volume (6-month holding period), Liu (2006)	Lm ⁶ 12	Prior 6-month turnover-adjusted number of zero daily trading volume (12-month holding period), Liu (2006)
Lm ¹² 1	Prior 12-month turnover-adjusted number of zero daily trading volume (1-month holding period), Liu (2006)	Lm ¹² 6	Prior 12-month turnover-adjusted number of zero daily trading volume (6-month holding period), Liu (2006)
Lm ¹² 12	Prior 12-month turnover-adjusted number of zero daily trading volume (12-month holding period), Liu (2006)	Mdr1	Maximum daily return (1-month holding period), Bali, Cakici, and Whitelaw (2011)
Mdr6	Maximum daily returns (6-month holding period), Bali, Cakici, and Whitelaw (2011)	Mdr12	Maximum daily return (12-month holding period), Bali, Cakici, and Whitelaw (2011)
Ts1	Total skewness (1-month holding period), Bali, Engle, and Murray (2015)	Ts6	Total skewness (6-month holding period), Bali, Engle, and Murray (2015)
Ts12	Total skewness (12-month holding period), Bali, Engle, and Murray (2015)	Isc1	Idiosyncratic skewness per the CAPM (1-month holding period)
Isc6	Idiosyncratic skewness per the CAPM (6-month holding period)	Isc12	Idiosyncratic skewness per the CAPM (12-month holding period)
Isff1	Idiosyncratic skewness per the FF 3-factor model (1-month holding period)	Isff6	Idiosyncratic skewness per the FF 3-factor model (6-month holding period)
Isff12	Idiosyncratic skewness per the FF 3-factor model (12-month holding period)	Isq1	Idiosyncratic skewness per the q -factor model (1-month holding period)
Isq6	Idiosyncratic skewness per the q -factor model (6-month holding period)	Isq12	Idiosyncratic skewness per the q -factor model (12-month holding period)
Cs1	Coskewness (1-month holding period), Harvey and Siddique (2000)	Cs6	Coskewness (6-month holding period), Harvey and Siddique (2000)

Cs12	Coskewness (12-month holding period), Harvey and Siddique (2000)	Srev	Short-term reversal, Jegadeesh (1990)
β^-1	Downside beta (1-month holding period) Ang, Chen, and Xing (2006)	β^-6	Downside beta (6-month holding period) Ang, Chen, and Xing (2006)
β^-12	Downside beta (12-month holding period) Ang, Chen, and Xing (2006)	Tail1	Tail risk (1-month holding period) Kelly and Jiang (2014)
Tail6	Tail risk (6-month holding period) Kelly and Jiang (2014)	Tail12	Tail risk (12-month holding period) Kelly and Jiang (2014)
$\beta^{lcc}1$	Liquidity beta (illiquidity-illiquidity) (1-month holding period), Acharya and Pedersen (2005)	$\beta^{lcc}6$	Liquidity beta (illiquidity-illiquidity) (6-month holding period), Acharya and Pedersen (2005)
$\beta^{lcc}12$	Liquidity beta (illiquidity-illiquidity) (12-month holding period), Acharya and Pedersen (2005)	$\beta^{lrc}1$	Liquidity beta (return-illiquidity) (1-month holding period), Acharya and Pedersen (2005)
$\beta^{lrc}6$	Liquidity beta (return-illiquidity) (6-month holding period), Acharya and Pedersen (2005)	$\beta^{lrc}12$	Liquidity beta (return-illiquidity) (12-month holding period), Acharya and Pedersen (2005)
$\beta^{lcr}1$	Liquidity beta (illiquidity-return) (1-month holding period), Acharya and Pedersen (2005)	$\beta^{lcr}6$	Liquidity beta (illiquidity-return) (6-month holding period), Acharya and Pedersen (2005)
$\beta^{lcr}12$	Liquidity beta (illiquidity-return) (12-month holding period), Acharya and Pedersen (2005)	Shl1	The high-low bid-ask spread estimator (1-month holding period), Corwin and Schultz (2012)
Shl6	The high-low bid-ask spread estimator (6-month holding period), Corwin and Schultz (2012)	Shl12	The high-low bid-ask spread estimator (12-month holding period), Corwin and Schultz (2012)
Sba1	Bid-ask spread (1-month holding period) Hou and Loh (2015)	Sba6	Bid-ask spread (6-month holding period) Hou and Loh (2015)
Sba12	Bid-ask spread (12-month holding period) Hou and Loh (2015)	$\beta^{Lev}1$	Leverage beta (1-month holding period) Adrian, Etula, and Muir (2014)
$\beta^{Lev}6$	Leverage beta (6-month holding period) Adrian, Etula, and Muir (2014)	$\beta^{Lev}12$	Leverage beta (12-month holding period) Adrian, Etula, and Muir (2014)

Table A2 : Insignificant Anomalies at the 5% Level with All-but-micro Breakpoints and Equal-weighted Returns, January 1967 to December 2014, 576 Months

We report the average returns (m) of the high-minus-low deciles and their t -statistic (t_m) adjusted for heteroscedasticity and autocorrelations. Table A1 provides a brief description of the symbols.

	Sue12	R^{11}_{12}	Rs12	Tes12	Nei12	52w1	Ile12	Bm ^{q1}	Bm ^{q6}	Dm	Dm ^{q1}	Dm ^{q6}	Dm ^{q12}	Am ^{q1}	Am ^{q6}	Am ^{q12}	Efp1	Efp6	Efp12
m	0.18	0.31	0.11	0.12	0.13	0.32	0.13	0.39	0.35	0.39	0.36	0.36	0.38	0.37	0.32	0.42	0.50	0.34	0.36
t_m	1.85	1.39	1.01	1.15	1.30	1.04	1.01	1.45	1.38	1.84	1.36	1.44	1.60	1.16	1.06	1.46	1.60	1.14	1.29
	Dp	Dp ^{q1}	Dp ^{q6}	Dp ^{q12}	Nop ^{q1}	Sr	Ocp ^{q6}	Ocp ^{q12}	Vfp	Ebp ^{q1}	Ebp ^{q6}	Ebp ^{q12}	Ndp ^{q1}	Ndp ^{q6}	Ndp ^{q12}	Ltg1	Ltg6	Ltg12	dNcl
m	0.18	0.20	0.12	0.15	0.45	-0.26	0.39	0.44	0.28	0.32	0.35	0.46	0.24	0.24	0.24	-0.52	-0.53	-0.50	-0.16
t_m	0.87	0.94	0.57	0.73	1.77	-1.73	1.43	1.76	1.07	0.96	1.08	1.52	1.05	1.15	1.26	-1.13	-1.20	-1.16	-1.90
	dSti	Roe12	Roa12	Rna	Pm	Ato	Pm ^{q6}	Pm ^{q12}	Gla	Ole	Ola	Fp1	Fp12	O ^{q6}	O ^{q12}	Z	Z ^{q1}	Z ^{q6}	Z ^{q12}
m	0.03	0.35	0.37	0.22	0.20	0.21	0.31	0.22	0.29	0.18	0.28	-0.48	-0.44	-0.23	-0.21	-0.12	-0.11	-0.19	-0.20
t_m	0.22	1.84	1.85	1.35	0.85	1.22	1.41	1.04	1.85	0.97	1.64	-1.67	-1.73	-1.43	-1.41	-0.54	-0.45	-0.82	-0.88
	Cr1	Cr6	Cr12	Tbi	Tbi ^{q1}	Tbi ^{q6}	Tbi ^{q12}	Bl	Bl ^{q1}	Bl ^{q6}	Bl ^{q12}	Sg ^{q1}	Sg ^{q6}	Rds	Rds ^{q1}	Rds ^{q6}	Rds ^{q12}	Rca	Bca
m	-0.28	-0.35	-0.38	0.15	0.11	0.12	0.13	0.05	0.31	0.23	0.16	0.28	-0.09	0.02	-0.14	0.12	0.07	0.41	0.30
t_m	-0.89	-1.08	-1.16	1.42	1.08	1.36	1.51	0.30	1.50	1.15	0.84	1.66	-0.55	0.06	-0.26	0.25	0.15	1.25	1.40
	Aop	Pafe	Crđ	Hs	Ha	He	Age1	Age6	Age12	D1	D2	D3	dSa	dGs	dSs	Etr	Lfe	Anal	Ana6
m	-0.13	0.06	0.15	-0.16	-0.13	-0.04	0.24	0.31	0.33	0.07	0.13	0.11	0.11	0.05	-0.12	0.01	0.04	-0.16	-0.12
t_m	-0.96	0.17	1.06	-1.06	-0.77	-0.20	0.96	1.09	1.50	0.55	1.66	0.85	1.42	0.48	-1.19	0.10	0.46	-1.07	-0.86
	Ana12	Tan	Tan ^{q1}	Tan ^{q6}	Tan ^{q12}	Kz	Kz ^{q1}	Kz ^{q6}	Kz ^{q12}	Ww	Ww ^{q1}	Ww ^{q6}	Ww ^{q12}	Sdd	Cdd	Vcf1	Vcf6	Vcf12	Cta1
m	-0.06	0.14	0.32	0.29	0.21	0.03	0.13	0.11	0.07	0.03	-0.09	-0.07	-0.10	-0.19	-0.05	-0.50	-0.48	-0.47	-0.02
t_m	-0.41	0.91	1.77	1.75	1.34	0.20	0.68	0.59	0.35	0.13	-0.29	-0.24	-0.35	-1.36	-0.23	-1.73	-1.79	-1.79	-0.06
	Cta6	Cta12	Gind	Acq	Eper	Esm	Evr	Etl	Ecs	Frm	Fra	Alm	Ala ^{q1}	Ala ^{q6}	Ala ^{q12}	Dls1	Dls6	Dls12	Dis1
m	-0.03	-0.08	-0.02	-0.07	-0.08	-0.04	0.14	0.16	0.10	0.11	0.04	0.13	0.03	-0.03	-0.17	-0.36	-0.14	-0.12	-0.41
t_m	-0.11	-0.31	-0.11	-0.32	-0.71	-0.33	1.62	1.82	1.60	0.76	0.37	0.89	0.09	-0.12	-0.66	-1.83	-0.84	-0.83	-1.80
	Dis6	Dis12	Dlg1	Dlg6	Dlg12	$R_n^{[11,15]}$	$R_n^{[16,20]}$	Ob	Me	Iv	Ivff1	Ivff6	Ivff12	Ivc6	Ivc12	Ivq6	Ivq12	Tv1	Tv6
m	-0.36	-0.25	-0.22	-0.16	-0.15	-0.23	-0.15	0.07	-0.20	-0.55	-0.63	-0.52	-0.45	-0.54	-0.46	-0.53	-0.46	-0.67	-0.52
t_m	-1.72	-1.25	-0.90	-0.68	-0.64	-1.74	-1.31	0.43	-1.13	-1.49	-1.95	-1.72	-1.51	-1.77	-1.53	-1.77	-1.56	-1.91	-1.59
	Tv12	Sv6	Sv12	β_1	β_6	β_{12}	β^{FP}_1	β^{FP}_6	β^{FP}_{12}	β^D_1	β^D_6	β^D_{12}	Tur1	Tur6	Tur12	Cvt1	Cvt6	Cvt12	Cvd1
m	-0.46	-0.16	-0.13	-0.09	-0.05	-0.12	-0.29	-0.21	-0.20	-0.12	0.01	-0.06	-0.44	-0.48	-0.47	0.07	0.08	0.11	0.10
t_m	-1.43	-1.24	-1.25	-0.25	-0.14	-0.35	-0.80	-0.61	-0.59	-0.55	0.07	-0.36	-1.71	-1.87	-1.90	0.52	0.65	0.90	0.67
	Cvd6	Cvd12	Pps1	Pps6	Pps12	Amil	Lm ¹ 1	Lm ¹ 6	Lm ¹ 12	Mdr6	Mdr12	Ts1	Ts6	Ts12	Isc1	Isc6	Isc12	Isff1	Isff6
m	0.10	0.12	0.08	0.15	0.07	0.28	0.08	0.42	0.41	-0.40	-0.33	-0.12	-0.02	-0.01	-0.03	0.00	0.02	0.04	0.02
t_m	0.74	0.89	0.31	0.63	0.31	1.87	0.38	1.96	1.93	-1.44	-1.20	-1.32	-0.29	-0.23	-0.37	0.01	0.38	0.52	0.28
	Isff12	Isq1	Isq6	Isq12	Cs1	Cs6	Cs12	β^{-}_1	β^{-}_6	β^{-}_{12}	Tail1	Tail6	Tail12	β^{lcc}_1	β^{lcc}_6	β^{lcc}_{12}	β^{lrc}_1	β^{lrc}_6	β^{lrc}_{12}
m	0.02	0.11	0.01	0.02	-0.12	-0.02	-0.01	-0.53	-0.52	-0.42	0.18	0.20	0.22	0.17	0.16	0.11	0.10	0.08	0.14
t_m	0.45	1.42	0.14	0.49	-1.19	-0.35	-0.38	-1.51	-1.49	-1.28	1.01	1.25	1.55	1.07	1.10	0.75	0.40	0.31	0.58
	β^{lcr}_1	β^{lcr}_6	β^{lcr}_{12}	Shl1	Shl6	Shl12	Sba1	Sba6	Sba12	β^{Lev}_1	β^{Lev}_6	β^{Lev}_{12}							
m	0.02	0.01	0.04	-0.44	-0.41	-0.35	-0.19	-0.20	-0.14	0.32	0.27	0.24							
t_m	0.25	0.07	0.51	-1.50	-1.54	-1.39	-0.91	-1.08	-0.81	1.62	1.37	1.24							